

IMPROVED DELAYED NEUTRON GROUP PARAMETERS FOR MOLTEN SALT REACTORS

BY

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DISSERTATION

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Abstract

Fluid-fueled molten salt reactors (MSRs) have garnered interest since the Generation IV International Forum in the early 2000s, where the MSR was designated as one of six advanced reactors of interest. A unique consideration in MSRs are delayed neutron precursors (DNPs), which move from areas of high importance in the core to areas of lower importance in the core and ex-core regions. This causes the effective delayed neutron fraction to decrease, impacting reactor behavior in multiple ways. DNPs are important for modeling and simulation of MSRs.

Many studies have investigated how the effective delayed neutron fraction changes based on the movement of fuel salt in MSRs. The movement of the DNPs is modeled using DNP groups, typically six groups, with each group k consisting of a decay constant λ_k and yield of delayed neutrons per fission event $\bar{\nu}_{d,k}$. However, these studies do not account for the effects of fuel salt movement and chemical removal rates on the DNP group parameters, λ_k and $\bar{\nu}_{d,k}$, used in the transient simulations. Neglecting these MSR-specific phenomena reduces accuracy in the DNP group parameters in MSR simulations.

Fuel salt recirculation and chemical removal rates are important to consider, as simulations use the DNP group parameters to enable safety, security, and safeguards by design. By using DNP group parameters with greater accuracy, models can more accurately reflect physical phenomena. Having greater accuracy in predictive models is key for successful MSR deployment.

This work resolves DNP group parameters for MSRs using the microscopic method, also known as the aggregate or summation method. This method entails using individual DNP data to reconstruct the DNP groups by summing the delayed neutron count rate contribution of each DNP in simulated irradiation experiments. Pulse irradiation experiments occur at very short time scales relative to MSR physical phenomena and are used to generate the short-lived DNP group parameters. Saturation, or infinite, irradiation experiments occur at time scales several times longer than the longest DNP half-life and are used to generate the longer lived DNP group parameters. Because the saturation irradiation lasts on the order of minutes, the thermal-hydraulic and chemical effects in MSRs have a non-negligible impact. Therefore, the longer-lived DNPs' group parameters will vary, while the shorter-lived DNPs' group parameters will remain constant.

A non-linear least squares solver is used to generate an improved set of DNP group parameters. The method is verified by comparing the improved DNP group parameters to the data used to generate those parameters, as the total delayed neutron yields and average half-lives should be nearly identical. Verification of the parameters is conducted via comparison with total delayed neutron yields and average half-lives from the literature.

The methodology used to generate these parameters can be replicated for other MSR conditions, enabling the formation of more accurate sets of DNP group parameters for MSRs. The proposed work includes: development and evaluation of a more sophisticated model for generating DNP group parameters; validation via comparison with the Molten Salt Reactor Experiment (MSRE) during pump start-up and coast-down; generation of delayed neutron group spectra; evaluation of various safety, security, and safeguards-related transients using improved DNP group parameters; and comparison of how different datasets affect the DNP group parameters.

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Chapter 1

Introduction

One generation IV reactor design is the fluid-fueled molten salt reactor (MSR) [91]. MSRs are advanced reactors intended to improve safety and efficiency over traditional reactor designs. Instead of a solid fuel, MSRs use a liquid fuel composed of a molten salt in which some fissile nuclide, such as ^{235}U , is dissolved. Molten salt fuels offer many benefits: high operating temperature, low operating pressure, high thermal conductivity, and high heat capacity. Other benefits include a strong, expansion driven negative temperature coefficient of reactivity, high resource utilization through higher burn-up, reduced cost of fabricating and transporting fuel elements, and improved neutron economy due to online fission product removal [167].

MSRs have several barriers to deployment which must be addressed before they are commercialized. Corrosion and material degradation pose an issue, as the high temperature salts corrode materials. Another deployment barrier involves reactor safety, security, and safeguards by design. MSRs have distinct designs compared with traditional light water reactors, such as requiring bulk rather than discrete item material control and accounting. To properly design MSRs, models must be able to simulate them accurately. One specific phenomenon of critical importance in MSRs is the circulating delayed neutron precursors (DNPs).

DNPs are nuclides that undergo β^- decay, producing a daughter, called an emitter, which emits a neutron [123]. Without properly modeling the DNPs, reactor simulations may predict incorrect power responses, reactivity feedback, control element worth, and reactor controllability [44, 45, 96]. Reactor physics simulations typically do not model DNPs explicitly, but instead represent them using a smaller number of artificial groups, which preserve the same physical effect at much lower computational cost.

These artificial groups are referred to as DNP groups, and each group k contains two parameters: a decay constant λ_k and an average yield of delayed neutrons per fission event $\bar{\nu}_{d,k}$. Although they are artificial, the use of six groups is common for capturing the general delayed neutron emission rate from all the DNPs present in the system. The DNP group parameters provide sufficient data to calculate the rate of delayed neutrons emitted by group k over a given time period.

Table 1.1 shows a brief historical timeline of DNP group parameter modeling advancements. Early work, shown from 1946 through 1957, consisted of fissile sample irradiation in reactors to calculate DNP group parameters. From 1965 onward, investigation started in more detail on individual DNPs and how those DNPs could be mapped onto DNP group parameters. DNP modeling and group parameter formulation advanced rapidly thanks to increased experimental data regarding half-lives and emission probabilities of individual DNPs, which enabled more accurate group parameter formulation.

However, these formulations of DNP group parameters did not account for how MSR fuel movement and

chemical removal affects the DNP group parameters. For example, if an MSR chemically removed significant quantities of DNPs, that removal would need to be captured in the group parameters; these chemical removals are element-dependent, which the artificial group structure does not capture.

There is a gap in the literature, which can be closed by capturing the effects of chemical removal and recirculation on the DNP group parameters. Although researchers have analyzed DNPs in MSRs, that work has not investigated this gap [202, 206, 100, 174, 95]. Most of these studies investigate DNPs movement during transients. However, accurate transient models require accurate DNP group parameters. Thus, the next step in group parameter development is DNP group parameters that accurately reflect recirculation and chemical removal in MSRs.

Table 1.1: Timeline of advancements in DNP modeling.

Year	Advancement
1939	Roberts et al. discover delayed neutrons [147]; Bohr and Wheeler explain delayed neutrons [20].
1946	Burgy et al. conduct initial spectral measurements [30].
1947	Hughes et al. initial use of six DNP groups at Argonne heavy water pile [78].
1957	Keepin et al. generate improved DNP group parameters [89, 196].
1965	Amiel identifies seven DNPs that account for 80% of delayed neutrons [6].
1969	Marmol compiles experimental data on emission probability and half-life for 34 DNPs [46].
1972	Tomlinson maps 38 DNPs onto six groups [187].
1977	Saphier et al. construct delayed neutron spectra from 20 DNPs [162].
1982	Rudstam generates DNP group parameters and spectra using 67 DNPs based on half-lives [154].
1983	England et al. generate DNP group parameters and spectra using 105 DNPs based on half-lives [54].
1988	Brady generates DNP group parameters and spectra using 271 DNPs with least squares [24]. Group parameters included in ENDF/B-VI.8 and remain unchanged through ENDF/B-VIII.0 [197, 28, 153, 22, 53, 23].
1999	Parish et al. state DNP group parameters in Evaluated Nuclear Data File (ENDF) are in need of revision [129].
2014	The International Atomic Energy Agency (IAEA) Coordinated Research Project (CRP) on Development of a Reference Database for β -delayed neutron emission data [118].
2020	Leconte et al. use more than 400 DNPs to combine data with macroscopic experimental measurements [99].
2025	This study develops DNP group parameters for use in MSRs.

This research aims to generate improved DNP group parameters that account for MSR recirculation and chemical removal, verify the parameters with the literature, validate the parameters by simulating transients and comparing with experimental data, and evaluate the impact of the improved parameters on safety, safeguards, and security by design. For example, chemical removal of DNPs decreases the number of delayed neutrons emitted per fission event on average. This reduces the total delayed neutron yield, decreases the effective delayed neutron fraction, shortens the reactor period, and reduces reactor controllability. Current DNP group parameters in MSRs simulations may overestimate the number of delayed neutrons in the system, resulting in non-conservative assessments of controllability.

The recirculation and chemical removal in MSRs can be incorporated into fissile sample irradiation simulations at varying levels of fidelity, with associated differences in cost and accuracy. This document describes three models that simulate irradiation of a fissile sample, from which DNP group parameters can be calculated. These models use different simplifications, but all of them simulate a incident neutron source that irradiates a target fissile sample consisting of a single fissile nuclide. After irradiation, the concentrations of the DNPs are extracted and used to calculate the delayed neutron count rate of the sample. Then, some selected number of DNP groups with decay constants λ_k and yields $\bar{v}_{d,k}$ are used to fit that delayed neutron count rate.

The least accurate and fastest model is the 0D scaled model, which has been completed for the preliminary results. This model neglects time dependence, decay chains, and cross-sections while scaling the flux and chemical removal rates based on the fraction of fuel salt in different regions of the primary loop. The intermediate approach is the 0D flow model, which is part of the proposed work. This model accounts for time dependence of DNP concentrations and decay chains while applying the flux and chemical removal rates only in the relevant parts of the primary loop. It is referred to as the 0D flow model because it captures the effects of the fuel salt flow by using residence times. The most accurate and slowest approach is the spatially resolved model, which is discussed as potential future work. This model accounts for time and space dependence of DNP concentrations across the entire primary loop. Two types of irradiations are considered: saturation irradiations, in which long-lived DNPs dominate, and pulse irradiations, in which short-lived DNPs dominate.

The preliminary results cover impactful DNPs, DNP group parameter sensitivity studies, and 0D scaled model verification. The results show the total delayed neutron yield ($\bar{v}$) and average half-life ($\tau_{1/2}$) values are within the propagated experimental uncertainty bounds between the raw DNP data and the DNP group parameters. The match between the raw data and the group parameters shows the effect of each DNP is captured in the DNP group parameters. Sensitivity studies reveal each DNP affects multiple DNP group parameters, not only the group with the most similar half-life. Verification with the literature shows the 0D scaled model works well for longer-lived groups but poorly for shorter-lived groups. The 0D scaled model only simulates saturation irradiations, which provide less data for shorter-lived DNPs than pulse irradiations.

The proposed work starts with 0D flow model development, followed by a verification stage to ensure improved DNP group parameters are correct for a static case. The work then proposes 0D flow model validation using a transient simulation of the Molten Salt Reactor Experiment (MSRE) compared to experimental data to ensure improved DNP group parameters are physically accurate. The improved group spectra are proposed to be included in DNP group parameter formulation and to be verified for a static case. Finally, the work proposes simulation of MSRE transients for various safety, security, and safeguards scenarios using improved and pre-existing DNP group parameters to study their impact.

Section 5.4 presents potential future work. Future work includes spatially resolved model development, uncertainty tracking of MSR physical phenomena in DNP group parameters, and investigation of the effects of varying the incident neutron energy spectrum on DNP group parameters.

Chapter 2

Background and Motivation

This chapter provides the background and motivation for generating more accurate DNP group parameters for MSRs and implementation of those parameters for safety, security, and safeguards design purposes. An overview including information on DNPs and MSR fuel depletion is followed by relevant context motivating the problem, including safety, security, and safeguards by design. The DNP group parameters in MSRs differ from traditional group parameters in stationary systems because the fuel circulates and undergoes online chemical reprocessing. These effects can be captured in spatially resolved MSR fuel depletion simulations. The existing methodology for DNP group parameter generation does not account for the recirculation and chemical removal present in MSR fuel depletion [89, 189, 111, 13, 194, 24, 78, 108, 109, 137, 43, 137, 156, 197, 22, 99]. This work presents more accurate DNP group parameters that account for recirculation and chemical removal. These improved DNP group parameters are particularly important for safety, security, and safeguards related concerns.

2.1 Delayed Neutron Precursors

Before delving deeper into the DNP group parameters, it is useful to first understand DNPs conceptually because they are an important consideration in nuclear reactors [44]. DNPs are nuclides which decay via β^- decay, producing a daughter, called an emitter, which emits a neutron [123]. In this work, the term DNP will be solely used rather than referring to emitters, as the delay of the neutron emission is effectively only due to the β^- decay [20, 128]. Distinguishing between the emitter and the DNP is not relevant for this work.

Equation (2.1) shows a simplified equation for the reactor period τ_e , calculated using a single DNP group and without the effect of prompt neutrons included [39]. The delay in neutron emission significantly affects the reactor period, or the time it takes for the reactor power to increase by a factor of e , and transient behavior in general.

$$\tau_e \approx \frac{\beta_{eff} - \rho}{\rho \lambda_{eff}} \quad (2.1)$$

where

β_{eff} = the effective delayed neutron fraction [-]

ρ = the reactivity [-]

λ_{eff} = the one DNP group decay constant [s^{-1}].

The effective delayed neutron fraction is discussed in more detail in Section 2.1.2. The one DNP group approximation is not used in this study, so this approximation is not discussed in greater detail. However, this simplified equation demonstrates how the reactor period depends greatly on the effective delayed neutron fraction, which itself is based on the DNPs.

Equation (2.2) shows the time-dependent rate of delayed neutron emission from a single DNP as:

$$\dot{n}_{d,i}(t) = P_{n,i} \lambda_i N_i(t), \quad (2.2)$$

where

$\dot{n}_{d,i}$ = the delayed neutron activity, or count rate, from DNP i [$n \cdot s^{-1}$]

$P_{n,i}$ = the average number of delayed neutrons emitted from
decay of nuclide i , or the emission probability [-]

λ_i = the decay constant of DNP i [s^{-1}]

$N_i(t)$ = the number of atoms of DNP i [*atoms*].

In a solid-fueled reactor, such as a light water reactor (LWR), spatial resolution is not necessary to properly model the DNPs, as the fuel remains stationary. In MSRs, the movement of the fuel salt necessitates spatial resolution for tracking DNP distributions and concentrations. Modeling space and time together largely increases computational cost, making problems involving DNPs in MSRs more challenging than handling DNPs in stationary fuels.

Properly modeling the DNPs in reactors is important, otherwise simulations may predict incorrect power responses, reactivity feedback, control element worth, and reactor controllability [44, 45, 96]. Section 2.1.3 shows how the power response depends on the DNPs. The reactor controllability varies based on the available number of DNPs. If there are fewer delayed neutrons available due to movement of the fuel salt, such as in an MSR, then the reactor period becomes shorter, making the reactor harder to control.

2.1.1 Measurement

Macroscopic and microscopic approaches are used for measuring DNPs [129, 156]. The macroscopic approach uses a fissile sample irradiation to get the delayed neutron count rate. The microscopic approach uses the data from many experiments combined together to simulate a fissile sample irradiation and generate the delayed neutron count rate. Although the microscopic approach can be more challenging due to dependence on the work of many individual experiments, it offers more detail on the delayed neutron count rate.

After the delayed neutron count rate is collected, both approaches then try to approximate the behavior of all the DNPs with only a few artificial DNPs, known as DNP groups. These groups consist of decay constants λ_k and yields $\bar{\nu}_{d,k}$. These groups have historically been formed based on the DNP half-lives or based on what parameters best fit the delayed neutron count rate.

Macroscopic Measurement of DNPs

The macroscopic approach entails irradiating a sample containing a single fissile nuclide at or near a given neutron energy. This delayed neutron count rate from this sample can then be measured and used to formulate DNP group

parameters. Analysis and collection of macroscopic experiment data is common in the literature [89, 189, 111, 13, 194, 24, 78, 108, 109, 40, 164].

Small samples are used to minimize neutron multiplication within the sample to avoid spectral effects [89]. Spectral effects are caused by differing neutron energy spectra between each DNP group and the incident irradiating neutrons [89]. Another reason for using a small sample is because of physical challenges associated with the experimental sample handling (rabbit) system [29].

Two different types of irradiation are typically used: pulse and infinite. Pulse irradiation irradiates the sample with a pulse shorter than the shortest DNP half-life, and captures the short-lived DNPs with more accuracy than it does the long-lived DNPs [24]. Infinite irradiation irradiates the sample for longer than several times the longest half-life, and it captures the long-lived DNPs with more accuracy than the short-lived DNPs [24]. After irradiation, the delayed neutrons from the sample are counted as a function of time. Figure 2.1 (recreated from [89]) shows an experimental setup for collecting delayed neutrons. The delayed neutron count rate as a function of time after irradiation, $\dot{n}_d(t)$, is given as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t), \quad (2.3)$$

where

$\dot{n}_d(t)$ = the delayed neutron count rate [$n \cdot s^{-1}$]

I = the total number of DNPs [-]

ϵ = the detector efficiency [-].

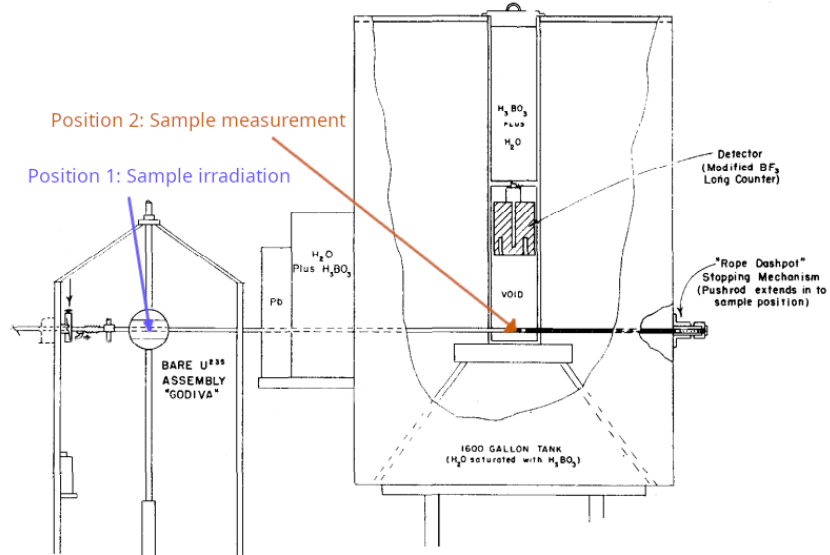


Figure 2.1: Simplified experimental setup for measuring delayed neutron emission from irradiation of a 3 gram sample in Godiva referenced from [89].

Once this delayed neutron count rate is generated, the data is used to generate DNP group parameters, λ_k

and $\bar{\nu}_{d,k}$, for that fissile nuclide and neutron energy. Three different approaches have been used to generate DNP group parameters from macroscopic data: exponential stripping (or peeling) [29, 88], least squares [24, 89], and half-lives of specific DNPs [162]. Some approaches have even used a combination, such as least squares fitting with specific half-lives [178]. The exponential stripping, or peeling, method is a more subjective form of least squares fitting, where exponential curves are subtracted away from the raw data from right to left [141]. The use of specific half-lives to form the DNP groups is less subjective and offers a more physical interpretation of the groups. The least squares approaches are the most objective approach and generate groups which physically do not embody a single DNP. Instead, they group together the DNPs and effectively capture the overall physical effect. For this work, a least squares approach is used, as it offers an objective formulation that can be generically applied. The specifics of the least squares approach used to form the DNP parameters is discussed in Section 2.1.2.

Challenges associated with collecting experimental data include experimental noise and signal obfuscation due to moderation [123]. Experimental noise becomes significant after long periods of time relative to the half-life of the longest-lived DNP, as the delayed neutron counts decrease exponentially. The short-lived groups cannot be measured immediately after irradiation; they must first be transported to a delayed neutron counter. This transport time results in missing very short-lived counts. After a pulse irradiation, the long-lived groups form a background that interferes with measurement of the short-lived groups [123]. Due to these effects, the delayed neutron measurements are most accurate for intermediate times, while the short-lived and long-lived DNPs have greater uncertainty.

Similarly, uncertainty arises from the measurement of the number of fissions. Although the number of fission events can be measured somewhat accurately using a fission chamber after irradiation or other techniques, the fission chamber does not account for source multiplication within the irradiated sample. Keepin et al. suggest a technique to correct for the sample self-multiplication. The sample self-multiplication require a correction factor of up to 3% in the number of fissions.

By the year 2000, 245 distinct sets of DNP group parameters had been formulated for 20 different fissile nuclides [177]. Spriggs and Campbell go into significant detail, discussing the various analysis methods, neutron energies, fissile nuclides, counting intervals, and sample compositions for many of these DNP group parameter formulations [177]. Tuttle also provides a detailed review of many macroscopic DNP group parameters pre-1974, including details on the irradiated sample, reactor, counter, and travel time of the sample between irradiation and measurement [189]. Tables 2.1 through 2.4 show some macroscopic DNP group parameters from the literature for thermal fission of ^{235}U [129].

Table 2.1: Macroscopic formulation of group yields from thermal fission of ^{235}U (recreated from [129]).

Source	Group [-]					
	1	2	3	4	5	6
Keepin et al. [89]	0.00052	0.00346	0.00310	0.00624	0.00182	0.00066
Charlton et al. [35]*	0.00054	0.00344	0.00319	0.00607	0.00186	0.00066
Saleh et al. [160]	0.00055	0.00363	0.00297	0.00593	0.00168	0.00063
Synetos and Williams [180]*	0.00051	0.00328	0.00214	0.00583	0.00270	0.00063
Tuttle [189]	0.00065	0.00362	0.00320	0.00692	0.00218	0.00044
Waldo et al.[194]	0.00055	0.00366	0.00327	0.00660	0.00192	0.00070

* Modified to include a sixth group by Parish et al. [129].

There is reasonable agreement between the different DNP group parameters, though there is variation among

Table 2.2: Macroscopic formulation of group yield uncertainties from thermal fission of ^{235}U (recreated from [129]).

Source	Group [-]					
	1	2	3	4	5	6
Keepin et al. [89]	0.00006	0.00025	0.00041	0.00047	0.00019	0.00013
Charlton et al. [35]*	0.00003	0.00005	0.00024	0.00013	0.00014	0.00014
Saleh et al. [160]	0.00005	0.00032	0.00027	0.00104	0.00015	0.00012
Synetos and Williams [180]*	0.00004	0.00020	0.00036	0.00037	0.00029	0.00012
Tuttle [189]	0.00007	0.00013	0.00041	0.00019	0.00021	0.00007
Waldo et al.[194]	0.00006	0.00041	0.00041	0.00038	0.00017	0.00009

* Modified to include a sixth group by Parish et al. [129].

Table 2.3: Macroscopic formulation of group half-lives from thermal fission of ^{235}U (recreated from [129]).

Source	Group [s]					
	1	2	3	4	5	6
Keepin et al. [89]	55.89897	22.72614	6.24457	2.30281	0.60802	0.23028
Charlton et al. [35]*	55.45177	22.43195	6.02737	2.18658	0.49159	0.23028
Saleh et al. [160]	55.45177	22.65187	6.24457	2.31049	0.63013	0.23028
Synetos and Williams [180]*	56.35343	22.35959	7.00149	2.72893	0.77016	0.23028
Tuttle [189]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911
Waldo et al.[194]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911

* Modified to include a sixth group by Parish et al. [129].

Table 2.4: Macroscopic formulation of group half-life uncertainties from thermal fission of ^{235}U (recreated from [129]).

Source	Group [s]					
	1	2	3	4	5	6
Keepin et al. [89]	1.35239	0.67061	0.22503	0.09181	0.08000	0.02219
Charlton et al. [35]*	0.44361	0.50817	0.10482	0.06208	0.03138	0.02219
Saleh et al. [160]	3.99253	0.14805	0.39380	0.03851	0.01719	0.02219
Synetos and Williams [180]*	0.45816	0.28851	0.77794	0.21488	0.09413	0.02219
Tuttle [189]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545
Waldo et al.[194]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545

* Modified to include a sixth group by Parish et al. [129].

them. Not only did each approach use a slightly different experimental methodology via different energy spectra, data collection methods, sample transfer time between irradiation and measurement, sample size, and sample composition, but there are also differences in how the data was processed. There are several ways that the data was processed from delayed neutron count rates into DNP group parameters among the works in these tables: simultaneous non-linear least squares fitting of saturation and pulse irradiation data, separate non-linear least squares fitting of pulse and saturation irradiation data and subsequent combination, non-linear least squares of an intermediate amount of time between a pulse and saturation, and exponential stripping.

Keepin et al. use both a simultaneous fitting of DNP group parameters from pulse and saturation irradiation data as well as a separate fitting of four groups from each irradiation. The separate fitting of four groups is normalized at

the second group, and Keepin et al. report that this method gives better results than the simultaneous fit [89].

Charlton et al. use the equation for an intermediate amount of time that is not infinite or a pulse, as shown in Equation (2.7). They also only use five groups, as the sample took half a second to move from irradiation to the detector where delayed neutron counts could be measured. The shortest-lived DNPs that would make up group six were not included, but Parish et al. added group six values for their own analysis [129]. Charlton et al. use a least squares solve to determine the DNP group parameters, according to Spriggs and Campbell [177]. Synetos and Williams use the same approach as Charlton et al. [180, 178].

Saleh et al. use the equation for an intermediate amount of time, similar to Charlton et al., where the equation is described in Equation (2.7). However, instead of using a least squares solve, they use exponential stripping [160].

Tuttle formulates suggested values based on reviewing the literature and applying empirical correlations [189].

Waldo et al. apply a transit time correction using an empirical correlation, and the DNP group parameters are fitted using a least squares solve [194]. It is not made clear if pulse or saturation irradiations are used, as the conclusions indicate that the DNP group parameters are less applicable for short times or short irradiations, implying that a saturation irradiation was used.

Microscopic Measurement of DNPs

The other approach used to measure DNPs is the microscopic – also referred to in the literature as the aggregate or summation – approach, which uses data from individual nuclides summed together to form a simulated delayed neutron count rate. This count rate can then be used to generate DNP group parameters, the same as in the count rate from the macroscopic method.

This approach was investigated in detail in the 1990-2000s [137, 43, 156, 197, 22], with progress through the 2010's from more data becoming available [118, 81]. Leconte et al. used this approach as recently as 2020 [99]. This approach requires experiments on individual DNP nuclides, from which emission probabilities and half-lives can be measured. Typically, this involves rapid chemical extraction or isotope separation and magnetic tape to rapidly measure short lived nuclides, such as in the study by Björnstad et al. where an aluminized mylar tape is used [18]. The magnetic tape is used to transport samples or remove longer-lived decay products [171, 157]. Experimental methods include the neutron- β^- coincidence technique, separate counting of neutrons and β particles, separate counting of neutrons and γ rays, ion counting, fission yield from a set number of fissions, gamma counting of a long-lived daughter, and normalization of the neutron counter using a known precursor [157].

Other experimental methods used to study DNPs include on-line mass spectrometry, helium-3 spectrometry, proton-recoil detectors, the time-of-flight method, and neutron scintillators [44]. On-line mass spectrometry combined with a ^3He ionization chamber and a proton-recoil detector allows for measurement of the delayed neutron energy spectrum [170, 17]. The time-of-flight technique consists of measuring the flight time of a neutron over a known distance to calculate the velocity, including a neutron scintillator [17, 38, 64, 37].

This microscopic measurement data is exceptionally useful, as it allows for treatment of DNPs on a nuclide-by-nuclide basis. This is particularly important for MSRs, where the spatial DNP concentration depends on half-lives, fission yields, thermo-chemistry, and thermal-hydraulics. With data on individual nuclides, higher fidelity treatment of DNPs becomes possible.

There have been many new measurements in the past several decades investigating these nuclides. The IAEA has compiled and evaluated this data in a publicly accessible form [50, 49]. This data, when combined with a depletion simulation, can be used to model the DNPs explicitly rather than using groups. This is particularly useful for understanding what nuclides have the greatest impact, as well as for simulating experimental configurations which have yet to be thoroughly investigated for DNP group fitting.

Tables 2.5 through 2.8 show some microscopic DNP group parameters from the literature for thermal fission of ^{235}U [129]. There is reasonable agreement between the different DNP group parameters, though it should be noted that Brady and England may be closer to ENDF/B-VI.8 because ENDF/B-VI.8 uses the Brady and England results in its formulation.

Table 2.5: Microscopic formulation of group yields from thermal fission of ^{235}U (recreated from [129]).

Source	Group [-]					
	1	2	3	4	5	6
ENDF/B-VI.8 [153]	0.00058	0.00302	0.00288	0.00646	0.00265	0.00111
Mills et al. [119]	0.00056	0.00325	0.00196	0.00658	0.00287	0.00122
Brady and England [22]	0.00068	0.00341	0.00292	0.00611	0.00310	0.00158

* Modified to include a sixth group by Parish et al. [129].

Table 2.6: Microscopic formulation of group yield uncertainties from thermal fission of ^{235}U (recreated from [129]).

Source	Group [-]					
	1	2	3	4	5	6
ENDF/B-VI.8 [153]	-	-	-	-	-	-
Mills et al. [119]	0.00004	0.00023	0.00020	0.00051	0.00035	0.00012
Brady and England [22]	0.00004	0.00019	0.00016	0.00034	0.00017	0.00009

* Modified to include a sixth group by Parish et al. [129].

Table 2.7: Microscopic formulation of group half-lives from thermal fission of ^{235}U (recreated from [129]).

Source	Group [s]					
	1	2	3	4	5	6
ENDF/B-VI.8 [153]	51.72740	21.19716	5.73797	2.28913	0.81595	0.24295
Mills et al. [119]	55.27489	22.80089	7.67860	2.77148	1.07381	0.28177
Brady and England [22]	52.11633	21.32761	5.68620	2.18727	0.70114	0.23462

* Modified to include a sixth group by Parish et al. [129].

Table 2.8: Microscopic formulation of group half-life uncertainties from thermal fission of ^{235}U (recreated from [129]).

Source	Group [s]					
	1	2	3	4	5	6
ENDF/B-VI.8 [153]	-	-	-	-	-	-
Mills et al. [119]	0.09918	0.11723	0.37861	0.11259	0.08521	0.01020
Brady and England [22]	-	-	-	-	-	-

* Modified to include a sixth group by Parish et al. [129].

ENDF/B-VI.8 pulls from Brady and England, as well as other works by England [153, 22, 53, 23]. This work and

Brady and England's approach entails collecting 271 DNPs, generating a delayed neutron count rate curve, and then using the two four group aggregation approach described by Keepin et al. [89, 22].

Mills et al. use two different approaches: fitting all parameters simultaneously as described by Keepin et al. and using the Keepin et al. half-lives while fitting only the yields [119, 89]. The values presented here are from the approach where all parameters are fit simultaneously for all DNP group parameters.

Comparison of Macroscopic and Microscopic Approaches

Huynh and Jouanne offer a very useful comparison between the macroscopic and microscopic approaches via different datasets [81]. Specifically, they compare the total delayed neutron yield from various data sources from macroscopic measurements to the calculated total delayed neutron yield from microscopic measurements. This data can also be compared to the previously shown tables, from which the total delayed neutron yield can be calculated. The total data compilation is presented in Table 2.9.

Table 2.9: Total delayed neutron yields of 100 fissions from thermal fission of ^{235}U (based on [81]).

Approach	Data	Delayed Yield [-]
Microscopic	JEF-2.2 [123]	1.708 ± 0.12
Microscopic	JEF-3.1.1 [161]	1.477 ± 0.08
Microscopic	ENDF/B-VII.1 [33]	1.905 ± 0.10
Microscopic	ENDF/B-VI.8 [153]	1.670
Microscopic	Mills et al. [119]	1.645 ± 0.07
Microscopic	Brady and England [22]	1.780 ± 0.05
Macroscopic	JEF-2.2 [123]	1.655
Macroscopic	JEF-3.1.1 [161]	1.620
Macroscopic	ENDF/B-VII.0 [32]	1.585
Macroscopic	Keepin et al. [89]	1.580 ± 0.07
Macroscopic	Charlton et al. [35]	1.577 ± 0.03
Macroscopic	Saleh et al. [160]	1.645 ± 0.11
Macroscopic	Synetos and Williams [180]	1.510 ± 0.06
Macroscopic	Tuttle [189]	1.700 ± 0.05
Macroscopic	Waldo et al.[194]	1.670 ± 0.07

Table 2.9 shows that the total delayed neutron yield for thermal fission of ^{235}U ranges from 0.01477 to 0.01905 (or 100 times that for 100 fissions, which makes the data more legible). The microscopic values range from 1.477 to 1.905, capturing the minimum and maximum of the range of values seen in the table. The macroscopic values range from 1.510 to 1.700, presenting a tighter range of values. Another point of interest is that there does not appear to be a consistent difference between the microscopic and macroscopic values for the same dataset. For instance, JEF-2.2 has a larger microscopic yield, but JEF-3.1.1 has a larger macroscopic yield. The differences in the macroscopic yields do not change as much as the microscopic yields, indicating larger differences in the individual DNP data that affected their concentrations, emission probabilities, or half-lives.

2.1.2 DNP Group and Kinetics Parameters

In Section 2.1.1, the various methods for experimentally understanding delayed neutron precursors were discussed. The next stage of understanding these precursors is to understand how they are used in kinetics models, as well as the various ways they are represented. The microscopic and macroscopic experimental approaches provide the necessary data to create more easily handled forms. By summing the contributions from each DNP in the

microscopic approach, the macroscopic and microscopic approaches can be said to provide data in the form of a delayed neutron count rate over time. This section discusses the least squares approach to convert this data into several DNP groups.

The DNPs are commonly represented in six groups. These groups were developed originally for computational efficiency and due to a lack of data for individual precursors. Brady and England found that increasing the number of groups from six to nine caused no significant differences in a point kinetics calculation for step changes in reactivity [22]. Their study shows that increasing the number of DNP groups alone does not provide an increase in accuracy for point kinetics simulations.

Representing the DNPs in group form also simplifies the calculation of reactor kinetics, as without the groups the hundreds of DNPs would each require their own equations. Tracking all the DNPs becomes particularly computationally expensive when tracking all their concentrations over both space and time.

Various kinetics parameters are introduced in Section 2.1.3, though without detail regarding the purpose of each variable. This section provides more information on what these kinetics and delayed neutron group parameters physically represent and how they are calculated. The DNP group parameters directly affect many of the kinetics parameters, and both sets of data are used in kinetics and dynamics problems. Table 2.10 goes through each of the terms of importance for this section.

Table 2.10: DNP group parameters and kinetics parameters along with their definitions.

Parameter	Description	Units
λ_k	The decay constant for DNP group k	$[s^{-1}]$
λ_i	The decay constant for DNP i	$[s^{-1}]$
$\bar{\nu}_{d,k}$	The delayed neutron yield for DNP group k	$[-]$
$\bar{\nu}_{d,i}$	The delayed neutron yield for DNP i	$[-]$
χ_k	The spectra of DNP group k	$[-]$
χ_i	The spectra of DNP i	$[-]$
$P_{n,i}$	The delayed neutron emission probability for nuclide i	$[-]$
$\bar{\tau}$	The mean half-life	$[s]$
$\bar{\nu}_d$	The total delayed neutron yield	$[-]$
β_k	The delayed neutron fraction of DNP group k	$[-]$
β_i	The delayed neutron fraction of DNP i	$[-]$
β	The total delayed neutron fraction	$[-]$
β_{eff}	The effective delayed neutron fraction	$[-]$
Λ	The mean neutron generation time	$[s]$
Λ_{eff}	The effective mean neutron generation time	$[s]$

The DNP groups come with associated parameters, such as the group decay constants, λ_k , and the group yields, $\bar{\nu}_{d,k}$. The group yields are also known as group abundances, and can be found represented using a_k , b_k , $\nu_{d,k}$, and α_k in the literature [89, 196, 128, 22]. These group yields can be converted to group delayed neutron fractions, β_k , by dividing the group yields by the average number of neutrons emitted per fission event, as shown in Equation (2.4) as:

$$\beta_k = \frac{\bar{\nu}_{d,k}}{\bar{\nu}}. \quad (2.4)$$

The group yields represent the number of delayed neutrons produced per fission in group i , and when summed yield the total delayed neutron yield per fission. The β_k and λ_k terms are the defining parameters for the DNP

groups, along with the group spectra, χ_k , for DNP group k . With this data, the other group parameters can be derived, such as the total delayed neutron yield, $\bar{\nu}_d$, and delayed neutron fraction β .

Although the groups are useful, there are times when individual DNP data are needed. For example, in 1986, Perry et al. demonstrate that kinetics calculations yield different results when using individual nuclides compared to the six-group parameters [135]. In 1989, Brady and England also use the individual nuclides to compare with group data [22], and updated the six-group parameters based on the explicit DNPs. Updating the six-group parameters enabled greater consistency between experimental results and simulations.

Moving from the group parameters to the kinetics parameters, there is a convenient analogy which can be used to better understand some of the kinetics parameters. The effective multiplication factor, k_{eff} , is an important term in reactor physics. This term represents the scaling of the number of neutrons over each generation in a finite reactor system. The infinite multiplication factor, k_{∞} , represents the multiplication without considering leakage terms. This is analogous to the delayed neutron fraction, β , and mean generation time, Λ , terms, where the effective form of the term requires information on the specific reactor geometry.

The adjoint flux, $\phi^\dagger$, is important for calculating the effective form of many kinetics parameters. The adjoint flux is not a physical quantity, but rather an importance, or weighting, function, describing the importance of the neutron flux in the reactor. For example, areas further from the center of the core are less important than those closer to the center, which is reflected by a decrease in the effective delayed neutron fraction in MSRs as DNPs move away from the center of the core due to fuel salt advection. The adjoint flux can be calculated by taking the adjoint of the Boltzmann transport equation, or by using the iterated fission probability (IFP) method, which provides an importance term proportional to the adjoint flux [183, 122, 188].

The delayed neutron fraction, β , represents the ratio of delayed neutron production over all neutron production [117]. Related to the delayed neutron fraction is the effective delayed neutron fraction, β_{eff} . This term is more sophisticated, as rather than being a physical property of a fissile nuclide, it depends on the composition and geometry of the reactor. For MSRs, it also accounts for the movement of the DNP in different parts of the reactor.

A straightforward way to interpret the effective delayed neutron fraction is that it is the delayed neutron fraction corrected for the actual composition and geometry of the reactor. More specifically, the effective delayed neutron fraction represents the ratio between the delayed-neutron induced fission to the total number of fission events weighted by the spectrum and adjoint neutron flux [152]. Alternatively, it can be thought of as the ratio of the spectrum weighted and adjoint weighted delayed neutron production over all spectrum weighted and adjoint weighted neutron production [117]. For MSRs, the movement of the fuel salt also affects the effective delayed neutron fraction, as the integrated adjoint neutron flux is reduced due to movement of the DNPs out of the core and into less neutronically-important regions.

The mean neutron generation time, Λ , is the mean time between generation of a neutron and fission induced by that neutron [152]. This term also has an effective form, Λ_{eff} , which is the adjoint weighted form. The effective mean neutron generation time can vary significantly, on the order of hundreds of percent, from the mean neutron generation time [191]. In contrast, the effective delayed neutron fraction, β_{eff} , varies by approximately 20% from the delayed neutron fraction [191].

Calculation of DNP Group Parameters

The DNP group parameters, τ_k and $\bar{\nu}_{d,k}$, are important for simulating transients. The parameters are formulated based on the fitting of the delayed neutron count rate, as discussed in Section 2.1.1. Equation (2.3) shows the exact formulation for the delayed neutron count rate as a function of time measured by a detector for I individual DNPs

for any number of fissile nuclides, recreated here as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t).$$

Equation (2.5) instead gives the delayed neutron count rate measured by the same detector for K DNP groups from a pulse irradiation as:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}. \quad (2.5)$$

For an infinite, or saturation, irradiation, Equation (2.6) is used rather than Equation (2.5), given as [24]:

$$\dot{n}_d(t) \approx \epsilon \dot{F}_s \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t}. \quad (2.6)$$

Equation (2.7) shows the intermediate case, where the irradiation time, T , is neither infinite nor a pulse [197]. This equation gives the delayed neutron count rate as:

$$\dot{n}_d(t, T) \approx \epsilon \dot{F}_s \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} (1 - e^{-\lambda_k T}). \quad (2.7)$$

The terms in Equations (2.5), (2.6), and (2.7) are defined as follows:

- K = the number of DNP groups [-]
- λ_k = the decay constant of DNP group k [s^{-1}]
- F_s = the total number of fission events in the pulse irradiation [-]
- $\dot{F}_s$ = the fission rate for a saturation irradiation [s^{-1}]
- $\bar{v}_{d,k}$ = the group yield for group k [-] .

More information on the derivation of these equations can be found in Appendix A. These equations are specifically for a single fissile nuclide, and the derivation can be found in the work of Wilson and England [197].

The pulse irradiation approach gives better statistics on the short-lived DNPs, while the longer-lived DNPs are better represented by saturation irradiation. Brady discusses this, and uses both pulse and saturation irradiation to form groups [24]. Specifically, Brady uses the four long-lived groups from the saturation irradiation and the four short-lived groups from the pulse irradiation renormalized at the second DNP group.

Once the delayed neutron count rate is acquired, an over-determined non-linear least squares approach is used to calculate the group yields and decays. For example, Brady used STEPIT, an algorithm for finding local minima of a continuous function, to handle the least squares solve with both pulse and infinite irradiation experimental data [24, 34]. Waldo et al. also solve the least squares problem, though the specific approach used is not specified [194]. Using Equation (2.5), (2.6), or (2.7), the general form of the solve is given as:

$$A\vec{x} \approx \vec{b}. \quad (2.8)$$

Since the system is overdetermined, no exact solution exists; only a residual minimization, as shown in Equation (2.12), is possible.

Based on the irradiation type used, the coefficient matrix will change, as shown in Equations (2.5), (2.6), and

(2.7). Equation (2.9) gives the coefficient matrix for a pulse irradiation as:

$$A = \epsilon F_s \begin{bmatrix} \lambda_1 e^{-\lambda_1 t_0} & \lambda_2 e^{-\lambda_2 t_0} & \dots & \lambda_n e^{-\lambda_n t_0} \\ \lambda_1 e^{-\lambda_1 t_1} & \lambda_2 e^{-\lambda_2 t_1} & \dots & \lambda_n e^{-\lambda_n t_1} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1 e^{-\lambda_1 t_m} & \lambda_2 e^{-\lambda_2 t_m} & \dots & \lambda_n e^{-\lambda_n t_m} \end{bmatrix}. \quad (2.9)$$

Note that this term changes for a saturation irradiation, removing the λ_i term and changing the fission term from net fissions to the fission rate.

Within this least squares equation, the solution vector contains only the group yields, given in Equation (2.10) as:

$$\vec{x} = \begin{bmatrix} \bar{v}_{d,1} \\ \bar{v}_{d,2} \\ \vdots \\ \bar{v}_{d,K} \end{bmatrix}. \quad (2.10)$$

The target vector contains the delayed neutron count rate data, shown in Equation (2.11) as:

$$\vec{b} = \begin{bmatrix} n_d(t_0) \\ n_d(t_1) \\ \vdots \\ n_d(t_m) \end{bmatrix}. \quad (2.11)$$

Using ordinary least squares, with a non-relative residual, would lead to early time step count rates dominating due to their larger magnitudes. One way to resolve this is to take the natural logarithm of the target vector $\vec{b}$, such that the count rate evolution is linear in time. Another way to resolve this is to use a relative residual, given by Equation (2.12) as:

$$r = \min_x \left\| \frac{\vec{b} - A\vec{x}}{\vec{b} + \epsilon} \right\|_2^2, \quad (2.12)$$

where

$$\begin{aligned} \epsilon &= \text{a small value to avoid division by zero} \\ &= 1 \times 10^{-12}. \end{aligned}$$

The Euclidean norm, $\|\cdot\|_2$, is defined in Equation 2.13 as:

$$\|\vec{x}\|_2 = \sqrt{x_1^2 + \dots + x_n^2}. \quad (2.13)$$

By solving this non-linear system of equations, the group decay constants, λ_k , and yields, $\bar{v}_{d,k}$, can be generated. Once this group data is generated, and if there are multiple fissioning nuclides, F , then Equation (2.14) can be used to determine the K group decay constants using [44] as:

$$\lambda_k = \frac{\sum_F \lambda_k^{(F)} \beta_{k,eff}^{(F)}}{\sum_F \beta_{k,eff}^{(F)}}, \quad (2.14)$$

where

$$\begin{aligned}\lambda_k^{(F)} &= \text{the decay constant of DNP group } k \text{ from fission of nuclide } F \\ \beta_{k,eff}^{(F)} &= \text{the effective delayed neutron fraction of DNP group } k \text{ from fission of nuclide } F.\end{aligned}$$

This approach is effectively a fission-weighting, which can be written in a more general manner. The DNP group parameters are generated for a given fuel composition, weighted by the volumetric fission rate fraction of each fissile nuclide, F , similar to Equation (2.14). The fission-weighting term is given as:

$$w_F = \frac{\int \Sigma_f^{(F)}(\vec{r}, E) \phi(\vec{r}, E) dE d\vec{r}}{\sum_{F'} \int \Sigma_f^{(F')}(\vec{r}, E) \phi(\vec{r}, E) dE d\vec{r}}, \quad (2.15)$$

the delayed neutron yield of group k is given as:

$$\bar{v}_{d,k}^{(FW)} = \sum_F \bar{v}_{d,k}^{(F)} w_F, \quad (2.16)$$

and the decay constant of group k is given as:

$$\lambda_k^{(FW)} = \sum_F \lambda_k^{(F)} w_F, \quad (2.17)$$

where

$$\begin{aligned}\bar{v}_{d,k}^{(F)} &= \text{the DNP group } k \text{ delayed neutron yield from nuclide } F \text{ [-]} \\ \lambda_k^{(F)} &= \text{the DNP group } k \text{ decay constant from nuclide } F \text{ [s}^{-1}\text{]} \\ \phi &= \text{the neutron flux [cm}^{-2} \cdot \text{s}^{-1}\text{]} \\ \bar{v}_{d,k}^{(FW)} &= \text{the fission-weighted DNP group } k \text{ delayed neutron yield [-]} \\ \lambda_k^{(FW)} &= \text{the fission-weighted DNP group } k \text{ decay constant [s}^{-1}\text{]} \\ \Sigma_f^{(F)} &= \text{the macroscopic fission cross section for nuclide } F \text{ [cm}^{-1}\text{]}.\end{aligned}$$

The fission-weighted values can also be multiplied by an importance term to account for the difference between the delayed and prompt neutron fission spectra [143]. This is used for kinetics simulations that do not explicitly model DNPs in the transport solve, such as in point reactor kineticss (PRKs). Equation (2.16) becomes:

$$\bar{v}_{d,k} = \bar{v}_{d,k}^{(FW)} \frac{\int \chi_{d,k}(E) \phi^*(E) dE}{\int \chi(E) \phi^*(E) dE}, \quad (2.18)$$

and Equation (2.17) becomes:

$$\lambda_k = \lambda_k^{(FW)} \beta_{eff} \left(\sum_{k=1}^K \frac{\beta_k}{\lambda_k} \right)^{-1}, \quad (2.19)$$

where

- β_{eff} = the effective delayed neutron fraction [-]
- β_k = the delayed neutron fraction of group k [-]
- $\chi_{d,k}$ = the delayed neutron fission spectrum for DNP group k [-]
- χ = the total fission spectrum [-]
- ϕ^* = the adjoint flux [$cm^{-2} \cdot s^{-1}$].

Calculation of DNP Group Spectra and Concentrations

The equations thus far have not included the energy spectra, but instead have focused on the delayed neutron count rate. Including the energy in the delayed neutron count rate requires the energy spectra, $\chi_{d,k}(E)$, of the relevant group to be included, as shown in Equation (2.20) as:

$$\dot{n}_d(E, t) \approx \epsilon \tilde{F}_s \sum_{k=1}^K \chi_{d,k}(E) A_k e^{-\lambda_k t}, \quad (2.20)$$

where

- $\tilde{F}_s$ = the generic fission term [- or s^{-1}]
- A_k = the generic matrix term [s^{-1} or -]
- $\chi_{d,k}(E)$ = the delayed neutron energy spectrum of DNP group k [-].

This is the general form of the saturation and pulse irradiation equations, where A_k represents $\bar{v}_{d,k} \lambda_k$ for a pulse irradiation or $\bar{v}_{d,k}$ for saturation irradiation. $\tilde{F}_s$ is the total number of fissions for a pulse irradiation or the fission rate for saturation irradiation.

Another consideration in calculating group parameters is constructing group concentrations from individual nuclide data. This is important for situations where reducing the problem from several hundred DNPs to a few DNP groups reduces computational cost. A specific use case for this is explicitly modeling all DNPs spatially and then using those spatially resolved concentrations with an existing six-group structure. In such a case, the DNPs must be combined in such a manner to preserve their contributions to each DNP group. The reverse is also true, where a given six-group spatially resolved concentration can be separated into individual DNPs spatially resolved concentrations.

The methods for calculating the DNP group spectra and the group concentrations directly overlap. Two different grouping methods are considered for calculation of group spectra and group concentrations: based on half-lives [54, 154] and based on fractional contribution to adjacent groups [24].

The grouping based on half-lives requires comparing each nuclide's half-life to a constant half-life bin structure formulated from analysis of individual DNPs. The groupings based on half-lives are given in the top two sets in Table 2.11. The bottom set of half-lives is from ENDF/B-VIII.0 and is used with the fractional contribution approach [28].

Table 2.11: Half-lives of DNP groups from various sources.

Source	τ_1 [s]	τ_2 [s]	τ_3 [s]	τ_4 [s]	τ_5 [s]	τ_6 [s]
England et al. [54]	$\tau > 31$	$31 \geq \tau > 9$	$9 \geq \tau > 3$	$3 \geq \tau > 0.8$	$0.8 \geq \tau > 0.3$	$0.3 \geq \tau$
Rudstam [154] [*]	$\tau > 30$	$30 \geq \tau > 10$	$10 \geq \tau > 4$	$4 \geq \tau > 1.4$	$1.4 \geq \tau > 0.4$	$0.4 \geq \tau$
ENDF/B-VI.8 [153]	$\tau = 51.98$	$\tau = 21.17$	$\tau = 5.74$	$\tau = 2.29$	$\tau = 0.82$	$\tau = 0.24$

^{*} Rudstam uses $\tau_1 = 55.6$, but to ensure all DNPs are included, this has been modified.

England and Wilson chose their half-life bins based on the half-lives given in ENDF/B-V, while specifying that of the reasonable choices for bounding values, “No one was completely satisfactory” [54]. Rudstam selected half-life bins into six groups, although it is clarified that “...the grouping is somewhat arbitrary” [154]. The half-life grouping method thus results in a series of comparisons; if a nuclide is within a given group’s bounds, the concentration of that nuclide is added to that group.

For the fractional grouping, the half-life selection is less arbitrary and is instead based on the DNP group parameters. The ENDF/B-VI.8 data comes from the ^{235}U half-lives from the (n, ν_d) reaction [153, 22, 53, 23]. These delayed neutron data have remained unchanged through ENDF/B-VIII.0 [28, 153]¹.

The fractional grouping is more complex, but assigns nuclides to groups in a more rigorous manner. The method uses two groups, k and $k + 1$, for each nuclide i . These groups are determined via Equation (2.21) as:

$$\lambda_k < \lambda_i < \lambda_{k+1}. \quad (2.21)$$

Nuclides which don’t fit in these bounds will have the entire fractional contribution go into the closest group. Otherwise, the nuclide will have some fraction of its concentration or spectra go to each DNP group. The fractional contribution, or weight, of nuclide i to each group is represented with $f_{i,k}$ and $f_{i,k+1}$. The total fraction must be equal to 1 as shown in Equation (2.22) as:

$$f_{i,k} + f_{i,k+1} = 1. \quad (2.22)$$

To solve for the fractions, the following least-squares minimization problem is used [24]:

$$\int_0^\infty \left(\lambda_i e^{-\lambda_i t} - f_{i,k} \lambda_k e^{-\lambda_k t} - f_{i,k+1} \lambda_{k+1} e^{-\lambda_{k+1} t} \right)^2 dt. \quad (2.23)$$

Solving this equation for where the partial derivatives are zero for each fraction, $f_{i,k}$ and $f_{i,k+1}$, combined with Equation (2.22), yields the overdetermined system of equations as:

$$A = \begin{bmatrix} \lambda_k & \frac{2\lambda_k \lambda_{k+1}}{\lambda_k + \lambda_{k+1}} \\ \frac{2\lambda_k \lambda_{k+1}}{\lambda_k + \lambda_{k+1}} & \lambda_{k+1} \\ 1 & 1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} \frac{2\lambda_k \lambda_i}{\lambda_k + \lambda_i} \\ \frac{2\lambda_i \lambda_{k+1}}{\lambda_i + \lambda_{k+1}} \\ 1 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} f_{i,k} \\ f_{i,k+1} \end{bmatrix}. \quad (2.24)$$

Because the number of rows is greater than the number of columns in the matrix, the left-inverse is used. This system of equations is then solved as shown in Equation (2.25) as:

¹The ENDF/B-VIII.1 data is also the same as ENDF/B-VI.8, but it is currently unpublished.

$$\vec{x} = (A^T A)^{-1} A^T \vec{b} \quad (2.25)$$

With the fractional contribution terms, $\vec{x}$, the concentration of a given nuclide can then be multiplied by each fraction to show how much it contributes to that group. By summing the contributions of each nuclide, the group concentrations can be calculated, and conversely, the nuclide contributions can be calculated from the group concentrations. Alternatively, the spectra can be calculated by using the same approach but with individual DNP spectra instead of concentrations.

Calculation of Kinetics Parameters

The other relevant parameters to be calculated are simpler and do not require solving least squares problems. The k^{th} DNP group delayed neutron fraction is calculated in Equation (2.4), recreated here as:

$$\beta_k = \frac{\bar{v}_{d,k}}{\bar{v}}.$$

A useful term for comparing DNP group parameters is the average half-life [138, 139, 99]. Parish et al. also use the average half-life, but with a non-standard formulation which does not include the natural log of two [129]. Equation (2.26) gives the average half-life as:

$$\bar{\tau} = \ln(2) \sum_k^K \frac{\bar{v}_{d,k}}{\lambda_k}. \quad (2.26)$$

Equation (2.27) shows how the total delayed neutron yield, or average number of delayed neutrons per fission event, $\bar{v}_d$, can be calculated [68]:

$$\bar{v}_d = \sum_k^K \bar{v}_{d,k}. \quad (2.27)$$

This yield can also be approximately represented for an individual nuclide by integrating Equation (2.2) over all time and normalizing to a single fission, where the concentration over time is assumed to be:

$$N_i(t) = N_i(0)e^{-\lambda_i t}. \quad (2.28)$$

This yields Equation (2.29), where the total number of delayed neutrons emitted from a single nuclide i is given as:

$$\int_0^\infty \dot{n}_{d,i}(t) dt = P_{n,i} N_i(0). \quad (2.29)$$

After dividing by the total number of fission events, Equation (2.30) shows that the total delayed neutron yield can be written as [81, 113]:

$$\bar{v}_d = \sum_{i=1}^I P_{n,i} CFY_i, \quad (2.30)$$

where

CFY_i = the cumulative fission yield of nuclide i .

This also means that the delayed neutron yield of each contributing DNP can be calculated.

The delayed neutron fraction can be written as:

$$\beta = \frac{\bar{v}_d}{\bar{v}} \quad (2.31)$$

The mean neutron generation time can be calculated as:

$$\Lambda = \frac{l}{k}, \quad (2.32)$$

where

$$\begin{aligned} l &= \text{the prompt neutron lifetime [s]} \\ k &= \text{the neutron multiplication factor [-]}. \end{aligned}$$

(2.33)

This can be calculated using a time-dependent version of a Monte Carlo code along with PRK equations [152].

Finally, the effective parameters are introduced: the effective delayed neutron fraction, β_{eff} , and the effective mean neutron generation time, Λ_{eff} . The parameter β_{eff} is the spectrum- and adjoint-weighted ratio of delayed-neutron-induced fissions to the total induced fissions from all neutrons [152]. This is given in Equation (2.34) as:

$$\beta_{eff} = \frac{\int \Phi^\dagger(\vec{r}, E, \vec{\Omega}) \chi_d(E) \beta \bar{v}(E) \Sigma_f(\vec{r}, E) \Phi(\vec{r}, E, \vec{\Omega}) dE d\vec{\Omega} d\vec{r}}{\int \Phi^\dagger(\vec{r}, E, \vec{\Omega}) \chi(E) \bar{v}(E) \Sigma_f(\vec{r}, E) \Phi(\vec{r}, E, \vec{\Omega}) dE d\vec{\Omega} d\vec{r}}, \quad (2.34)$$

where

$$\begin{aligned} \Phi^\dagger &= \text{the adjoint neutron flux [n} \cdot \text{cm}^{-2} \cdot \text{s}^{-1}] \\ \chi_d &= \text{the delayed neutron energy spectrum [-]} \\ \chi &= \text{the neutron energy spectrum [-]} \\ \beta &= \text{the delayed neutron fraction [-]} \\ \bar{v} &= \text{the average number of neutrons released per fission [-]} \\ \Sigma_f &= \text{the macroscopic fission cross section [cm}^{-1}] \\ \Phi &= \text{the neutron flux [n} \cdot \text{cm}^{-2} \cdot \text{s}^{-1}]. \end{aligned}$$

Λ_{eff} can also be calculated explicitly by its definition, which is the time between the birth of a neutron and the subsequent absorption that induces fission. This is shown in Equation (2.35) as [191]:

$$\Lambda_{eff} = \frac{\int \Phi_0^\dagger \frac{1}{\bar{v}} \Phi_0 dE d\vec{\Omega} d\vec{r}}{\int \Phi_0^\dagger F_0 \Phi_0 dE d\vec{\Omega} d\vec{r}} \quad (2.35)$$

where

$$\begin{aligned} \Phi_0^\dagger &= \text{the critical adjoint neutron flux [n} \cdot \text{cm}^{-2} \cdot \text{s}^{-1}] \\ \Phi_0 &= \text{the critical neutron flux [n} \cdot \text{cm}^{-2} \cdot \text{s}^{-1}] \\ v &= \text{the average neutron speed [cm} \cdot \text{s}^{-1}] \\ F_0 &= \text{the total fission operator [cm}^{-1}]. \end{aligned}$$

The fission operator is used because only the absorptions that induce fission are of interest.

There are other methods used to calculate β_{eff} and Λ_{eff} in the literature [191]. These include the direct integration, or adjoint weighted, method [90]; the eigenvalue, or perturbation, method [74]; and the iterated fission probability method [19]. Λ_{eff} can also be calculated using the pulsed method [152]. For β_{eff} , there are also the prompt, Meulekamp, and Spriggs methods which can be used [152, 117]. Aufiero et al. have conducted a comparison of the Meulekamp, iterated fission probability, and prompt methods. Their results show that the iterated fission probability and prompt methods both yield similar results while the Meulekamp method is 200 pcm higher for the Molten Salt Fast Reactor (MSFR) [9].

For the effective parameters to account for the movement of the fuel salt, there is typically another step required in which the static parameters, calculated with no movement, are multiplied by another term. Aufiero et al. use an analytical approach with importance-weighted spatial decay probabilities to account for the movement of the fuel salt, a deterministic approach which includes computational fluid dynamics (CFD) techniques for precursor transport, and a Monte Carlo approach which adjusts the position according to the fuel velocity field and decay constant when the emission of a delayed neutron is sampled [9]. Another approach is to account for the decay of the DNPs outside of the core, as shown by Nestor [124]. Generally, an advection term is introduced to account for the circulation of the DNPs, after which the spatial distribution is incorporated into the effective parameter calculation.

2.1.3 Kinetics and Dynamics Methods

There are several different methods which have been used for analyzing transients in MSRs. These methods include multi-group neutron diffusion, PRK schemes, and quasi-statics [198]. These methods are generally coupled, either tightly or loosely, to a thermal-hydraulics solver. In general, although these methods have varying accuracy and costs, none of them include MSR specific phenomena when generating the DNP group parameters.

Multi-Group Diffusion

Multi-group diffusion kinetics methods are among the most accurate, and computationally expensive, approaches applied to transient problems, surpassed only by transport methods. Hurwitz and Ehrlich provide a standard formulation of multi-group diffusion written in 1953 [80].

Examples of software that use fluid-flow-velocity field coupled multi-group include GenFOAM and Griffin/Pronghorn coupling using Multiphysics Object-Oriented Simulation Environment (MOOSE) [59, 181]. Other examples can be found in the review by Wooten et al. [94, 60, 93] and elsewhere in the literature [202, 200, 125, 131]. These studies typically use Picard iteration with multi-group neutron diffusion coupled with thermal-hydraulics to accurately capture the multi-physics effects of the reactor. However, this accuracy also comes with a higher computational cost than simpler methods. The general form of the coupling is given as [198]:

$$\begin{aligned}
\frac{1}{\bar{v}_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\
& + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (\nu \Sigma_f)_{g'} \phi_{g'} \\
& + \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g,
\end{aligned} \tag{2.36}$$

and,

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (\bar{v}_g \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \left(\frac{\nu_T}{Sc_T} \nabla C_i \right), \quad (2.37)$$

where

- $\bar{v}_g$ = the average number of neutrons produced by fission of group g [-]
- ϕ_g = the scalar neutron flux of group g [$n \cdot cm^{-2} \cdot s^{-1}$]
- D_g = the group g diffusion term [cm]
- G = the total number of energy groups [-]
- $\Sigma_{s,g' \rightarrow g}$ = the macroscopic scattering cross section from group g' to group g [cm^{-1}]
- $\chi_{p,g}$ = the fraction of prompt neutrons born in group g [-]
- $\chi_{d,g,i}$ = the fraction of delayed neutrons from precursor group i born in group g [-]
- C_i = the concentration of DNPs in group i [$atoms \cdot cm^{-3}$]
- $\vec{u}$ = the velocity profile [$cm \cdot s^{-1}$]
- $\Sigma_{a,g}$ = the macroscopic absorption cross section for group g [cm^{-1}]
- ν_T = the eddy viscosity [$cm^2 \cdot s^{-1}$]
- Sc_T = the turbulent Schmidt number [-].

These equations solve neutron transport using multi-group neutron diffusion and solve for the DNP precursor concentrations over space and time. The precursor concentrations account for the flow field via the velocity vector and turbulent mixing using the Schmidt number and eddy viscosity. The ratio of the eddy viscosity to the turbulent Schmidt number represents diffusivity via turbulent mixing. Because the non-turbulent diffusion in molten salts is small relative to other spatial effects, it is not included in this equation.

Modified Point Kinetics

PRK equations are a simpler approach to modeling a transient, with the basic form given as:

$$\frac{dn(t)}{dt} = \frac{\rho(t) - \beta_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_i(t), \quad (2.38)$$

and,

$$\frac{dC_i(t)}{dt} = \frac{\beta_{eff,i}}{\Lambda} n(t) - \lambda_i C_i(t), \quad (2.39)$$

where

- n = the relative neutron density [-]
 - ρ = the reactivity [-]
 - β_{eff} = the effective delayed neutron fraction [-].
- (2.40)

The derivation of the PRK equations can be found in the work by Henry, with a fluid-fueled derivation in the work by Lapenta et al. [74, 97]. Modified point kinetics are used for MSRs, as standard PRK equations do not properly

handle the movement of the fuel salt.

PRK generally struggle to capture spatial effects, such as flow variation, which is important for MSRs. PRK do, however, give a correct prediction for general evolution trends [198]. Wooten et al. provide four different publications which use modified point kinetics in MSR analysis [69, 116, 173, 201].

The work by Guo et al. analyzes unprotected loss of heat sink (ULOHS) and unprotected loss of flow (ULOF) transients in the Molten Salt Actinide Recycler and Transmuter reactor (MOSART). The PRK equations are modified into the delayed point reactor kinetics equations, which model precursor concentrations in-core and ex-core, with the primary set of equations given as:

$$\frac{dn(t)}{dt} = \frac{\rho - \tilde{\beta}_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_{i,c}(t), \quad (2.41)$$

$$\begin{aligned} \frac{dC_{i,c}(t)}{dt} = & \frac{\beta_i}{\Lambda} n(t) - \left(\lambda_i + \frac{1}{\tau_c} \right) C_{i,c}(t) \\ & + \frac{1}{\tau_l} \left(\frac{V_l}{V_c} \right) C_{i,l}(t - \tau_l), \end{aligned} \quad (2.42)$$

$$\frac{dC_{i,l}(t)}{dt} = \frac{1}{\tau_c} \left(\frac{V_c}{V_l} \right) C_{i,c}(t) - \left(\lambda_i + \frac{1}{\tau_l} \right) C_{i,l}(t). \quad (2.43)$$

These equations, shown in Equations (2.41) through (2.43) (recreated from [69, 173]), are used in standard software, such as in the Reactor Excursion and Leak Analysis Program (RELAP). Shi et al. simulate a load demand change, primary flow transient, secondary flow transient, and reactivity transient in the Molten Salt Breeder Reactor (MSBR) [173]. The modified point kinetics approach enables modeling of ex-core decay of DNPs, capturing effects during power operation and loss-of-flow conditions, but it does not accurately capture all oscillatory effects [198].

Intermediate terms used in Equations (2.41) through (2.43) are shown in Equations (2.44) through (2.47) as:

$$\tilde{\beta}_{eff} = \beta_{static} - \beta_{loss}, \quad (2.44)$$

$$\beta_{loss} = \frac{\Lambda(t)}{n(t)} \sum_{i=1}^I \lambda_i C_{i,l}, \quad (2.45)$$

$$\tau_c = \frac{M_c}{\dot{M}(t)}, \quad (2.46)$$

$$\tau_l = \frac{M_l}{\dot{M}(t)}. \quad (2.47)$$

The terms in Equations (2.41) through (2.47) are defined as follows:

- V_l = the volume of the loop [cm^3]
- V_c = the volume of the core [cm^3]
- β_{static} = the value of β_{eff} without flow [-]
- M_c = the mass in the core [kg]
- M_l = the mass in the loop [kg]
- $\dot{M}(t)$ = the mass flow rate [$kg \cdot s^{-1}$].

A similar method is also used by Suzuki and Shimazu for analyzing a reactivity initiated accident in the FUJI-12 reactor [179]. That work uses a simpler form of the equations, without dividing the precursor concentrations into

core and loop regions.

The work by Merle-Lucotte et al. uses the "IPK method," which may stand for the "Improved Point-Kinetic method," though this is not explicitly stated [116]. This method uses a fixed mesh for neutronics and a mobile mesh for fluid properties. Merle-Lucotte et al. found that the PRK method without any modifications was able to match the IPK method for general trends and end-states, but was unable to capture reactivity oscillations properly. The IPK method was able to capture these oscillations, but assumes uniform fission and instantaneous heat transfer [116].

The work by Zanetti et al. extends the TRACE code, a finite volume code used for LWR transients [201]. The DNP concentration, dependent on both space and time, is discretized with an upwind scheme in space and a backward Euler method in time. Adjoint-weighted quantities discretized with the trapezoidal rule are then passed to the PRK equation. The set of DNP concentration and power, or neutron density, equations are resolved using LU factorization.

Another approach to modify the PRK is to alter the kinetics parameters based on a phase-space integration weighted with the adjoint flux [98]. Alternatively, effective flow-inclusive group yields can be calculated, as shown by Haubenreich and used in PRK equations [73].

The modified point kinetics approach provides a less computationally expensive approach to treating the DNPs while preserving general behavior. Although vortices and turbulent diffusion may not be physically captured, capturing the recirculation of the DNPs between the in-core and ex-core regions provides a higher level of fidelity than standard PRK equations.

Quasi-Statics

Quasi-static methods seek to maintain the accuracy of multi-group diffusion approaches while also having the cost savings of PRK. The general approach of these methods is to update PRK parameters with occasional multi-group diffusion simulations to capture changes in flux and DNP spatial distributions [198]. The general formulation used for quasi-static methods is very similar among the various publications that use it [51, 181]. A complete derivation of the quasi-static method can be found in the work by Henry, who discusses the shape and amplitude flux functions, and the work by Devooght, who discusses the generalized quasi-static method [74, 48]. There are also other studies which discuss the quasi-static method in detail [12].

Wooten et al. introduce several different publications which have used quasi-statics [198]. One of these, from 2005 by Rineiski et al., covers the SIMMER code, which couples neutronics, thermal-hydraulics, and structure codes [146]. The neutron flux is constrained in the work by a term containing the phase-space integration and the weighting function used. The neutron flux is treated as:

$$\Phi(\vec{x}, t) = \phi_A(t)\psi(\vec{x}, t), \quad (2.48)$$

where

$$\begin{aligned} \Phi &= \text{the neutron flux } [n \cdot cm^{-2} \cdot s^{-1}] \\ \phi_A &= \text{the neutron flux amplitude function } [n] \\ \psi(\vec{x}, t) &= \text{the neutron flux shape function } [cm^{-2} \cdot s^{-1}]. \end{aligned}$$

The neutron flux shape function can be approximated as time-independent, but Rineiski et al. do not use this approximation.

Rineiski et al. solve "shape" steps to handle the neutronics and DNP spatial concentrations. The equations

used in the "shape" steps are recreated here in Equations (2.49) and (2.50). These equations are solved by time-discretizing Equation (2.49) into a set of steady-state-like equations, which are then treated using the DANTSYS module of SIMMER [5]. These primary equations are given as:

$$\begin{aligned} \frac{1}{\bar{v}} \frac{\partial \psi(\vec{x}, t)}{\partial t} + \frac{1}{\bar{v} \phi_A} \frac{d\phi_A}{dt} \psi(\vec{x}, t) + M(t) \psi(\vec{x}, t) = \\ (1 - \beta) \chi_p F(t) \psi(\vec{x}, t) + \frac{1}{\phi_A} \chi_d \sum_{i=1}^I \lambda_i C_i(\vec{x}, t) + \frac{1}{\phi_A} Q(\vec{x}, t), \end{aligned} \quad (2.49)$$

$$\frac{\partial C_i(\vec{x}, t)}{\partial t} = \phi_A \beta_i F \psi(\vec{x}, t) - \lambda_i C_i(\vec{x}, t) - \nabla(\vec{u}(\vec{x}, t) C_i(\vec{x}, t)). \quad (2.50)$$

Within these "shape" steps are also sub-steps referred to as "reactivity" steps. At each of these "reactivity" steps, the PRK parameters and flux amplitude are calculated. The equations used in the "reactivity" steps are recreated here as:

$$\frac{d\phi_A}{dt} = \left(\frac{\rho - \beta_{eff}}{\Lambda} \right) \phi_A + q_{eff} + \sum_{i=1}^I \lambda_i (c_{s,i} + c_{m,i}), \quad (2.51)$$

$$\frac{dc_{s,i}}{dt} = \frac{\beta_{eff}}{\Lambda} \phi_A - \lambda_i C_{s,i}, \quad (2.52)$$

where,

$$\rho = \frac{\langle W(\vec{x}), (\chi F(t) - M(t)) \psi(\vec{x}, t) \rangle}{\langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}, \quad (2.53)$$

$$c_{m,i} = \frac{\langle W(\vec{x}), \chi_d C_{m,i} \rangle}{\Lambda \langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}. \quad (2.54)$$

Intermediate terms used in Equations (2.51) through (2.54) are given as:

$$\frac{\partial C_{m,i}(\vec{x}, t)}{\partial t} = -\nabla(\vec{u}(\vec{x}, t) C_{s,i}(\vec{x}, t)) - \lambda_i C_{m,i}(\vec{x}, t) - \nabla(\vec{u}(\vec{x}, t) C_{m,i}(\vec{x}, t)), \quad (2.55)$$

$$\frac{\partial C_{s,i}(\vec{x}, t)}{\partial t} = \phi_A(t) \beta_i F(t) \psi(\vec{x}, t) - \lambda_i C_{s,i}(\vec{x}, t). \quad (2.56)$$

The $C_s(\vec{x}, t)$ term is computed at steady state by setting the left-hand side of both Equation (2.49) and (2.50) to zero as well as the fluid velocity to zero at $t=0$. This allows Equation (2.55) to be solved. The terms in Equations (2.48)

through (2.56) are defined as follows:

- $M(t)$ = the neutron disappearance operator [cm^{-1}]
- $F(t)$ = the neutron production operator [cm^{-1}]
- χ_p = the prompt neutron energy spectra [-]
- $C_i(\vec{x}, t)$ = the concentration of DNP group i [$atoms \cdot cm^{-3}$]
- $Q(\vec{x}, t)$ = the external neutron source [$n \cdot cm^{-3} \cdot s^{-1}$]
- $\vec{u}(\vec{x}, t)$ = the DNP velocity [$cm \cdot s^{-1}$]
- $C_{s,i}(\vec{x}, t)$ = the "standard" precursor concentration of group i [$atoms \cdot cm^{-3}$]
- $C_{m,i}(\vec{x}, t)$ = the "movable" precursor concentration of group i [$atoms \cdot cm^{-3}$]
- ρ = the reactivity [-]
- $c_{s,i}$ = the PRK "standard" precursor concentration of group i [$atoms$]
- $c_{m,i}$ = the PRK "movable" precursor concentration of group i [$atoms$]
- q_{eff} = the PRK external neutron source [$n \cdot s^{-1}$]
- Λ = the neutron generation time [s]
- $W(x)$ = the weighting function (the default in SIMMER
is an adjoint flux computed at steady-state conditions) [-].

The quasi-statics approach provides an intermediate approach between modified point kinetics and fluid-flow-velocity field coupled multi-group diffusion. Quasi-static methods use occasional multi-group diffusion with PRK parameters, offering higher accuracy than PRK methods and less cost than full multi-group diffusion.

2.2 Molten Salt Reactor Fuel Depletion

Liquid-fueled MSRs are nuclear reactors which have the fissile material dissolved in a circulating molten salt. This fuel salt circulates through the in-core region, where most fissions occur, and through external piping in the ex-core region, where chemical processing occurs, after which it re-enters the in-core region. The salt is both the fuel and coolant in an MSR, where fission heat is generated in the in-core region and removed through the heat exchanger in the ex-core region.

There are many different MSR designs in the literature, as shown in Table 2.12 [36]. These designs feature a circulating salt containing fissile material. In particular, the circulating fuel salt allows for online chemical reprocessing, which causes depletion to vary spatially, leads to movement of DNPs, and requires close coupling of thermal-hydraulics to neutronics for realistic modeling.

Table 2.12: A collection of various MSR designs publicly available in the literature.

MSR Name	Source
FUJI series	[63]
Actinides Molten Salt TransmutER (AMSTER)	[192]
MOSART	[82]
Fast Spectrum Molten Salt Reactor (FS-MSR)	[76]
European MSFR	[115]
Small Mobile Molten Salt Reactor (SM-MSR)	[31]
MSBR	[148]
MSRE	[149]
Accelerator-Driven Neutron Applications (ADNA) Tier-1 reactor	[21]
Single fluid and two fluid denatured molten salt reactors (DMSRs)	[52, 4]
Thorium molten salt reactor (TMSR)	[127, 114]

Many studies have investigated online chemical reprocessing of MSRs, as shown in Table 2.13, which has been recreated from [163]. Table 2.13 shows which MSR depletion simulations have modeled DNPs in the "DNP" column, showing most models do not include DNP movement in depletion simulations. The table is sorted by the "Reprocessing" column, which signifies the method(s) used in the different MSR models, discussed in Section 2.2.1. It is also sorted by the publication year in the "Year" column. The table shows there are many MSR concepts for which depletion was modeled. The study by Valdez et al. investigates two MSR designs, specifically the MSBR and the Molten Salt Demonstration Reactor (MSDR). These models used a variety of spatial refinement levels, shown in the "Model" column, ranging from unit cells to full 3D models.

Table 2.13: Molten salt reactor models analyzed with methods approximating online reprocessing (recreated from [163]).

Publication	Reprocessing	Model	Reactor	DNP	Year
Valdez et al. [85]	Continuous	Unit Cell	Several	No	2020
Zhuang et al. [204]	Continuous	Full 3D	MSFR	No	2020
Xia et al. [199]	Continuous	Full 3D	MSFR	No	2019
Zhou et al. [203]	Continuous	Full 3D	MSBR	Yes	2018
Aufiero et al. [10]	Continuous	Full 3D	MSFR	No	2013
Rodriguez-Vieitez [150]	Continuous	Unit Cell	ADNA	No	2002
Rykhlevskii [159]	Batchwise	Full 3D	MSBR	No	2019
Betzler et al. [16]	Batchwise	Unit Cell	MSBR	No	2017
Ridley and Chvala [145]	Batchwise	Full 3D	Modified DMSR	No	2017
Ahmad et al. [4]	Batchwise	Unit Cell	DMSR	No	2015
Park et al. [130]	Batchwise	Full 3D	MSBR	No	2015
Sheu et al. [172]	Batchwise	Full 3D	MOSART	No	2013
Nuttin et al. [126]	Batchwise	Full 3D	MSBR	No	2005
Fiorina et al. [58]	Mixed	2D Slice	MSFR	No	2012
Nagy et al. [121]	Mixed	Unit Cell	Modified MSBR	Yes	2008
Merle-Lucotte et al. [114]	Mixed	Full 3D	TMSR	No	2007
Bauman et al. [70]	Mixed	2D Slice	MSBR	Yes	1971

Although these studies incorporate the effects of online chemical reprocessing on depletion, they do not explicitly model the spatial effect on depletion. Instead, a simple approximation is used to handle the in-core and ex-core regions while accounting for the entire primary salt volume. The reason why simulations do not include fully resolved spatial resolution in depletion calculations is due to the computational cost.

2.2.1 Depletion Simulations

There are many approaches to simulating depletion with different computational costs and levels of accuracy. Building a computational tool to handle depletion requires two components: the depletion matrix and a numerical solver.

Simulating depletion requires specific information, such as decay constants, fission yields, and fission and transmutation cross-section data. The fission and transmutation cross-section data can come from a transport calculation, while the other data can be read from a data library [101]. Simulating depletion entails numerically

solving the Bateman equations; a generalized form of which is given as [101]:

$$\begin{aligned} \frac{dN_j}{dt} = \sum_{i \neq j} [(\gamma_{i \rightarrow j} \sigma_{f,i} \phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \phi) N_i] \\ - (\lambda_j + \sigma_j \phi + \lambda_{r,j}) N_j + \dot{N}_{f,j}, \end{aligned} \quad (2.57)$$

where

- N_j = the atomic density of isotope j [$atoms \cdot cm^{-3}$]
- $\gamma_{i \rightarrow j}$ = the fractional fission product yield of j from the fission of isotope i
- $\sigma_{f,i}$ = the microscopic fission cross-section of isotope i [cm^2]
- ϕ = the spectrum-averaged scalar flux in the fuel region [$n \cdot cm^{-2} \cdot s^{-1}$]
- $\lambda_{i \rightarrow j}$ = the decay constant of decay $i \rightarrow j$ [s^{-1}]
- $\sigma_{i \rightarrow j}$ = the microscopic transmutation cross-section of reaction $i \rightarrow j$ [cm^2]
- N_i = the atomic density of isotope i [$atoms \cdot cm^{-3}$]
- λ_j = the decay constant of isotope j [s^{-1}]
- $\lambda_{r,j}$ = the removal constant for removal of isotope j [s^{-1}]
- σ_j = the microscopic total transmutation cross-section of isotope j [cm^2]
- $\dot{N}_{f,j}$ = the feed rate for isotope j [$atoms \cdot cm^{-3} \cdot s^{-1}$].

Equation (2.57) includes a continuous reprocessing method, where the chemical removal rate, $\lambda_{r,j}$, is part of the differential equation. There are different ways of modeling this reprocessing, including batchwise methods where material is only removed at certain time intervals [163]. For this work, the continuous reprocessing method is used.

The matrix form of Equation (2.57) is given as:

$$\frac{d\tilde{N}}{dt} = A\tilde{N}, \quad (2.58)$$

where

- $\tilde{N}$ = the vector of material composition [$atoms \cdot cm^{-3}$]
- A = the depletion matrix [s^{-1}].

The depletion matrix contains the gain and loss terms for each nuclide in the material of interest, which vary continuously as a function of time and depend on the concentrations as well. If there are transfers between two different materials, such as the fuel salt N_f sending chemically removed nuclides to a waste tank N_w , then Equation (2.59) gives the concentrations as:

$$\begin{bmatrix} \dot{N}_f \\ \dot{N}_w \end{bmatrix} = \begin{bmatrix} A_f - T_{f \rightarrow w} & 0 \\ T_{f \rightarrow w} & A_w \end{bmatrix} \begin{bmatrix} N_f \\ N_w \end{bmatrix}, \quad (2.59)$$

where

A_f = the fuel salt depletion matrix [s^{-1}]

A_w = the waste tank depletion matrix [s^{-1}]

$T_{f \rightarrow w}$ = the transfer matrix of nuclides from the fuel salt to waste tank [s^{-1}].

There are several approaches to solving these equations, such as the Transmutation Trajectory Analysis (TTA) method and the Chebyshev Rational Approximation Method (CRAM) [102]. Other codes may use external software to compute depletion, such as REM, an in-house code used by Nuttin et al. [126] to implement 4th order Runge–Kutta. Codes may also use a power series approximation of the matrix exponential, as discussed by Aufiero et al. [10].

2.2.2 Stationary and Circulating Fuel Comparison

For stationary fuel, the Bateman equation as shown previously in Equation (2.57) is sufficient to simulate depletion. However, for the purposes of a streamlined comparison with circulating fuel, the Bateman equation can be rewritten as:

$$\frac{dN_j}{dt} = S_j(t) - \lambda_{net,j} N_j(t), \quad (2.60)$$

where

S_j = the source term for nuclide j ,

containing fission, decay, and transmutation sources [$atoms \cdot cm^{-3} \cdot s^{-1}$]

$\lambda_{net,j}$ = the loss term for nuclide j ,

containing decay and transmutation losses [s^{-1}].

Over the course of a depletion step, the depletion matrix is assumed to be constant, so the source term varies in time only due to the effects of the changing compositions of other nuclides. The loss term, $\lambda_{net,j}$, does not vary with time in this equation.

This form of the Bateman equation does not have space or energy resolution, which makes it simpler to understand and compare with the circulating fuel version. When solving this equation, the common approach is to have different material groupings, or blocks, in different spatial regions. Each of these material blocks will have different compositions and neutronic properties, such as flux, used for that entire material. This way, the spatial resolution is represented by the material blocks, and the system of ordinary differential equations can be solved rapidly.

Circulating fuel is more complex, as instead of solving a system of ordinary differential equations, a system of partial differential equations must be solved. The simplified static Bateman equation can be modified to include advection and diffusion terms, yielding:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t)), \quad (2.61)$$

where

D = the mass diffusivity [$cm^2 \cdot s^{-1}$].

The depletion matrix is assumed to have no spatial resolution in this equation. Because of this assumption, only the concentrations and material blocks provide spatial resolution to the problem. Spatial refinement can be added to this equation based on the specific approach, such as using a spatially resolved neutron flux.

Since the main challenge in solving this problem is its high computational cost, many studies have proposed methods to mitigate this issue. One common approach, the scaled flux method, is computationally inexpensive [15]. However, this method does not model the in-core and ex-core regions in a physical manner. The scaled flux method scales the flux by the fraction of fuel salt in-core compared to ex-core. This approach does not model the in-core and ex-core regions explicitly, but instead assumes the salt is perfectly mixed in the primary loop. This approximation does not introduce significant error in the concentrations of nuclides unless there are large chemical removal rates or near-stationary flow rates [165]. An example implementation of this approach is given in Appendix B.

Another approach has shown how spatial resolution can be investigated using a system of ordinary differential equations to solve the Bateman equation. This approach can estimate the steady-state concentration of certain nuclides across a flow loop, but it does not account for temporal resolution [168].

Alternatively, a more accurate treatment was recently demonstrated through MOOSE-based coupling of Griffin, a deterministic neutron transport code, with Pronghorn, a thermal fluids code, allowing for spatial and temporal resolution of certain nuclides [182]. Tano et al. present an approach where only insoluble, medium-lived, and high concentration nuclides are tracked with pre-computed spatial distributions, which only need to be updated if the medium-lived isotopes are not in equilibrium after the thermal-hydraulics perturbation.

Spatially and temporally resolved approaches, such as the one used by Tano et al., are more computationally expensive than those that only require solving a system of ordinary differential equations. However, Tano et al. have shown that including circulation of the fuel salt in depletion directly affects the equilibrium concentration of ^{135}Xe by approximately 45% compared to a spatially unresolved approach without flux scaling [182]. This indicates there may be a need for spatially resolved depletion simulations to get accurate fuel salt compositions.

Tano et al. and Betzler have both described fully modeling advection in depletion as computationally intractable [182, 15]. This is because tracking all nuclides over space and time requires far more computational work than handling a traditional time dependent system of ordinary differential equations (ODEs). This spatially resolved system of partial differential equations (PDEs) requires accuracy in the short time scale flow effects. These small time steps cause multi-year depletion simulations to be exceedingly expensive.

Numerical stability of the problem, when using an explicit-time solving scheme, uses the Courant number, C :

$$C = \frac{u\Delta t}{\Delta z}, \quad (2.62)$$

where

Δt = the time step size [s]

Δz = the length between spatial nodes [cm].

The Courant number must be less than or equal to 1 for the Courant–Friedrichs–Lewy convergence condition in an explicit finite differencing scheme [41]. This equation limits the time step size based on the flow rate and spatial mesh size. Although using an implicit time solver can work around this condition, the overall computational cost remains high due to the number of nuclides, the time step size, spatial mesh resolution, and the overall amount of time simulated in depletion.

For a short depletion simulation, such as a saturation irradiation of a fuel sample, capturing these effects is computationally tractable. Because the depletion lasts only for several minutes instead of several years, the

computational cost is orders of magnitude smaller. A high-fidelity, spatially resolved depletion simulation of an MSR sample irradiation experiment is feasible to generate improved DNP group parameters.

2.3 Safety, Security, and Safeguards

2.3.1 Safety

An important part of ensuring safe reactor operation is making sure safety-related transient events do not disrupt safe reactor operations. These transients can be caused by postulated initiating events (PIEs), which are defined in the IAEA Safety Standard Series No. NS-R-1 [1]. Of particular importance are PIEs that potentially lead to design basis accidents (DBAs). Although No. SSR-2/1 has superseded No. NS-R-1, the PIEs and DBAs for MSRs have not changed [2]. Although there are not specific events listed for MSRs, guidance is provided on categories of events are of interest.

These events include equipment failures, such as pumps failing or pipes rupturing; human error, such as improper maintenance or faulty operator actions; external events, such as an earthquake; and combinations of events. For combinations of events, the IAEA specifies that these events should make sense in combination and in the time frame modeled. For example, if a particular event lasts a long time or has a long lasting effect, then that event could possibly overlap with another unrelated event. Alternatively, dependent events could potentially occur at the same time.

Safety-related transients for MSRs include ULOHS, ULOF, unprotected transient over power (UTOP), and fuel salt over cooling (FSOC) [104, 42]. An ULOHS transient is when the heat exchanger experiences a failure, possibly caused by a pump failure in the intermediate loop, causing a loss of cooling of the reactor. For the MSFR, Brovchenko et al. have shown that slow ULOHS transients that take place over a minute or longer minimize the increase in fuel salt temperature [26]. An ULOF transient is when the flow rate in the primary loop becomes zero, caused by pump failure in the primary loop. An UTOP transient is a reactivity insertion event, which could be caused by various means, such as removing poisons, adding fissile material, or temperature changes. A FSOC transient is when the fuel salt temperature decreases, which could be caused by an increased heat removal rate from the heat exchanger.

The current work is important for safety-related transients, as the MSR specific DNP group parameters will impact the results of these transients. Depending on the properties of the specific reactor, which affect the group parameters, the static DNP group parameter transient simulation may generate overly or insufficiently conservative results.

2.3.2 Security

The goal of security, or physical protection (PP), is to prevent actors from acquiring information, stealing nuclear material, or committing sabotage [185]. Security is enforced through physical protection systems, which detect, delay, and neutralize the adversarial actor before they can complete their objective [7, 65]. There are several different categories which are important to understanding security: the actor type, their capabilities, their objectives, and their strategies. The main consideration for security in this work is protection of the public by preventing: theft of hazardous radioactive material and sabotage with the intent to damage facilities, disrupt operations, or release radioactive material. The majority of the information presented in this section comes from The Proliferation Resistance and Physical Protection Evaluation Methodology Working Group of the Generation IV International Forum [185].

The actor type involved in PP can be an individual, a group, a large organization, or a non-host state combined with any of the former. These actors may be composed of some combination of insiders and outsiders. Insiders are employees or contract staff who may be sympathetic with or suborned by the hostile actor. Outsiders can include terrorist groups, agents of proliferant states, or lone individuals. Based on the specifics of the actor type, their capabilities will vary significantly.

Actor capabilities can be classified into five categories: knowledge, skills, weapons and tools, number of actors, and dedication. The knowledge of the actor will vary depending on if they have insider knowledge or not, and how much information that insider knows. The skills of the actor will affect their strategy they use to achieve their goals, though a large organization or non-host state can be assumed to have access to a wide variety of skills. The weapons and tools of the actor will affect their strategy, but access to an insider would grant more tools accessed with less effort. The number of actors will vary with the specific strategy of that actor as well as their tactics, as a more discrete approach would require fewer actors than a forceful approach. The dedication of the actor is important to consider, as an insufficiently motivated actor may be deterred with sufficient security measures.

The objectives of the actor may include: disruption of operations, radiological release, nuclear explosives, Radiological Dispersal Devices (RDDs), and/or information theft. The objectives of the actor will influence their other decisions, particularly their strategies. If an actor's objective requires force, then the capabilities and strategies implemented by that actor will vary from one whose objective can be accomplished covertly. Assuming the actor is rational, they will select a strategy which optimizes the probability of successfully completing their objective.

The actor's strategies include their modes of attack and tactics. The modes of attack vary significantly, including cyber attacks, waterborne attacks, direct or diversionary attack using light anti-tank weapons, and more. The tactics actors can use include stealth, deceit, and overt force. The stealth and deceit tactics are more covert, including avoiding or deactivating security systems for the stealth tactic or using false identification or authorization for the deceit tactic. The overt force tactic ties more directly into many of the modes of attack, which may itself also serve as a diversion for a different tactic.

For this work, the specific actor capabilities and strategies are not considered. Additionally, only a single type of actor and objective combination is considered – an actor with access to an insider and the objective to disrupt operations through sabotage. This work investigates the formulation of new DNP group parameters, which impact the reactor behavior during a transient. There may be security issues that arise during a sabotage-initiated transient due to the recalculated kinetics and DNP group parameters.

2.3.3 Safeguards

Preventing the spread of nuclear material for weapons is the goal of nuclear safeguards. Safeguards are defined as the nuclear reactor characteristics that impede "...the diversion or undeclared production of nuclear material and the misuse of technology by the Host State seeking to acquire nuclear weapons or other nuclear explosive devices" [185]. This definition agrees with the definition made at the IAEA international workshop in 2002 [83]. In this work, proliferation resistance (PR) is used as a way to understand and measure certain criteria used in safeguards. PR is a spectrum, so it should not be considered necessary to optimize the PR, but rather as a way to evaluate strengths and weaknesses in the design relating to safeguards.

There are six measures for PR: technical difficulty, cost, time, fissile material type, detection probability, and detection resource efficiency. In this work, the measures analyzed include time and detection probability. The time measure is analyzed by conducting diversion scenarios over various time frames to determine the effects on diversion fingerprints. A very low PR time is 0-3 months, while a very high PR time is >30 years [185]. The detection probability measure goes hand-in-hand with time, as a protracted diversion scenario is likely to have a

reduced detection probability. A very low PR detection probability ranges from 0 to 5 percent, whereas a very high PR detection probability ranges from 95 to 100 percent [185].

The Department of Energy (DOE) released an updated nuclear material control and accountability (MCA) order in February of 2023 that cancels DOE Order 474.2 Chg 4, which was last updated in 2016 [77]. Based on the more recent order, special nuclear material (SNM) is classified as shown in Table 2.14.

Table 2.14: The SNM classification table is recreated from [77].

Material Type	Accountable Quantity [g]	Weight Field Used for Element	Weight Field Used for Isotope
^{235}U	1.0	total uranium	^{235}U
^{233}U	1.0	total uranium	^{233}U
^{242}Pu	1.0	total plutonium	^{242}Pu
$^{239-241}\text{Pu}$	1.0	total plutonium	$^{239}\text{Pu} + ^{241}\text{Pu}$
^{238}Pu	0.1	total plutonium	^{238}Pu
$^{241}\text{Am}^{**}$	1.0	total americium	^{241}Am
$^{243}\text{Am}^{**}$	1.0	total americium	^{243}Am
$^{237}\text{Np}^{**}$	1.0	total neptunium	-

^{**} Americium and neptunium are treated as SNM if they are separated.

Table 2.14 is used along with the graded safeguards table from the updated Order to properly classify the attractiveness level and categorization of SNM. These categories include weapons, pure products, high-grade materials, low-grade materials, and all other materials, ranging from attractiveness level "A", the most attractive, to "E", the least attractive. For this work, only materials of attractiveness level "C" and below are relevant. High-grade materials, with attractiveness level "C", include fuel elements and assemblies. Low-grade materials, with attractiveness level "D", include solutions with target concentrations of 1-25 g/L. All other materials, with attractiveness level "E", are made up of highly irradiated forms, solutions with concentrations <1 g/L, or uranium containing <20% ^{235}U or <10% ^{233}U .

The quantities provided in Table 2.14 show the minimum reportable, or accountable, quantities for each material type. These quantities make up the lower limit of Category IV materials for "B" through "E" attractiveness level materials. The SNM Category depends on the quantity of material as well as the attractiveness level of that material. Higher Category material has more stringent requirements due to their greater strategic importance, with Category I SNM having the most requirements. For example, if material is classified as attractiveness level "C", then any quantity of plutonium or $^{233}\text{U} \geq 6$ kg is considered Category I SNM. Alternatively, if the same material contains ^{235}U , separated ^{237}Np , or ^{241}Am and ^{243}Am , then an amount ≥ 20 kg is considered Category I SNM.

If the material in the reactor is classified as attractiveness level "D", then any quantity ≥ 16 kg is considered Category II, as Category I quantities do not exist for this attractiveness level. Material of attractiveness level "D" do not have a listed Category I quantity for material containing ^{235}U , separated ^{237}Np , or ^{241}Am and ^{243}Am .

Materials of attractiveness level "E" do not have Category I, II, or III quantities. Instead, reportable quantities are considered Category IV.

Significant quantities (SQs) of material are also of interest, of which there are two categories: direct use and indirect use. Direct use nuclear material consists of ^{233}U , highly enriched uranium (HEU) (^{235}U enriched over

Table 2.15: SQs of various materials are given along with their use category.

Material	Use	Significant Quantity [kg]
Plutonium	Direct	8
HEU	Direct	25
^{233}U	Direct	8
^{235}U	Indirect	75
Natural Uranium	Indirect	10,000
Depleted Uranium	Indirect	20,000

20%), and any amount of ^{239}Pu with less than 80% ^{238}Pu . Indirect use nuclear material consists of thorium and low enriched uranium (LEU) (^{235}U enriched at less than 20%). SQs of the different material categories are given in Table 2.15 [185, 3, 11].

This information is important for the current work, as it provides values for quantities of diverted material to analyze. The exact classification, materials, and quantities to use will vary based on the specifics of the fuel salt and diversion scenario. However, this information provides a set of data which can be directly referenced to determine the diversion rates.

For example, a fuel salt categorized as direct use with a plutonium diversion would have 8 kilograms of plutonium as a SQ. If that fuel salt also had an attractiveness level of "D", then 16 kilograms of plutonium would be Category II. Then, the diversion time could vary over a range of months to years, based on various PR times. A method of detection for these diversion scenarios is via change in reactivity due to the change in delayed neutron precursors and fissile concentration. The detection probability is then estimated based on this measurable reactivity change. Additionally, these diversions could be detected based on measured delayed neutron count rates [166].

Chapter 3

Methodology

This chapter details the tools and methods used to both generate the improved DNP group parameters and to validate them. Three models are covered in this chapter: the 0D scaled model, which is already complete; the 0D flow model, which is proposed; and the spatially resolved model, which is discussed for potential avenues of future research. Although each model varies in its fidelity and cost, the general methodology is very similar: generate concentrations of DNPs from irradiation of a sample, convert those concentrations into a delayed neutron count rate, and then use non-linear least squares to fit DNP group parameters to the count rate curve. Validation of the DNP group parameters requires solving kinetics problems that make use of the parameters. There are also transient problems of interest that affect safety, security, and safeguards by design.

3.1 Tools

Several software tools are used to evaluate and generate the improved DNP group parameters. The in-house Python package Molten Salt Delayed Neutron (MoSDeN) collects and consolidates data, OpenMC parses nuclear data and will be used for higher-fidelity analysis, and the MOOSE application Moltres evaluates the parameters through transient simulations [190, 151, 106]. Preliminary results use MoSDeN to generate the improved DNP group parameters with support from OpenMC, while the proposed work will use OpenMC more heavily in simulations and Moltres for transient simulations. Each tool is described in detail in its corresponding sub-section.

3.1.1 MoSDeN

MoSDeN is a Python package designed to enable generation of DNP group parameters using the microscopic approach for any given reactor of interest. MoSDeN allows for varying levels of fidelity in the generation, ranging from only using nuclear data and broad assumptions to a high-fidelity multiphysics simulation with neutronics and thermal-hydraulics. The MoSDeN package relies on the SciPy, NumPy, Matplotlib, pandas, and Uncertainties packages [193, 72, 79, 184, 56]. Many features of MoSDeN also require OpenMC to function properly [151]. MoSDeN automates data collection and processing, simulates pulse and/or saturation irradiations, converts the concentrations into a delayed neutron count rate, and generates group parameters.

The preliminary work on MoSDeN uses the 0D scaled model, while the proposed work will make use of the 0D flow model. Both models are discussed in more detail in Section 3.2, as is the spatially resolved model, which is proposed for future work.

3.1.2 OpenMC

OpenMC is an open-source Monte Carlo neutral particle transport code. Its Python Application Programming Interface (API) offers functionality that simplifies interfacing with MoSDeN, an in-house developed open-source Python package. The sample irradiation process is straightforward and configurable, as users can adjust parameters through an input file. OpenMC also offers useful feature in depletion, where a zero-power depletion step is rapidly calculated since only decay is tracked. This feature allows resolving ex-core residence time depletion steps quickly. Additionally, OpenMC's transfer rate functionality during depletion enables simulation of chemical removal in an MSR.

For this work, MoSDeN by default uses the ENDF/B-VII.1 based depletion chains provided by OpenMC and ENDF/B-VII.1 cross section data. The thermal spectrum chain applies to neutron irradiation below 1 *eV*, the fast spectrum chain applies to neutron irradiation above thermal and below 5 *MeV*, and anything above that is not treated in this work [177]. The models by default assume monoenergetic neutrons irradiate the sample.

3.1.3 Moltres

Moltres is an open-source MOOSE application that simulates MSRs by solving the multi-group neutron diffusion and Navier-Stokes equations [106]. MOOSE is an open-source, parallel, finite element framework for multiphysics applications [67]. MOOSE includes many pre-built applications that couple physical phenomena. The HEAT CONDUCTION and NAVIER-STOKES modules are particularly useful in this work, as they can couple with Moltres for general thermal-hydraulic modeling [134, 105, 136, 132]. Moltres will simulate the MSRE with different DNP group parameters. Moltres will also validate the improved DNP group parameters in this work.

The MSRE reactivity data can be used for validation, and was previously modeled with Moltres [106]. The MSRE core consists of flow through channels, simplifying its modeling compared to open-pool designs due to the restricted flow path. Since the initial demonstration, Park et al. have noted general agreement in results, attributing differences to simplifications and assumptions in the geometry [132]. Additionally, Park has published a dissertation detailing 1-D, 2-D, and 3-D neutronic models of the MSRE [131].

Moltres requires various data to model various physical phenomena in a reactor. The delayed neutron modeling requires delayed neutron fractions, β_k , and delayed neutron decay constants, λ_k , which are fission and adjoint-weighted. Neutronics modeling requires cross-sections, prompt and delayed fission spectra, and other data to properly model those physical effects. Moltres provides scripts which can parse various transport solvers, such as OpenMC and Serpent, to automatically collect this information and format it for use in Moltres.

There are two main capabilities in Moltres which are of interest. Moltres couples three physical phenomena to model an MSR: neutron flux, DNPs, and thermal-hydraulics. This allows Moltres to model transient simulations of MSRs, which will be used to validate the improved DNP group parameters. Moltres can also simulate transients using the MSR-specific DNP group parameters, and it can generate improved parameters with higher fidelity than the other methods. This is of interest for future work, which may use higher-fidelity multi-physics to generate improved DNP group parameters.

Neutron Flux Modeling

Neutron flux modeling determines the energy of neutrons interacting with the sample and the number of interactions. It determines the fission rate and the resulting fission product concentrations, including DNPs. The time-dependent multi-group neutron diffusion equation appears in Equation (2.36), recreated here:

$$\begin{aligned}
\frac{1}{v_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\
& + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (v \Sigma_f)_{g'} \phi_{g'} \\
& - \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g
\end{aligned}$$

Modeling the neutron flux requires accounting for the DNPs. DNPs contribute to the neutron flux and possess their own energy spectra, necessitating their inclusion as a separate term.

DNP Modeling

The time dependent DNP concentration is given in Equation (2.37) and repeated here:

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (v \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{v_T}{S c_T} \nabla C_i$$

This is the primary equation of interest when validating the improved DNP group parameters. A discussion of turbulence and how it affects the DNP group parameters and transient simulations is given in Appendix C.

For the future work involving formulation of improved DNP group parameters within Moltres for the spatially resolved model, this equation is less helpful. This is because it requires DNP group parameters, which Moltres should be generating. To generate the DNP group parameters, this equation is replaced by (2.61), recreated here:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t))$$

The spatially resolved model will use spatially resolved depletion over the irradiation simulation. The change to the equation results in tracking concentrations of the individual DNPs instead of tracking concentrations of DNP groups.

Thermal-Hydraulic Modeling

Moltres offers time-dependent, incompressible, buoyancy-driven flow [131]:

$$\rho \frac{\partial \vec{u}}{\partial t} = -\rho(\vec{u} \cdot \nabla) \vec{u} - \nabla p + \mu \nabla^2 \vec{u} + \rho \alpha \vec{g} (T - T_{\text{ref}}), \quad (3.1)$$

$$\nabla \cdot \vec{u} = 0, \quad (3.2)$$

where

$$\begin{aligned}
\rho &= \text{fluid density } [kg \cdot m^{-3}] \\
p &= \text{pressure } [Pa] \\
\vec{u} &= \text{flow rate } [m \cdot s^{-1}] \\
\mu &= \text{dynamic viscosity } [Pa \cdot s] \\
\alpha &= \text{coefficient of thermal expansion } [K^{-1}] \\
\vec{g} &= \text{gravitational force vector } [N] \\
T &= \text{fluid temperature } [K] \\
T_{\text{ref}} &= \text{reference temperature at which the nominal} \\
&\quad \text{density is provided } [K].
\end{aligned}$$

The temperature distribution is given by [131] as:

$$\rho c_p \frac{\partial T}{\partial t} = -\rho c_p \vec{u} \cdot \nabla T + \nabla \cdot (k \nabla T) + Q_f - Q_s, \quad (3.3)$$

with the fission heat source term defined as:

$$Q_f = \sum_{g=1}^G \epsilon_g \Sigma_g^f \phi_g, \quad (3.4)$$

where

$$\begin{aligned}
c_p &= \text{specific heat capacity of molten salt } [J \cdot kg^{-1} \cdot K^{-1}] \\
k &= \text{effective thermal conductivity of molten salt } [W \cdot m^{-1} \cdot K^{-1}] \\
Q_f &= \text{fission heat source } [W \cdot m^{-3}] \\
\epsilon_g &= \text{average fission energy released by neutrons in group } g [J] \\
Q_s &= \text{heat sink/removal } [W \cdot m^{-3}].
\end{aligned}$$

With the combination of the flux modeling, precursor modeling, and thermal-hydraulic modeling, Moltres can almost simulate MSRs. However, there is one additional MSR phenomenon required to complete the model, the flow of the fuel salt between the in-core to ex-core regions.

Ex-core and In-core Coupling

Moltres treats the ex-core region as a tightly-coupled 1D pipe geometry with coupling using Picard iterations and the inlet/outlet boundary conditions. The specific equations to treat the variables of interest in this geometry can be derived from their full forms. For example, the flux and eddy viscosity are zero in the ex-core region, leading to the precursor concentration in the ex-core region:

$$\frac{\partial C_i}{\partial t} = -\lambda_i C_i - u_x \frac{\partial C_i}{\partial x}. \quad (3.5)$$

3.2 DNP Group Generation

Current DNP group parameters are tabulated by fissile nuclide and incident neutron energy. For the improved DNP group parameters, it is challenging to continue using tabulated parameters due to the number of variables involved in fuel salt recirculation and chemical removal rates. Fully detailing the flow with a large number of parameters is computationally infeasible. Instead, the flow can be simplified to two parameters: an in-core average residence time, τ_{in} , and an ex-core residence time, τ_{ex} .

This approach captures fuel salt advection around the primary loop and approximates turbulent and diffusive mixing. The terms derive from the averaged behavior of a fuel salt control volume, treating thermal-hydraulic effects, such as vortices, as changes in residence times. This method allows for approximate fitting of any circulating fuel reactor, requiring only the generation of relevant parameters for determining the associated DNP group parameters.

Chemical removal rates are challenging to represent with only a small number of parameters while containing a useful amount of information. For any given set of removal rates for a reprocessing scheme, DNP group parameters can be generated. This does not provide easily tabulated data for every MSR parameter, but offers greater accuracy for specific reprocessing schemes.

Online refueling is not of interest in this work as it only affects the fission-weighting of the DNP group parameters, as discussed in Section 2.1.2. For example, if the irradiated sample is ^{235}U and the refuel is ^{239}Pu , then the addition of ^{239}Pu to the system would only change the fission-weighting and not the actual DNP concentrations from each fissile nuclide. Additionally, even if the same fissile nuclide is added, this could simply be treated as many identical individual samples circulating in the system. Similarly, if chemical removal removes some amount of fissile nuclide, that is also not important for this analysis. This is because the DNP group parameters are normalized per fission, so the only differences from adding or removing fissile nuclides are on the fission-weighting.

An irradiation time of approximately seven minutes suffices to reach equilibrium DNPs concentrations, as the saturation time frame is on the order of multiple half-lives of the longest-lived DNP. Elements with cycle times on the order of months to years exhibit negligible changes, as discussed in Section 4.2.1. Elements with changes on the order of seconds to days contribute sufficiently and must be included for accurate data generation.

The MSBR reprocessing scheme serves as the basis for analyzing the largest cycle times in this work. This is because the MSBR presents very large chemical removal rates, providing useful conservative estimates as to the impact these large removal rates could have on DNP group parameters. These chemical removal rates are given in Table 3.1. Rare earths, semi-noble metals, volatile fluorides, discard, and higher nuclides have small removal rates relative to the time scales of interest for saturation irradiations. Section 4.2.1 examines these longer cycle times, the inverse of the removal rates, to confirm their negligible impact. Section 4.2.1 scales the cycle times to analyze how reprocessing rates influence the group parameters.

Table 3.1: MSBR Online Reprocessing Removal Rates from [148].

Reprocessing Group	Element(s)	Removal Rate [s^{-1}]
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	2.32×10^{-7}
Rare Earths	Eu	2.32×10^{-8}
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	5.00×10^{-2}
Seminoble Metals	Zr, Cd, In, Sn	5.79×10^{-8}
Gases	Kr, Xe	5.00×10^{-2}
Volatile Fluorides	Br, I	1.93×10^{-7}
Discard	Rb, Sr, Cs, Ba	3.37×10^{-9}
Salt Discard	Th, Li, Be, F	3.37×10^{-9}
Protactinium	Pa	3.86×10^{-6}
Higher Nuclides	Np, Pu	1.98×10^{-9}

With the chemical reprocessing scheme in place, the next step to generate the improved DNP group parameters is to decide on the level of fidelity. The three levels of fidelity are the 0D scaled model, the 0D flow model, and the spatially resolved model. The 0D scaled model is completed, the 0D flow model is planned, and the spatially resolved model is a potential future development.

A Monte Carlo sampling approach is used to propagate uncertainties in the various inputs, such as in emission probabilities, half-lives, fission yields, reprocessing terms, or residence times. The sampling approach simply repeats the process for generating the DNP group yields, but with randomly sampled input data. The data itself must be perturbed, and not the delayed neutron count rate itself. This is because the uncertainties in the delayed neutron count rate are not independent from time step to time step, but instead they all depend on the underlying data which changes the shape of the delayed neutron count rate at all times.

For generation of the improved DNP group parameters, the main goal is to formulate them for individual fissile nuclides. For some of the analyses presented in this work, the fissions do not come from a single fissile nuclide. For these analyses, a fission-weighted approach is used, where the DNP group parameters are scaled based on the fraction of fissions that lead to those group parameters. Equations (2.16) and (2.17) show how the fission-weighted DNP group parameters can be calculated. The fission-weighted values can then be multiplied by an importance term to account for the difference between the delayed and prompt neutron fission spectra, as shown in Equations (2.18) and (2.19) [143].

3.2.1 0D Scaled Model

This is the lowest fidelity of the three models, and it primarily serves as a way to verify the microscopic approach is working as expected by comparing with existing literature. This approach can only model saturation concentrations, as there is no decay chain tracking nuclides decaying into each other. It is called the 0D scaled model because it uses the scaled flux method, discussed in Section 2.2.2. The cumulative fission yield (CFY) is used to calculate the

equilibrium DNP concentrations, with a scaling term based on the amount of fuel salt in the core. The CFY gives the amount of each nuclide that is formed from both fission and decay. An example solution with no reprocessing is shown in Equation (3.6) as [81]:

$$N_i = \frac{f_{in}CFY_i}{\lambda_i}, \quad (3.6)$$

where

f_{in} = the fraction of fuel-salt in the core
 CFY_i = the cumulative fission yield of nuclide i .

The derivation for this equation is given in Appendix B. It is important to note that the flux is scaled when calculating the fission rate as well, by the same f_{in} term that scales the cumulative fission yield. Effectively, this cancels the term out, resulting in no deviation in the results based on the in-core fuel salt fraction by reprocessing time.

No cross-sections are included in this model, nor is time dependent decay of nuclides into DNPs included. There is a possible approach to include the effects of cross-sections in this model, but this approach is not used as the 0D flow model will account for cross-sections [168, 14].

Figure 3.1 shows the how the different data is loaded into MoSDeN, including external data and user defined input data. The green circles represent input data required for generating the DNP group parameters. The red diamond represents MoSDeN, the tool takes in the data to simulate irradiation of the target fissile sample to generate the irradiated concentrations, the delayed neutron count rate, and finally the DNP group parameters that fit that count rate. The blue squares represent the outputs, in this case the DNP group parameters: the DNP group yields and the DNP group half-lives.

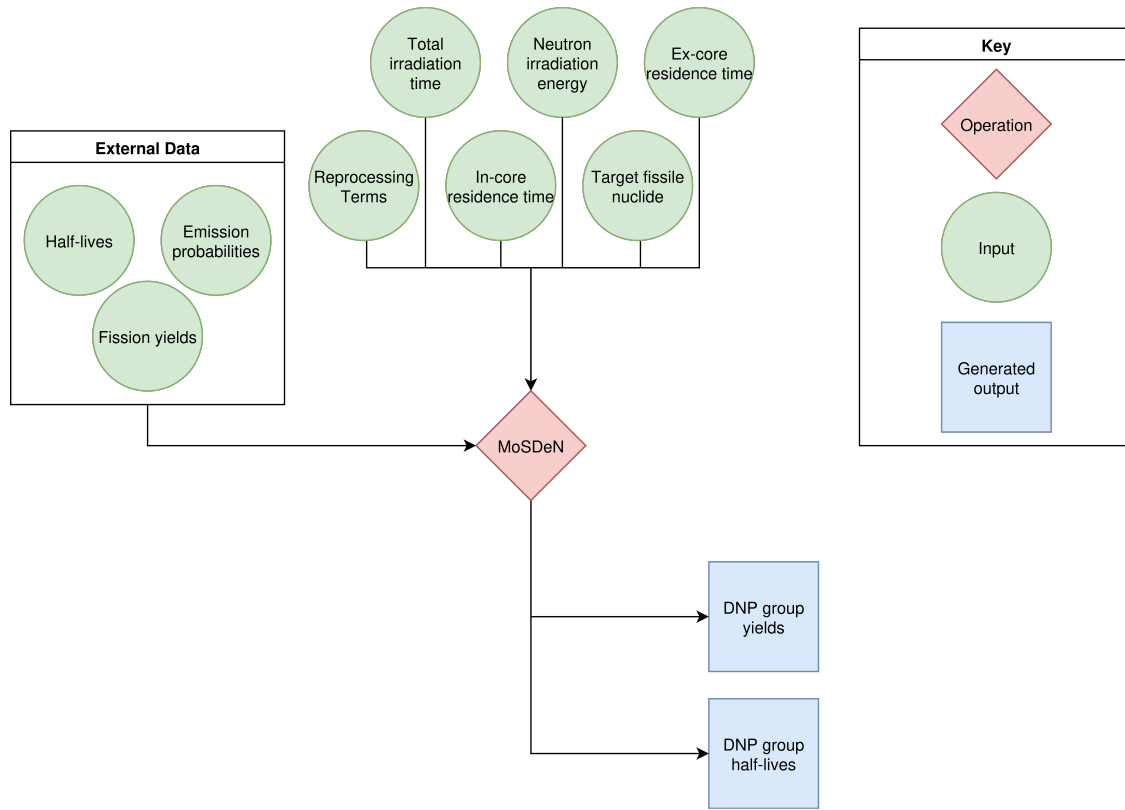


Figure 3.1: Flow diagram of 0D scaled model used to generate improved DNP group parameters.

After all this data is loaded into MoSDeN, it generates the concentration of each DNP. This concentration is then combined with emission probabilities and decay constants to form a delayed neutron count rate. A non-linear least squares solve then fits the improved DNP group parameters to the delayed neutron count rate, generating the group yields and half-lives, given as blue squares in Figure 3.1.

This model is only valid for saturation irradiation simulations, as the concentration calculated in Equation (3.6) is the equilibrium concentration. For a pulse irradiation, this approach is not valid. There are three potential options to handle pulse irradiations with the 0D scaled model: reduce net decay measurement time, use independent fission yields, or simply do not use the shortest-lived groups typically generated from a pulse irradiation since they are unlikely to be changed by MSR physical phenomena. Although these approaches warrant discussion, the 0D scaled model cannot accurately model a pulse irradiation. The 0D scaled model instead serves as a useful tool for qualitative analysis on the effects of recirculation and chemical removal on the longer-lived DNP group parameters.

The first approach, reducing the net decay time, does not improve the fundamental issue of insufficient data on short-lived DNPs, as the longer-lived DNPs will accumulate far more rapidly. Instead, this approach will overfit the short-lived groups, leading to less accurate results. The second approach, using only the shortest-lived groups with independent fission yields means that the decay chains will not be handled, but the initial pulse concentration should be accurate. However, the concentrations will quickly become less accurate as the decays of various nuclides into DNPs will not be treated, so this approach could only be potentially viable for the shortest lived group. The final approach would entail only using the longer-lived DNP group parameters from a saturation irradiation, while recognizing that the short-lived groups are less accurate. Because the MSR physical phenomena that cause non-negligible influence on the DNPs take at least several seconds to come into effect, it is reasonable to focus on the longer-lived groups with the 0D scaled model. The 0D flow model will provide more accurate data for the

shorter-lived DNPs.

3.2.2 0D Flow Model

The 0D flow model enables improved analysis of the general trends in nuclide evolutions. It treats the flow in a zero-dimensional manner using in-core and ex-core residence times, and uses OpenMC to track the full depletion matrix. The fission history function is a square wave, with a constant fission rate during the in-core residence time and no fission during the ex-core residence time. This approximation simplifies the analytical form of the fission history for creating the non-linear least squares problem, as shown in Appendix A.

Figure 3.2 shows the flow diagram for a fission history to simulate irradiation on a fissile sample circulating in the primary loop. The model passes eight parameters into OpenMC [151]: in-core residence time, ex-core residence time, neutron irradiation energy, target fissile nuclide, target density, total irradiation time, temperature, and reprocessing terms. The external data passed into OpenMC includes the emission probabilities and the depletion matrix. The depletion matrix contains information on half-lives, fission yields, cross-sections, branching ratios, and other useful data needed for OpenMC depletion. Section 3.3 provides further details on this process.

After OpenMC completes the fixed-source simulation, the final nuclide concentrations in the irradiated sample are saved for analysis in MoSDeN. MoSDeN combines the concentrations with data from the IAEA beta-delayed neutron emission database to generate a delayed neutron count rate [50]. A non-linear least squares fitting function from SciPy then calculates the group half-lives and yields using the Trust Region Reflective algorithm [193, 25]. These group parameters correspond to a specific nuclide, neutron energy, in-core residence time, ex-core residence time, reprocessing scheme, and reprocessing scalar. Section 3.4 provides further details on this block. Some additional factors to consider include self-shielding due to sample density (while avoiding self-multiplication), temperature-induced cross-section variations, and under- or over-irradiation, all of which can affect the results [89].

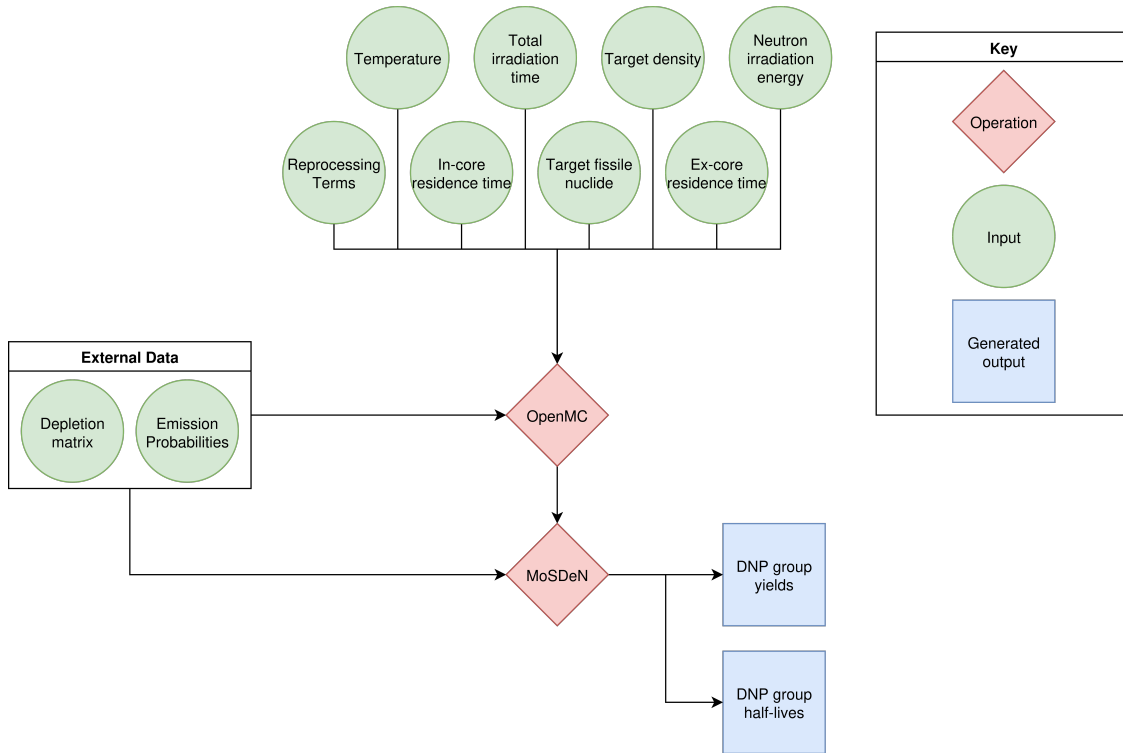


Figure 3.2: Flow diagram of 0D flow model used to generate improved DNP precursor group parameters.

The OpenMC model uses a three-gram spherical sample with vacuum boundary conditions. It treats the problem as a fixed-source simulation, with a uniform neutron source distributed evenly throughout the sphere. This approach is not physical, as a real neutron source would have a spectrum and originate externally. However, the exact neutron spectrum for irradiating samples to generate DNP group parameters is not provided in the literature. The source distribution also has negligible impact, as the sample is small and exhibits minimal self-multiplication. Thus, this approximation does not limit the effectiveness of the 0D flow model.

Another consideration for the 0D flow model is maintaining a consistent source particle rate. While not strictly necessary for group parameter generation since the values are per-fission, it simplifies the data analysis, as the tallies depend on the source particles. Equation (3.7) provides the source rate for the irradiation simulations:

$$\dot{S} = S_{net} \frac{\tau_{in} + \tau_{ex}}{\tau_{in} T}, \quad (3.7)$$

where

$$\begin{aligned} \dot{S} &= \text{the source particle rate } [n \cdot s^{-1}] \\ S_{net} &= \text{the total number of source particles } [n] \\ \tau_{in} &= \text{the in-core residence time } [s] \\ \tau_{ex} &= \text{the ex-core residence time } [s] \\ T &= \text{the net irradiation time } [s]. \end{aligned}$$

3.2.3 Spatially Resolved Model

The 0D flow model rapidly generates improved DNP group parameters, but it simplifies the treatment of flow. Instead of modeling the actual fuel salt flow, it estimates fuel circulation using residence times for each region. A spatially resolved model is useful for capturing the effects that the 0D flow model only capture with residence times for specific MSR geometries. Figure 3.3 shows the flow diagram for the spatially resolved model, which models the flow using Moltres.

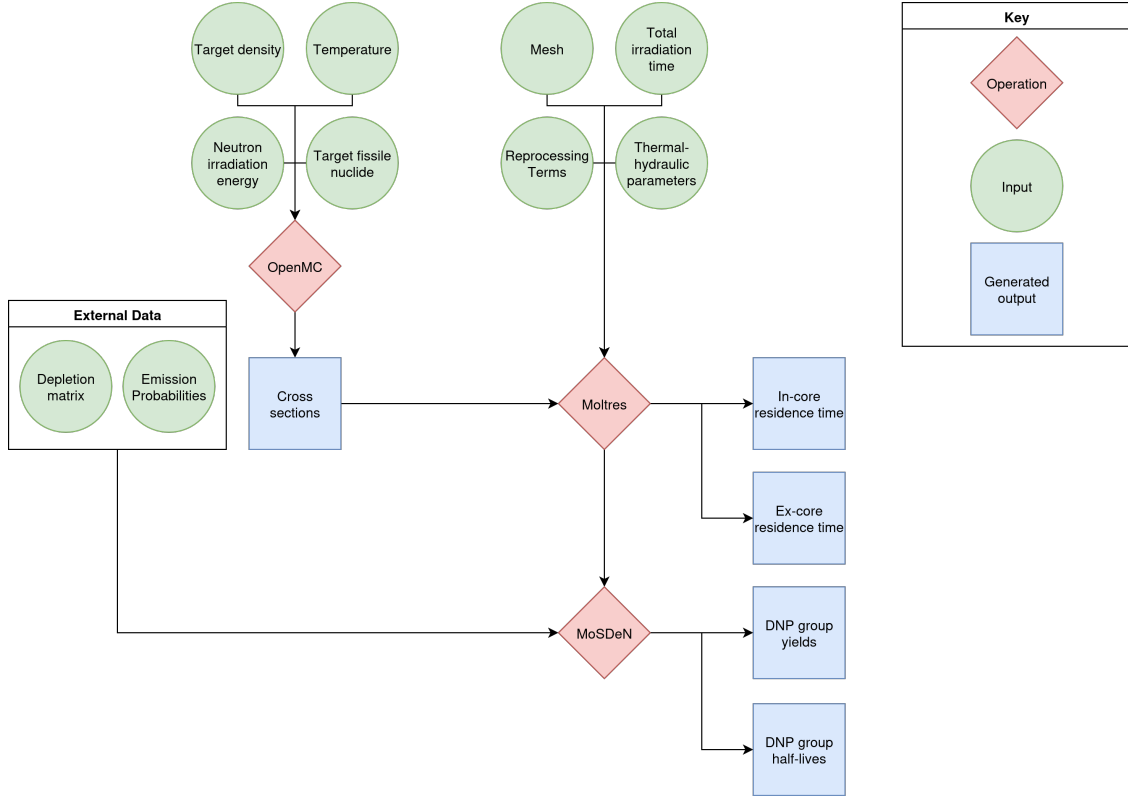


Figure 3.3: Flow diagram of spatially resolved flow model used to generate improved DNP group parameters.

The spatially resolved flow model introduces two new parameters absent in the 0D flow model. The first is the mesh, which defines the flow loop geometry. MOOSE provides built-in meshing tools, though complex geometries may require external tools such as Gmsh [66]. The second is the thermal-hydraulics parameters, which capture physical phenomena enabled by spatial resolution, as discussed in Section 3.1.3. Additionally, since the depletion happens in Moltres instead of OpenMC, OpenMC does not need as much data.

In the spatially resolved model, the in-core and ex-core residence times are outputs rather than inputs. These times derive from the movement of the irradiated sample over time in the fuel salt. The in-core and ex-core residence times vary as the irradiated sample diffuses and mixes further into the fuel salt. The net residence time, τ_{net} , remains constant unless the model changes, such as an increased temperature difference or induced flow rate. Equation (3.8) defines the in-core residence time as:

$$\tau_{in}(t) = \frac{\int_{V_{in}} \int_0^t N(\vec{x}, \tau) d\tau d\vec{x}}{\int_V \int_0^t N(\vec{x}, \tau) d\tau d\vec{x}} \tau_{net}, \quad (3.8)$$

where

V_{in} = the in-core volume [cm^3]

V = the total primary volume [cm^3]

$N(\vec{x}, t)$ = the time and space resolved sample concentration [$atoms \cdot cm^{-3}$]

τ_{net} = the net residence, or circulation, time of the fuel salt [s].

The ex-core residence time can be calculated in the same way, or alternatively as shown in Equation (3.9) as:

$$\tau_{ex} = \tau_{net} - \tau_{in}. \quad (3.9)$$

3.3 Delayed Neutron Count Rate Generation

Although the various models have been discussed, execution details were not. The following sub-sections provide specific information on the methodology behind the generation of the improved DNP group parameters. This includes a detailed discussion on the concentration generation and the data used with those concentrations to produce a delayed neutron count rate.

3.3.1 DNP Concentrations

The concentrations from the 0D scaled model are calculated using Equation 3.6, while the concentrations in the spatially resolved model can be calculated using Equation 2.61. The 0D flow model will use OpenMC to handle calculation of the concentrations over time.

The 0D flow model will use a fixed-source problem on a three-gram spherical sample of a fissile target in OpenMC. A uniformly distributed source will be placed inside the target with a vacuum boundary condition at the edge of the sphere. The sample must have a sufficiently large volume to ensure a high number of fissions, but excessive volume increases self-shielding and self-multiplication. Adjusting the density mitigates this issue: higher density increases self-shielding and self-multiplication, while lower density reduces fission probability and worsens statistical accuracy. By using a uniformly distributed neutron source within the sample, these effects are lessened. Additionally, since the fission counts can be taken directly from OpenMC tallies, self-multiplication and self-shielding can be accounted for directly without introducing error.

The fixed-source in this model will consist entirely of neutrons at a single energy, rather than a spectrum. For the proposed analyses, only thermal fission of ^{235}U is studied. This energy dictates fission cross sections, transmutation cross sections, and fission yields. Traditionally, tabulated group parameter data is formatted for a single energy or energy range, such as fast or thermal. However, macroscopic experiments, discussed in Section 2.1.1, use reactors with varying neutron energy spectra, not mono-energetic neutrons. This could be addressed by modeling a reactor in OpenMC and using a k-eigenvalue solve to generate a neutron energy spectrum. This analysis is considered outside the current scope, but is suggested as future work in Section 5.4. Photons can also generate delayed neutrons through photofission, though this is also considered outside the current scope [92]. There are also delayed neutrons from photons through non-fissile material, known as photoneutrons, also outside the current scope [86].

The neutron source distribution does not replicate the experimental process, where neutrons irradiate the sample externally rather than internally. However, an evenly distributed source eliminates self-shielding concerns. The exact spatial locations of fission events within the sample should negligibly affect the nuclide concentrations, which are the outputs of the OpenMC simulation. With a density of $10 \text{ g} \cdot \text{cm}^{-3}$, the spherical sample has a radius of approximately 0.4 cm .

After running the OpenMC simulation, the concentrations are collected from the depletion results for the final irradiation time step. These final concentrations are then converted into a delayed neutron count rate.

3.3.2 Concentrations to Count Rate

After receiving the concentrations from OpenMC, MoSDeN determines the concentration of each nuclide as a function of time during the measurement period. The concentrations are calculated based on half-life, as shown in Equation (3.10), where the concentration as a function of time is:

$$N(t) = N_0 e^{-\lambda t}. \quad (3.10)$$

Equation (3.10) shows the approach used in the 0D scaled model, and does not account for decay chains between nuclides. However, the general form of the equation can still be used, but with inclusion of the various decay chains and branching ratios. A simple way of implementing decay with the decay chain is using zero-power depletion simulations in OpenMC. This is used in the 0D flow model, though the 0D flow model also offers the option of using Equation (3.10). These concentrations are available at coarse time steps and linearly interpolated using SciPy, allowing MoSDeN to access more accurate concentrations at any time after irradiation [193]. The specific time steps at which concentrations are written will be investigated after development of the 0D flow model.

The concentrations are combined with Equation (2.3), recreated here:

$$\dot{n}_d(t) = \epsilon \sum_{k=1}^K P_{n,k} \lambda_k N_k(t).$$

The efficiency term, ϵ , is set to 1 in this work, as the simulated method includes every produced delayed neutron in the count rate. The contributions of each DNP over time are summed to generate the delayed neutron count rate.

3.4 DNP Group Reconstruction

Once the delayed neutron count rate is generated, it is used to generate the improved DNP group parameters, $\bar{\nu}_{d,k}$ and λ_k . Fitting the DNP group parameters to the delayed neutron count rate curve requires a non-linear least squares method, as both the group half-lives and yields must be determined. The equation describing the problem appears in Equation (2.5), recreated here:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{\nu}_{d,k} \lambda_k e^{-\lambda_k t}.$$

An important consideration for this process is the selection of time values for generating the DNP group parameters. Increasing the number of time nodes raises computational costs, but too few nodes result in a poorer fit [129]. Section 4.2.2 shows the sensitivity study for both the node spacing and the maximum time to measure delayed neutrons.

Another consideration for the non-linear least squares solve is the covariance, which is larger for the pulse irradiation than for the saturation irradiation. This results from multiplying the group yield by the group decay constant. Attempts to replace this multiplication with an independent term produces worse fits, as the multiplication is not independent and is necessary for accurate modeling.

The residual of the solve is given in Equation (2.12), recreated here as:

$$r = \min_x \left\| \frac{b - Ax}{b + \epsilon} \right\|_2^2.$$

Half-life bounds are set from 1 millisecond to 1000 seconds, while the yield bounds are from 0 to 1. Convergence

tolerances for the cost function, change in independent variables, and gradient norm are all set to 1×10^{-12} to ensure tight convergence.

The final consideration in DNP group reconstruction is uncertainty. This work tracks uncertainties using the uncertainties package [56]. These uncertainties fission yields, half-lives, and emission probabilities. The uncertainties are propagated through the count rate calculation, but for the non-linear least squares solve, Monte Carlo sampling is used. Each count rate is generated using randomly sampled fission yields, half-lives, and emission probabilities within their uncertainty bounds for each DNP. This enables formation of a normal distribution of group parameters, from which the mean and standard deviation of the mean can be extracted.

Once the DNP group parameters and their uncertainties are generated, the final step to obtain the finalized group parameters is to combine the saturation and pulse irradiation results. Saturation irradiation provides better statistics for longer-lived groups, while pulse irradiation is better for shorter-lived groups. Brady combines the results by using four long-period groups from saturation irradiation and four short-period groups from pulse irradiation. The groups are renormalized at the second group, where the two longest-lived group half-lives and yield ratios use the saturation results [24]. The remaining groups use the pulse irradiation results.

This work uses a similar but slightly simplified approach. The group parameters are first calculated for both pulse and saturation irradiation. Then, the two longest-lived groups use the saturation irradiation results, while the four shortest-lived groups use the pulse irradiation results.

3.5 Impact of DNP Groups on Transients

The impact of the DNP groups on transients will be evaluated using Moltres with fluid-flow-velocity field coupled multi-group diffusion kinetics, as discussed in Section 2.1.3. This approach provides high-fidelity results by coupling thermal-hydraulics and neutronics, a necessity for MSR modeling. Modeling every DNP in space and time would be prohibitively expensive, so they are instead modeled using DNP group parameters. Moltres allows direct comparison between static DNP group parameters, which do not account for MSR phenomena, and parameters that do.

The results of interest from this analysis include the reactivity and power responses to various initiating events. The initiating events and transients vary, as different reactor responses are studied. The events of interest depend on the perspective from which the reactor response is analyzed.

Specifically, there are four different perspectives of interest in this work: validation, safety, security, and safeguards. Each perspective involves a unique combination of motivation, proposed initiating event, and investigated response. These are discussed in detail in their respective sections.

There are some general results of interest, such as the effective delayed neutron fraction. This can be generated from Moltres using the prompt method, discussed in Section 2.1.2. Moltres uses fission-weighted DNP group parameters, and the effective multiplication factor will capture the spatial distribution of each DNP group concentration. Adjoint-weighting is not used for these parameters because Moltres explicitly captures the delayed neutrons in the diffusion solve. Moltres will provide an effective delayed neutron fraction that captures the effects of fuel circulation. The fission weighting values will be generated using fission rate tallies in OpenMC.

3.5.1 Validation

The motivation for validation is to confirm that the proposed methodology and results align with experimental data. This ensures the demonstrated improved DNP group parameters are applicable to actual MSRs. Discrepancies between experimental results and the model highlight potential weaknesses in the approach. If no discrepancies exist, confidence in the results increases for problems similar to those validated.

The validation of the DNP group parameters follows a multi-stage approach. The first stage compares the irradiated sample's delayed neutron count rate to the Keepin et al. experiment from 1957 [89]. If the count rates do not match, the issue lies in the data or the aggregation of individual DNPs into the count rate. If the count rate is accurately predicted, the formulation is considered valid. The second stage validates the improved DNP group parameters using MSRE pump start-up and coast-down experimental data to assess their accuracy in modeling reactivity changes from DNPs [142]. This experiment has been studied using Moltres and Quasimolto¹, allowing comparison with their results [144, 131]. However, previous models excluded the inlet and outlet plena in the MSRE, leading to discrepancies with experimental data. A model including the plena, combined with updated DNP group parameters, should yield results closer to the experimental data.

3.5.2 Safety

The motivation for studying safety is to protect the public and meet regulatory requirements. Safety is discussed in detail in Section 2.3.1, though the general takeaway is that many safety-related transients exist. The DNP group parameters are critical in these transients, as they influence the reactor period, power, and temperature.

The transients investigated are ULOHS, ULOF, UTOP, and FSOC. For each transient, the reactor period, power response, maximum temperature, minimum temperature, and reactivity response are recorded. The reactor period is critical, as it reflects the reactor's controllability. The power response shows how changes in the DNP group parameters affect other responses, such as temperature. It is important to make sure differences in the DNP parameters don't lead to temperature changes or oscillations which result in unsafe reactor conditions.

3.5.3 Security

Security is discussed in detail in Section 2.3.2, but the general takeaway is that bad actors may attempt to sabotage the reactor. This ties directly into the safety design study, as sabotage aims to create unsafe reactor conditions. Ensuring the MSR operates safely after sabotage is critical. If static DNP group parameters predict safe operation while improved parameters predict unsafe operation, correcting this discrepancy in the reactor design is essential.

Since the safety study covers most transients relevant to sabotage, additional considerations can be incorporated into these transients. As sabotage may initiate multiple safety-related transients simultaneously, it is reasonable to investigate such scenarios. With four unique transients, this results in six double-transient combinations, three triple-transient combinations, and one quadruple-transient combination, totaling ten security-related sabotage-initiated transients.

3.5.4 Safeguards

Safeguards are discussed in Section 2.3.3, but the main takeaway is the importance of preventing SNM diversion in MSRs. Detection methods, such as neutron detectors and surveillance systems, play a key role in safeguarding the reactor [205]. The improved DNP group parameters in this work affect the effective delayed neutron fraction and reactivity response. Using static DNP group parameters to detect SNM diversion predicts less accurate reactivity responses than the improved parameters. If the difference is significant, relying on static parameters may reduce the detection probability of a diversion scenario.

To investigate this, diversion scenarios are studied for a single significant quantity of ^{235}U and plutonium removed over the course of one year, simulated using OpenMC. The change in the fuel salt composition affects the

¹Quasimolto is an open-source multiphysics code that models multi-group neutron transport, DNP balance, and energy balance.

DNP group parameters, as discussed in Section 2.1.2. The importance and fission-weighted DNP group parameters can be used in Moltres, as discussed in Section 3.1.3, for simple step reactivity insertion transients, where the following four sets of data are compared: static group parameters without diversion, static group parameters with diversion, improved group parameters without diversion, and improved group parameters with diversion. If the difference between the static group parameters with and without diversion is smaller or similar to the difference between the static and improved group parameters, then this means detection of that diversion scenario requires models to account for the MSR physical phenomena when generating DNP group parameters.

Chapter 4

Preliminary Results

The preliminary results use the lowest-fidelity 0D scaled model to generate improved DNP group parameters. The results verify the DNP group parameters using stationary thermal fission of ^{235}U due to the abundance of data in the literature for this case. These preliminary results investigate individual DNPs, DNP group parameter sensitivities, and DNP group parameter verification.

Of the MSR physical phenomena considered, only chemical removal has an impact on the 0D scaled model. Varying the in-core and ex-core residence times does not affect the 0D scaled model because the scaling factor of the fission rate is identical to the scaling factor of the DNP generation rate. Because no decay chains are considered, the in-core and ex-core residence times do not have any effect. The proposed 0D flow model will be able to model the effects of residence time, as it accounts for time-dependent decays and parasitic absorptions. Although the 0D scaled model is rapid and enables more simulations, it does not incorporate fuel salt circulation into improved DNP group parameter generation.

An additional consideration is that the pulse irradiation results are not included. Whether combined via simultaneous solution of a larger matrix or solved separately and then combined, pulse irradiation results improve DNP group parameters for shorter-lived groups [89]. A simple investigation into the effects of pulse and saturation data is given in Appendix D. Parameters formulated for longer-lived groups are expected to be more accurate than for shorter-lived groups in the 0D scaled model, as it excludes pulse irradiation results. The 0D scaled model cannot model pulse irradiations because the cumulative fission yields do not provide accurate concentrations for non-equilibrium concentrations. Section 5.1.1 discusses this further.

The following results use the model parameters in Table 4.1, unless otherwise specified. The row labeled "sampled simulations" shows the number of non-linear least squares solves used in the Monte Carlo sampling. Each sampled simulation uses emission probabilities, half-lives, and fission yields within uncertainties from a normal distribution based on the nuclear data.

4.1 Important DNPs

Because the microscopic approach uses individual DNPs, it is possible to analyze the contribution of each DNP to the total delayed neutron yield. This is given in Equation (2.30), recreated here as:

$$\bar{v}_d = \sum_{i=1}^I P_{n,i} CFY_i.$$

Table 4.1: Default parameters for 0D scaled model simulations.

Parameter	Value
Fissile Nuclide	^{235}U [-]
Incident Neutron Energy	0.0253 [eV]
Fission Yields	ENDF/B-VII.1 [33]
Half-lives	IAEA [50]
Emission Probability	IAEA [50]
Final Decay Time	1200 [s]
Decay Time Spacing	Logarithmic
Decay Time Nodes	300
Sampled Simulations	1000
τ_{in}	9 [s] [71]
τ_{ex}	16 [s] [71]

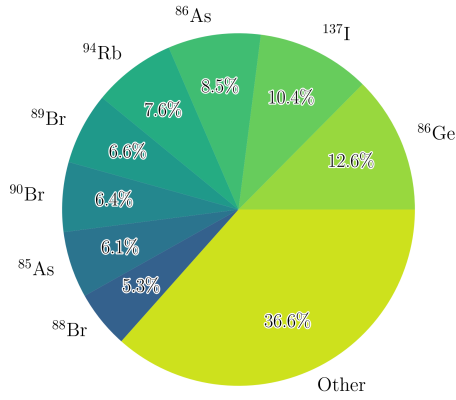
Another quantity of interest is the number of delayed neutrons from the existing concentration of nuclides, rather than the number of delayed neutrons per fission. This can be calculated by integrating the delayed neutron count rate, as shown in Equation (2.29), recreated here as:

$$\int_0^\infty \dot{n}_{d,i}(t) dt = P_{n,i} N_i(0).$$

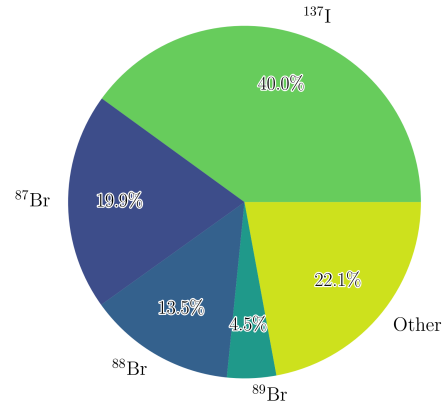
Figure 4.1 shows both approaches and how they differ. Figure 4.1(a) shows the delayed neutron yield contributions, which indicate the DNPs which emit the most delayed neutrons after a single fission event. Figure 4.1(b) shows each DNPs' contribution to the delayed neutron count rate for a saturation irradiation. The count rate contribution is weighed more heavily towards longer-lived DNPs because they build up during the saturation irradiation. Figure 4.1(c) shows the concentrations of each DNP, correlated with Figure 4.1(b). DNPs with a large concentration may have a small emission probability and contribute few delayed neutrons to the count rate, such as ^{141}Cs .

For the delayed neutron yield contributions in Figure 4.1(a), ^{86}Ge contributes the most, meaning that for each thermal fission event in ^{235}U , approximately 12.5% of delayed neutrons come from ^{86}Ge . ^{86}Ge is exceedingly short-lived, with a half-life of approximately 200 milliseconds. In Figure 4.1(b), it contributes less than 4% of the total delayed neutrons after the sample has been irradiated. Longer-lived DNPs, such as ^{137}I , accumulate during irradiation, dominating the delayed neutron count rate. This reinforces why pulse irradiation data are important, as saturation irradiation does not fully capture short-lived DNPs effects. Figure 4.1(c) shows the relative concentration of each DNP. The concentrations are generally consistent with the literature, though ^{141}Cs is an outlier [99]. This may result from the 0D scaled model simplifications, such as excluding parasitic absorption, or because other works ignore ^{141}Cs due to its extremely small emission probability.

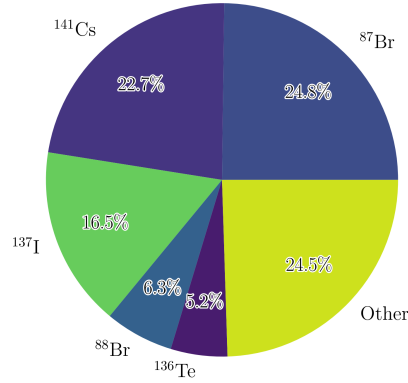
Figure 4.2 shows the delayed neutron count rate in Figure 4.2(a), as well as the integrated form in Figures 4.2(b) and 4.2(c), representing the total delayed neutron counts at each time. The DNP presented in this figure are based on the delayed neutron count rate; at each time step, the two DNPs with the largest delayed neutron count rate are added to the list of DNPs to be included in the figure. Figure 4.2(a) shows the ^{86}Ge count rate dominates at short time scales due to its short half-life, but it becomes less important after a tenth of a second. This again demonstrates how saturation irradiation enables better analysis of the longer-lived DNPs, but reduces the relative importance of shorter-lived DNPs. Figure 4.2(b) shows total counts varying over time on a log scale. ^{86}Ge initially dominates but quickly becomes unimportant compared to other DNPs. Figure 4.2(c) mirrors Figure 4.1(b), as both show ^{86}Ge contribution is minimal after saturation irradiation. The total delayed neutron count contribution is dominated by longer lived DNPs like ^{137}I and ^{87}Br .



(a) Delayed neutron yield contributions.



(b) Delayed neutron count rate contributions.



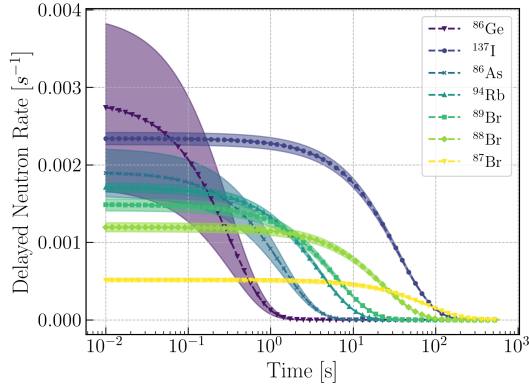
(c) DNP relative concentrations.

Figure 4.1: Delayed neutron contributions from each DNP after a thermal saturation irradiation of ²³⁵U using the 0D scaled model.

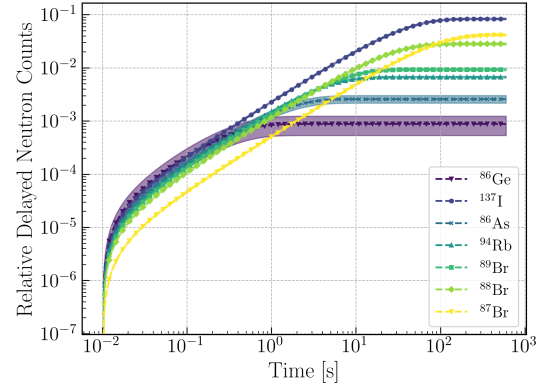
Another analysis of interest with the individual DNP data is calculation of the delayed neutron yield $\bar{\nu}_d$ and average half-life $\bar{\tau}$, shown in Table 4.2. These values are calculated and then compared with the six group DNP group parameters formulated from those DNPs. Ideally, these values would be identical, but the six-group fit cannot fully capture all decaying DNPs behavior, producing small differences. These differences are small relative to the uncertainty bounds for each term.

Table 4.2: The delayed neutron yield $\bar{\nu}_d$ and average half-life $\bar{\tau}$ calculated by summing individual DNPs and from the 6 DNP group parameters.

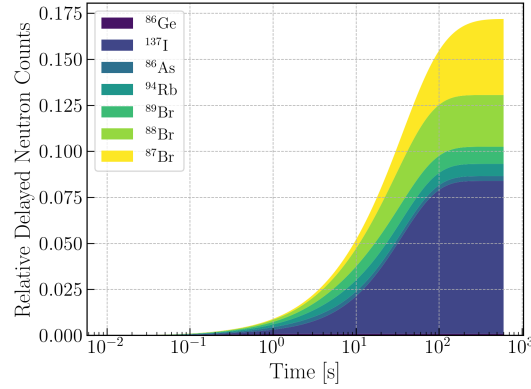
Term	DNP	Group
$\bar{\nu}_d$	0.0225 ± 0.0015	0.0224 ± 0.0013
$\bar{\tau}$ [s]	6.3852 ± 0.3801	6.4029 ± 0.3365



(a) The delayed neutron count rate.



(b) The total delayed neutron counts.



(c) A stack plot of the total delayed neutron counts.

Figure 4.2: Delayed neutron counts and count rate contributions after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

4.2 Sensitivity Studies

As part of this work, DNP group parameter sensitivity studies were performed to evaluate chemical removal, time nodes for the non-linear least squares problem, and nuclear data uncertainties. For the chemical removal analysis, the MSBR chemical reprocessing scheme is used. The MSBR reprocessing scheme has large chemical removal rates, providing a conservative impact of these chemical removal rates on the DNP group parameters. The time node evaluation includes the spacing between each node, the number of nodes, the total amount of time the sample should decay, and a combination of the total time and number of nodes. Finally, the nuclear data uncertainties include uncertainty in the fission yields, half-lives, and emission probabilities. To propagate these uncertainties to DNP group parameters, the non-linear least squares solver runs 1000 times using random normal samples of each DNP's fission yield, half-life, and emission probability.

4.2.1 Chemical Removal

As shown in Appendix B, the fraction of fuel salt in the ex-core region determines the scaling of the chemical removal rate. This analysis assumes an ex-core fraction, f_{ex} , of 0.5. This value is based on the core zone I, core zone II, and

annulus salt volumes relative to primary system salt volume in the MSBR [148]. Table 3.1 shows the MSBR chemical reprocessing scheme used in this analysis, recreated here.

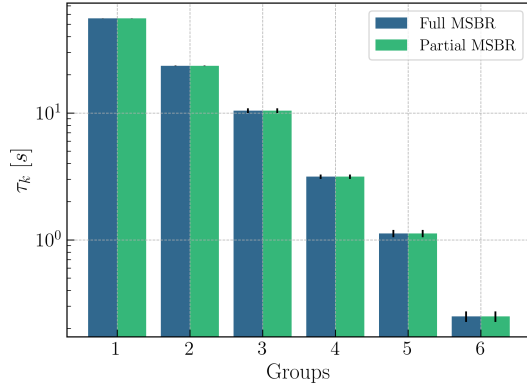
Table 4.3: MSBR Online Reprocessing Removal Rates from [148].

Reprocessing Group	Element(s)	Removal Rate [s^{-1}]
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	2.32×10^{-7}
Rare Earths	Eu	2.32×10^{-8}
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	5.00×10^{-2}
Seminoble Metals	Zr, Cd, In, Sn	5.79×10^{-8}
Gases	Kr, Xe	5.00×10^{-2}
Volatile Fluorides	Br, I	1.93×10^{-7}
Discard	Rb, Sr, Cs, Ba	3.37×10^{-9}
Salt Discard	Th, Li, Be, F	3.37×10^{-9}
Protactinium	Pa	3.86×10^{-6}
Higher Nuclides	Np, Pu	1.98×10^{-9}

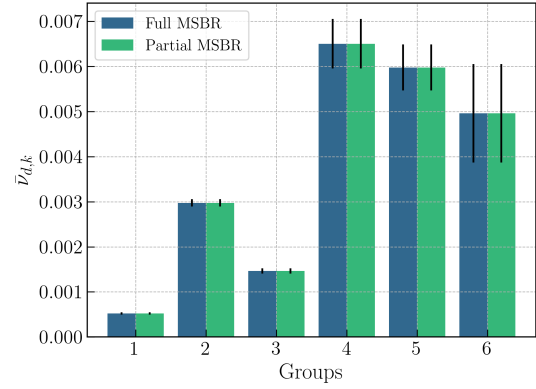
One consideration is that the 0D scaled model underestimates chemical removal rate effects. This is because it only chemically removes from the cumulative fission yield of the element in the reprocessing scheme, and does not account for decay chains which may have chemical removal rates included. For example, ^{137}I has a cycle time of 60 days, corresponding to a chemical removal rate of $1.9 \times 10^{-7} [s^{-1}]$. However, antimony and tellurium, which decay into iodine, have a chemical removal rate of $0.05 [s^{-1}]$, many orders of magnitude larger than the iodine removal rate. The 0D scaled model, with cumulative fission yields and no decay chain, does not capture this impact on the concentration of iodine.

Long Cycle Time Evaluation

Table 3.1 lists many elements to consider. Since the sample is irradiated only for a few minutes, it is useful to assess the significance of smaller removal rates, such as the 500-day cycle time of rare earths. This analysis considers only 20-second cycle time elements, the noble metals and gases. Figure 4.3 shows the difference between the full MSBR reprocessing scheme and the partial scheme considering only 20-second cycle time elements. The differences between each DNP group parameter, $\bar{v}_{d,k}$ and λ_k , are less than 0.001%. This analysis shows elements with cycle times longer than 20 seconds do not affect DNP group parameters for thermal saturation irradiation of ^{235}U using the 0D scaled model.



(a) Half-life τ_k for each DNP group k .

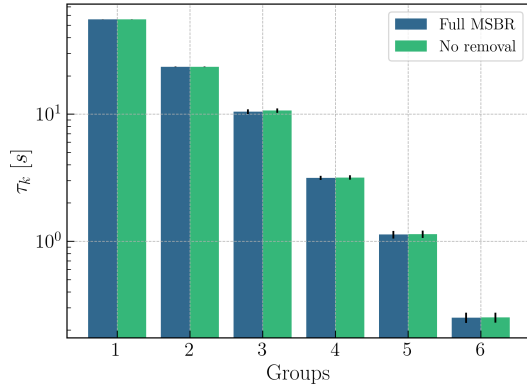


(b) Delayed neutron yield $\bar{v}_{d,k}$ for each DNP group k .

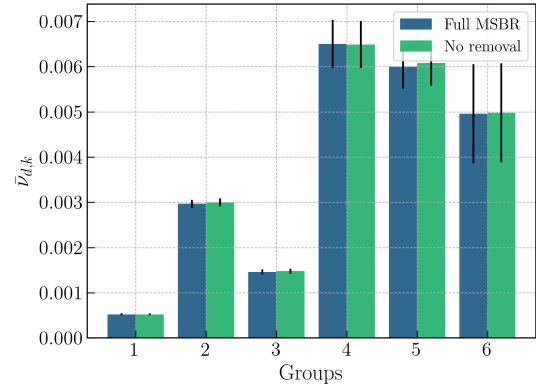
Figure 4.3: Delayed neutron contributions from each DNP with and without long cycle time chemical removal rates in the MSBR after a thermal saturation irradiation of ^{235}U using the 0D scaled model. The differences induced by exclusion of removal rates with cycle times longer than 20 seconds are minimal and well within uncertainty bounds.

Removal Rate Evaluation

It is also of interest to determine the sensitivity of the DNP group parameters to the chemical reprocessing scheme. The 0D scaled model underestimates chemical removal effects due to the use of cumulative fission yields without explicit nuclide tracking, but this analysis still demonstrates the general chemical reprocessing impact. The comparison of DNP group parameters with and without the MSBR reprocessing scheme is given in Figure 4.4. The figure shows the half-lives are similar, while yields without removal are slightly higher.



(a) Half-life τ_k for each DNP group k .



(b) Delayed neutron yield $\bar{v}_{d,k}$ for each DNP group k .

Figure 4.4: Delayed neutron contributions from each DNP with and without the MSBR chemical removal rates after a thermal saturation irradiation of ^{235}U using the 0D scaled model. Neglecting chemical removal rates generally leads to larger yields, though all differences are within uncertainty bounds.

Differences between average half-lives are negligible, while differences between the total delayed neutron yields are 0.57%, or approximately 12 pcm. This small difference likely results from the 0D scaled model underestimating the effect of chemical removal. Since chemical reprocessing consistently reduces total delayed neutron yield by removing DNPs, it is important to account for this factor.

4.2.2 Time Discretization

The discretization of the irradiation time is represented by some number of time nodes over the total decay time. The time node sensitivity is important, as many nodes may be required to fully capture the physics, but each additional node increases computational cost. Of interest are the node distribution, number of nodes, sample decay time, and combined decay time and number of nodes.

Distribution

Two time node distributions are compared: logarithmic spacing from 0.01 to 600 seconds and linear spacing from 0 to 600 seconds. With 300 time nodes, linear spacing does not capture behavior in the first two seconds but provides higher resolution at longer time scales. This longer time scale resolution is less useful, as many relevant DNPs quickly drop off at longer time scales, as discussed in Section 4.1.

Figure 4.5 compares DNP group parameters for logarithmic and linear spacing. Half-lives of shorter-lived groups are larger for linear spacing, consistent with expectations. Linear spacing has no data between zero and two seconds, and it places greater weight on longer-lived groups due to the data distribution. This causes shorter-lived groups to capture a larger yield, with yields shifted from groups two, three, and four to groups five and six. The total DNP yield differs by approximately 0.1%, so changing time node spacing does not alter macroscopic DNP group behavior. Instead, it affects the distribution of delayed neutrons over time, with logarithmic spacing more accurate by capturing shorter-lived DNPs effects.

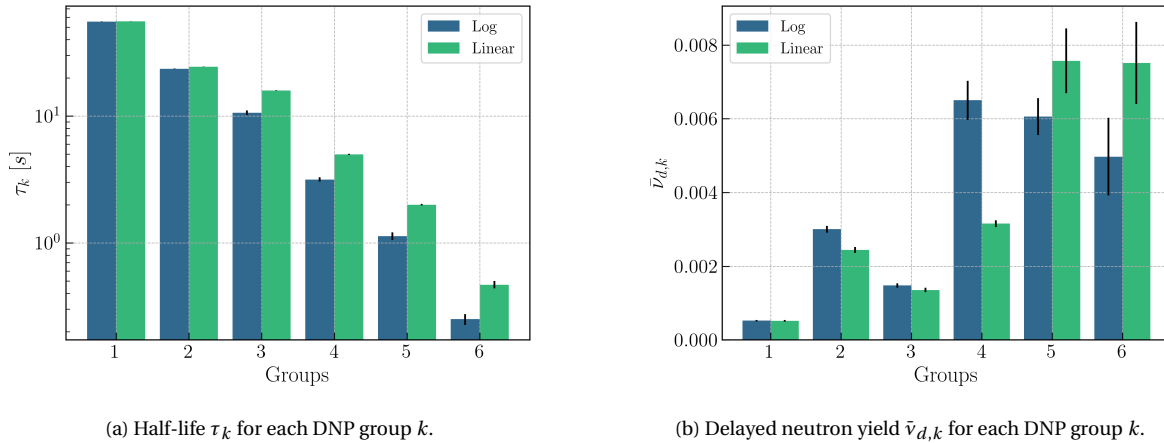


Figure 4.5: DNP group parameters for linear and logarithmic spacings of the time nodes after a thermal saturation irradiation of ^{235}U using the 0D scaled model. The logarithmic time node spacing better captures short-lived DNPs. The differences between the DNP group parameters for the different time node spacing is outside uncertainty bounds for every group except the longest-lived group one.

Number of Time Nodes

Figure 4.6 shows how varying the number of nodes affects the results. It appears there is not a significant difference between the results when varying the number of nodes. The main difference appears in the yields of groups four and five, with increasing nodes increasing the yield of group four and decreasing the yield of group five. There are also some smaller differences, such as the yield of group six decreasing and the yields of groups two and three increasing with an increasing number of nodes.

Physically interpreting these trends is challenging, as this is a numerical effect caused by different amounts of data being present. The DNP group parameters with the most nodes is the most accurate, as it provides the most accurate set of data for the DNP group parameters to use for least squares fitting. Based on this, one interpretation could be that the more significant differences in groups four and five are both approaching the correct value, but also affect each other. The logarithmic time node spacing means that the short-lived DNP groups are captured, and the long-lived DNP groups don't need as many data points to be captured. This indicates that the medium-lived DNP groups, particularly groups four and five, increase the number of time nodes required in the solve due to the logarithmic time node spacing.

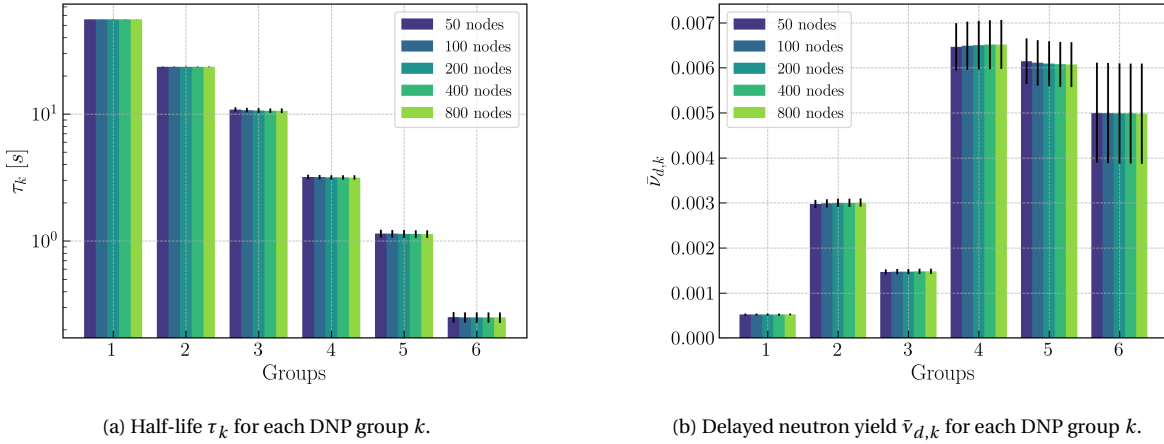


Figure 4.6: DNP group parameters for varying numbers of time nodes after a thermal saturation irradiation of ^{235}U using the 0D scaled model. Increasing the number of nodes increases the accuracy of the DNP group parameters, with differences in the yields that are within uncertainty bounds.

The simulation with the most nodes is most accurate, which allows calculation of percent differences in DNP group yields and the associated computational cost. This is shown in Table 4.4, as 50 nodes results in the group five yield being off by approximately 1%, and 200 nodes keeps differences below 0.02%. Computational cost remains fairly constant across these node numbers. These results indicate that 300 nodes provide sufficiently accurate yields, within 0.1%, while remaining in the constant computational cost regime.

Table 4.4: Comparison of how the number of nodes changes the computational cost and relative difference in yields compared to the 400 node case after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Nodes	Yield ($\bar{\nu}_{d,k}$) Difference [%]						Time [s]
	1	2	3	4	5	6	
50	0.30	0.90	0.53	0.75	0.99	0.59	297
100	0.17	0.49	0.25	0.39	0.46	0.77	297
200	0.01	0.02	0.00	0.01	0.01	0.09	300
400	0.00	0.00	0.00	0.00	0.00	0.01	304
800	-	-	-	-	-	-	310

Total Time Post-Irradiation

Increasing the total time after irradiation of the sample, or the total decay time, produces two simultaneous effects: decreasing time node density and increasing the number of nodes evaluated at longer times. The decrease in time

node density is because the same number of time nodes spread over a longer time frame will be less dense. The increase at time nodes at longer times can be understood by considering an extreme example of a distribution of 100 time nodes over 100 seconds compared to 100 time nodes over 100 years; even with logarithmic spacing, most of the nodes will be placed at a larger time in the 100 year case than the 100 second case. Because the longest-lived DNP¹, ⁸⁷Br, has a half-life of approximately one minute, it is reasonable to assume that the default final time of ten minutes offers sufficient data on all DNP decays.

Figure 4.7 shows this assumption is not accurate. The DNP group parameters, τ_k and $\bar{v}_{d,k}$, converge asymptotically as the total decay time increases, with the decay time of 1200 seconds, or 20 minutes, being the smallest decay time after which longer decay times do not change the results. The shorter total decay times inaccurately model DNPs, as they do not fully capture long-lived DNPs and have shorter group half-lives. These shorter half-lives correlate with an overestimation of some longer-lived DNP group yields.

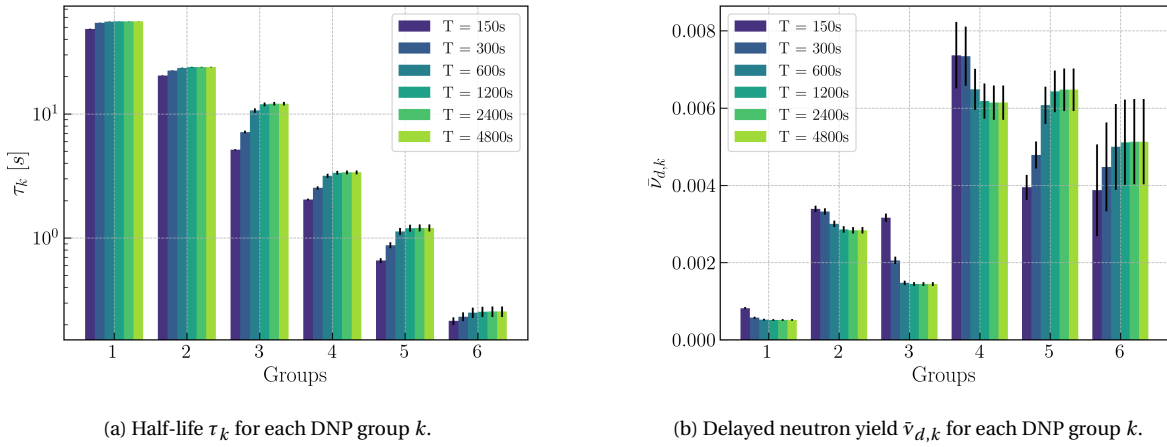


Figure 4.7: DNP group parameters for varying total decay times after a thermal saturation irradiation of ²³⁵U using the 0D scaled model. Collecting delayed neutron counts for longer after irradiation results in longer-lived groups with smaller yields in groups one through four and larger yields in groups five and six.

The total delayed neutron yields and average half-lives, shown in Table 4.5, provide another comparison between different decay times. The table shows differences are small and within uncertainty bounds.

Table 4.5: Comparison of how the net decay time, T changes the total delayed neutron yield and average half-life after a thermal saturation irradiation of ²³⁵U using the 0D scaled model.

T [s]	$\bar{v}_d$	$\bar{\tau}$ [s]
150	0.0225 ± 0.0015	6.3786 ± 0.3784
300	0.0225 ± 0.0014	6.3945 ± 0.3554
600	0.0225 ± 0.0013	6.3978 ± 0.3412
1200	0.0225 ± 0.0013	6.3977 ± 0.3415
2400	0.0225 ± 0.0013	6.3977 ± 0.3417
4800	0.0225 ± 0.0013	6.3977 ± 0.3417

¹The DNPs ²¹¹Tl, ²¹⁰Tl, and ²¹⁰Hg are longer-lived and have non-zero emission probabilities, but they live only tens of seconds longer and generally have negligible concentrations.

Time Node Density

This section evaluates DNP group parameter sensitivity to the time node density, number of time nodes, and total decay time by comparing different numbers of time nodes and decay times. Figure 4.8 shows how the DNP group parameters vary with time node density. For a linearly spaced set of time nodes, this could be represented by the time step.

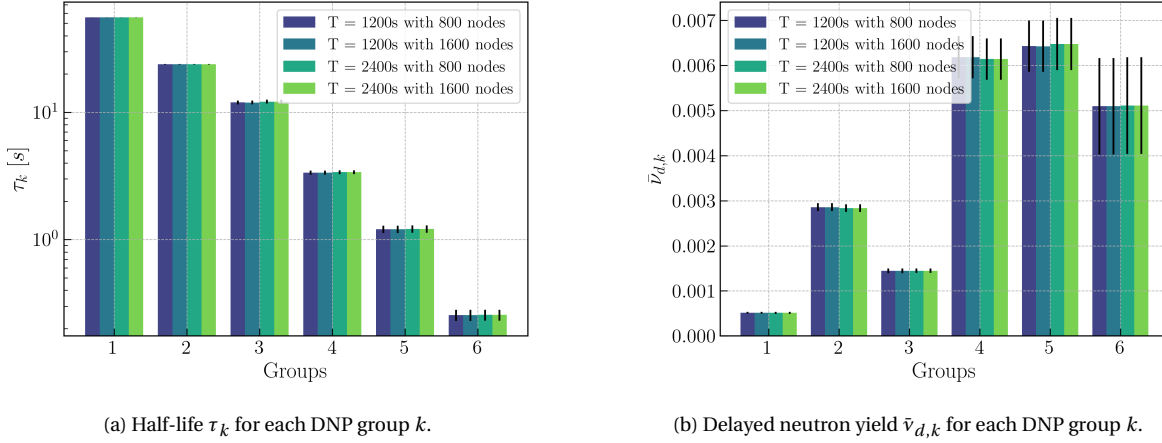


Figure 4.8: DNP group parameters, τ_k and $\bar{v}_{d,k}$, for varying node densities after a thermal saturation irradiation of ^{235}U using the 0D scaled model. Changing the number of nodes does not have a significant change on the DNP group parameters, whereas changes to the total decay time leads to significant differences.

The time node density of the 1200 second decay time with 800 nodes is two time nodes for every three seconds of decay; the time node density of the 2400 second decay time with 1600 nodes is the same. If the DNP group parameters primarily depend on the node density, then these two different sets of decay times and time nodes should give the same set of DNP group parameters. Instead, the results remain approximately constant with varying node numbers, with the main difference arising from the total decay time. This indicates the sensitivity analysis in Section 4.2.2 is valid and does not need to account for varying time node density.

4.2.3 Nuclear Data Uncertainties

This work uses a Monte Carlo approach to propagate the uncertainties in nuclear data and evaluate the sensitivity of the DNP group parameters to these uncertainties. For each nuclear data variable in the 0D scaled model, such as the ^{87}Br half-life, a random value is sampled from a normal distribution. This process is repeated for every DNP fission yield, half-life, and emission probability.

Count Rate

Figure 4.9 shows the count rate normalized to the Keepin et al. count rate. The red lines show count rates from sampled simulations, while the blue dashed line shows the DNP group parameters and uncertainty based on 1000 Monte Carlo non-linear least squares solves. The black X's indicate the mean count rate at each point. The other six-group DNP parameters plotted include Keepin (1.0, as the plot is normalized to Keepin values), Brady, and Synetos [89, 22, 180].

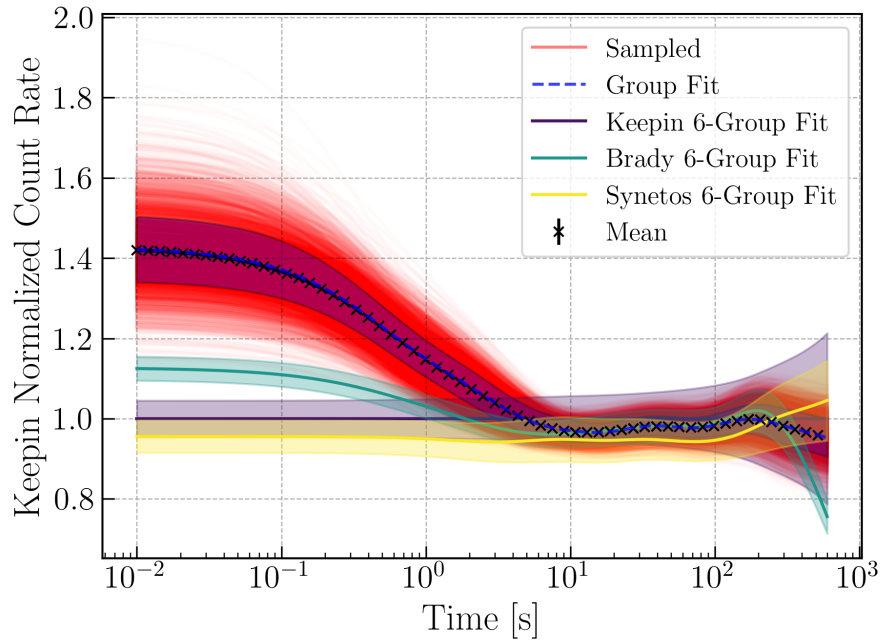


Figure 4.9: The delayed neutron count rate for thermal fission of ^{235}U normalized to the Keepin et al. six-group DNP group parameters. The current work, represented by the Sampled, Group Fit, and Mean data trends shows poor agreement at times less than one second and agreement within uncertainty bounds beyond three seconds.

Beyond approximately three seconds, the current DNP group parameters agree within uncertainty bounds with most literature results, with Brady parameters showing a noticeable difference at longer times. Short time scales are inaccurate in the 0D scaled model due to the lack of pulse irradiation data, as discussed in the chapter introduction. Adding this data is expected to improve the fit at earlier times. To demonstrate this, the first two groups are used while the shorter-lived groups three through six are replaced with the Keepin et al. DNP group parameters [89]. Figure 4.10 shows this modified set of DNP group parameters.

The modified DNP group parameters are not a perfect comparison, as the non-linear least squares solve adjusts each group's yield relative to the yields and half-lives of other groups. This indicates excluding pulse irradiation data also affects longer-lived groups, though less than it affects shorter-lived groups. Adjusting the shorter-lived groups improves agreement with the literature for the 0D scaled model, particularly at times less than one second.

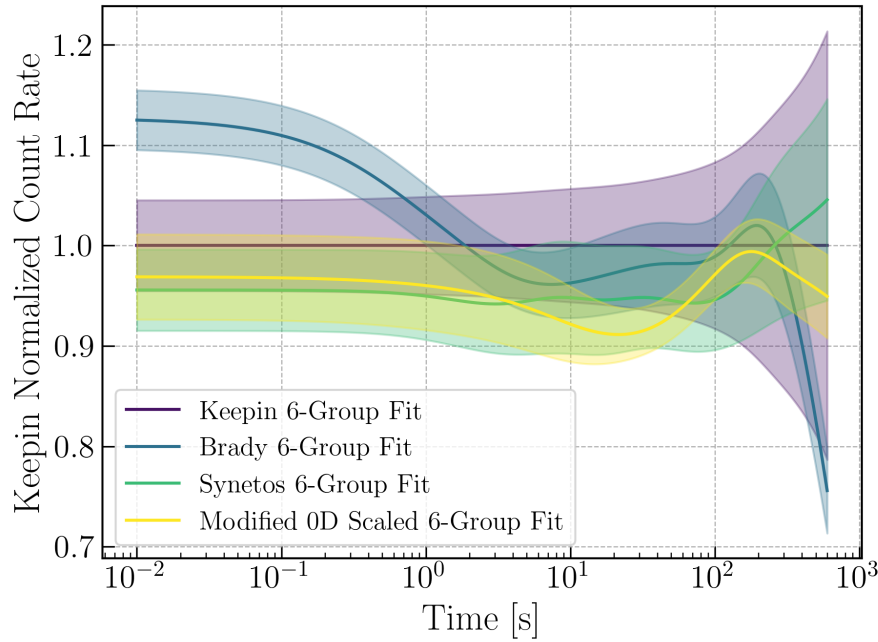


Figure 4.10: The delayed neutron count rate for thermal fission of ^{235}U normalized to the Keepin et al. six-group DNP group parameters. The modified 0D scaled model has overlapping uncertainty bounds with the Keepin et al. results at all points in time.

Individual DNPs

The Monte Carlo sampling of each DNP's nuclear data performed in this work enables sensitivity analyses on how each change affects the DNP group parameters. For example, because group one is dominated by ^{87}Br , changes to its half-life, emission probability, or fission yield noticeably affect the group one DNP half-life and yield.

The number of Monte Carlo simulations is increased to 5000 to provide sufficient data for each DNP, ensuring differences in DNP group parameters are statistically significant. To find the most sensitive DNPs that experience the most difference in DNP group parameters τ_k and $\bar{v}_{d,k}$, a linear regression fits the slope of the relative change in DNP group parameters to the relative change in the individual DNP nuclear data, while also calculating the Pearson correlation coefficient (PCC). The PCC quantifies the linear relationship between two variables: -1 indicates perfect negative correlation, +1 indicates perfect positive correlation, and 0 indicates no correlation [62]. Changes in the nuclear data include emission probability P_n , fission yields (affecting concentration N), and half-lives τ .

Correlations of interest for this analysis have absolute PCC values greater than 0.2. This threshold is somewhat arbitrary but indicates a DNP has a non-negligible correlation with a DNP group parameter. Table 4.6 presents the results of this analysis.

One of the most predictable sensitivities in Table 4.6 is ^{87}Br 's effect on group one half-life and yield, as it dominates this group. Specifically, ^{87}Br , with a half-life of 55.64 ± 0.15 seconds, has correlations with a PCCs greater than 0.2 that affect group parameters as follows: emission probability on group yield, concentration on group yield, and half-life on group half-life. Figure 4.11 shows how group one parameters vary with ^{87}Br data, as well as less strongly correlated data, such as ^{87}Br emission probability on the group three half-life. Although it may seem counterintuitive, changes to the ^{87}Br emission probability affect the group three half-life, showing a non-negligible correlation. This is because the DNP group parameters cannot be considered in isolation; they form a collective set

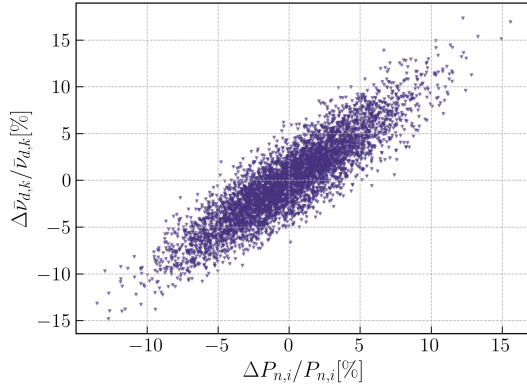
Table 4.6: PCCs with a magnitude larger than 0.2 for variation in DNP data which correspond to changes in DNP group parameters after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Nuclide	Group Value	DNP Value	PCC
^{84}Ga	$\tau_3[\text{s}]$	$N_i[-]$	-0.211724
^{84}Ga	$\tau_4[\text{s}]$	$N_i[-]$	-0.222001
^{84}Ga	$\tau_5[\text{s}]$	$N_i[-]$	-0.203050
^{85}As	$\bar{v}_{d,4}[-]$	$N_i[-]$	0.757084
^{85}As	$\tau_4[\text{s}]$	$N_i[-]$	-0.411166
^{85}As	$\bar{v}_{d,5}[-]$	$N_i[-]$	0.718393
^{85}As	$\tau_5[\text{s}]$	$N_i[-]$	0.448911
^{88}Br	$\bar{v}_{d,3}[-]$	$P_{n,i}[-]$	0.276294
^{88}Br	$\bar{v}_{d,3}[-]$	$\tau_i[\text{s}]$	-0.221999
^{88}Br	$\tau_3[\text{s}]$	$P_{n,i}[-]$	0.379647
^{88}Br	$\tau_3[\text{s}]$	$N_i[-]$	0.264940
^{138}I	$\bar{v}_{d,3}[-]$	$P_{n,i}[-]$	0.211254
^{137}I	$\bar{v}_{d,2}[-]$	$P_{n,i}[-]$	0.478664
^{137}I	$\bar{v}_{d,2}[-]$	$\tau_i[\text{s}]$	-0.270646
^{137}I	$\bar{v}_{d,2}[-]$	$N_i[-]$	0.746893
^{137}I	$\tau_2[\text{s}]$	$\tau_i[\text{s}]$	0.813782
^{137}I	$\tau_2[\text{s}]$	$N_i[-]$	0.252540
^{86}As	$\bar{v}_{d,5}[-]$	$N_i[-]$	0.512737
^{98}Y	$\bar{v}_{d,4}[-]$	$P_{n,i}[-]$	0.243371
^{86}Ge	$\bar{v}_{d,2}[-]$	$\tau_i[\text{s}]$	-0.238309
^{86}Ge	$\tau_2[\text{s}]$	$\tau_i[\text{s}]$	0.359997
^{86}Ge	$\bar{v}_{d,3}[-]$	$\tau_i[\text{s}]$	-0.637401
^{86}Ge	$\tau_3[\text{s}]$	$\tau_i[\text{s}]$	0.656167
^{86}Ge	$\bar{v}_{d,4}[-]$	$\tau_i[\text{s}]$	-0.407397
^{86}Ge	$\tau_4[\text{s}]$	$\tau_i[\text{s}]$	0.673363
^{86}Ge	$\bar{v}_{d,5}[-]$	$\tau_i[\text{s}]$	0.263800
^{86}Ge	$\tau_5[\text{s}]$	$\tau_i[\text{s}]$	0.662307
^{86}Ge	$\bar{v}_{d,6}[-]$	$P_{n,i}[-]$	0.830397
^{86}Ge	$\bar{v}_{d,6}[-]$	$N_i[-]$	0.464433
^{86}Ge	$\tau_6[\text{s}]$	$P_{n,i}[-]$	-0.306136
^{86}Ge	$\tau_6[\text{s}]$	$\tau_i[\text{s}]$	0.893433
^{87}Br	$\bar{v}_{d,1}[-]$	$P_{n,i}[-]$	0.896423
^{87}Br	$\bar{v}_{d,1}[-]$	$N_i[-]$	0.456307
^{87}Br	$\tau_1[\text{s}]$	$\tau_i[\text{s}]$	0.997442
^{137}Sb	$\bar{v}_{d,3}[-]$	$N_i[-]$	0.229445
^{137}Sb	$\tau_3[\text{s}]$	$N_i[-]$	-0.229578
^{137}Sb	$\tau_4[\text{s}]$	$N_i[-]$	-0.250857
^{137}Sb	$\tau_5[\text{s}]$	$N_i[-]$	-0.299691

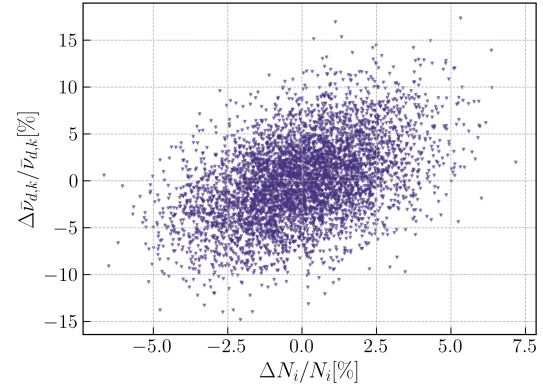
that best fits the delayed neutron count rate curve. Changes to that curve propagate to all group parameters, with varying magnitudes across groups and parameters.

Another DNP of interest in Table 4.6 is ^{86}Ge , an important contributor to delayed neutron yield with a half-life of 0.222 ± 0.022 seconds, as shown in Figure 4.1. Variation in ^{86}Ge data impacts many DNP group parameters, especially due to large relative uncertainty in its half-life of 9.9%. Figures 4.12 and 4.13 show each DNP group parameter's sensitivity to variations in ^{86}Ge data.

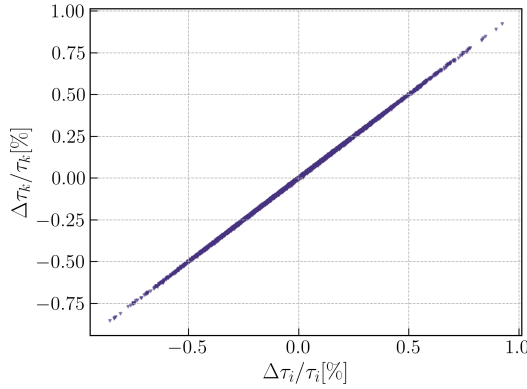
Figure 4.12 shows the largest group half-life sensitivities occur due to changes in ^{86}Ge half-life. For increasingly longer-lived groups, ^{86}Ge half-life has progressively less impact on group half-lives. This is intuitive, as a short-lived



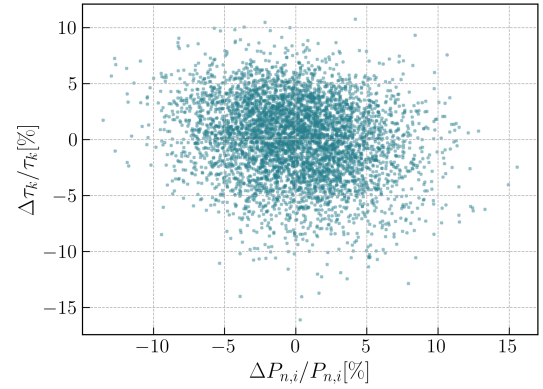
(a) DNP group 1 yield variation due to emission probability variation with a PCC of 0.90.



(b) DNP group 1 yield variation due to fission yield variation with a PCC of 0.46.



(c) DNP group 1 half-life variation due to half-life variation with a PCC of 1.00.



(d) DNP group 3 half-life variation due to emission probability variation with a PCC of -0.19.

Figure 4.11: DNP group parameter sensitivity to changes in ^{87}Br nuclear data after a thermal saturation irradiation of ^{235}U using the 0D scaled model. The group one PCC values are larger than that of group three because ^{87}Br 's half-life is closer to that of group one than the half-life of group three.

nuclide such as ^{86}Ge should have much more impact on short-lived groups than on long-lived groups. Group six half-life sensitivity to ^{86}Ge shows a lower bound with no data points, indicating asymmetry. Sensitivities typically exhibit symmetry about the line that best fits the data for a linear response. This asymmetry likely arises because many longer-lived DNPs exist, whereas ^{86}Ge is one of the few very short-lived DNPs dominating this time region. Because shorter-lived DNPs are absent in this time region, group six half-life decreases less than it increases.

Figure 4.13 shows changes in the group yields due to changes in the ^{86}Ge half-life. No clear trend exists from shorter to longer-lived groups, though groups three and four show negative correlations, suggesting a longer-lived ^{86}Ge decreases their yields while increasing their half-lives, with half-lives shown in Figure 4.12. These groups capture fewer short-lived DNPs as the ^{86}Ge half-life increases, both by increasing their group half-lives and decreasing their yields.

Another interesting sensitivity is the ^{86}Ge emission probability P_n on the group six yield, showing a narrow spread for small emission probabilities that widens for larger emission probabilities. Increasing the emission probability for the short-lived ^{86}Ge corresponds to an increase in the short-lived group 6 yield. This widening likely

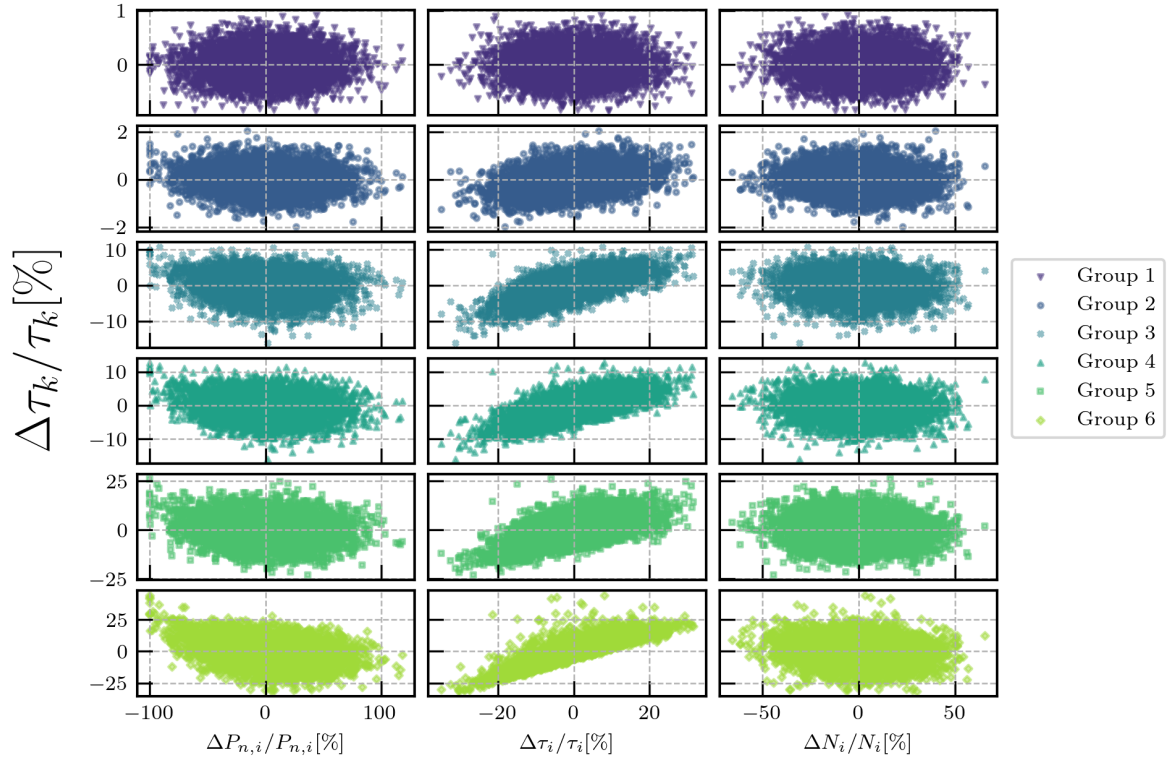


Figure 4.12: The difference in each DNP group half-life from the ^{86}Ge relative changes in emission probability, half-life, and concentration after a thermal saturation irradiation of ^{235}U using the 0D scaled model. The half-life of the shorter-lived groups is more sensitive to the half-life of ^{86}Ge , with less correlation for longer-lived groups.

arises from the absence of pulse irradiation data. Without the pulse irradiation data, short-lived group parameters τ_k and $\bar{v}_{d,k}$ are less accurate, so removing ^{86}Ge produces a more consistent group six yield. As ^{86}Ge becomes more impactful, possible group six yield values spread due to missing pulse irradiation data.

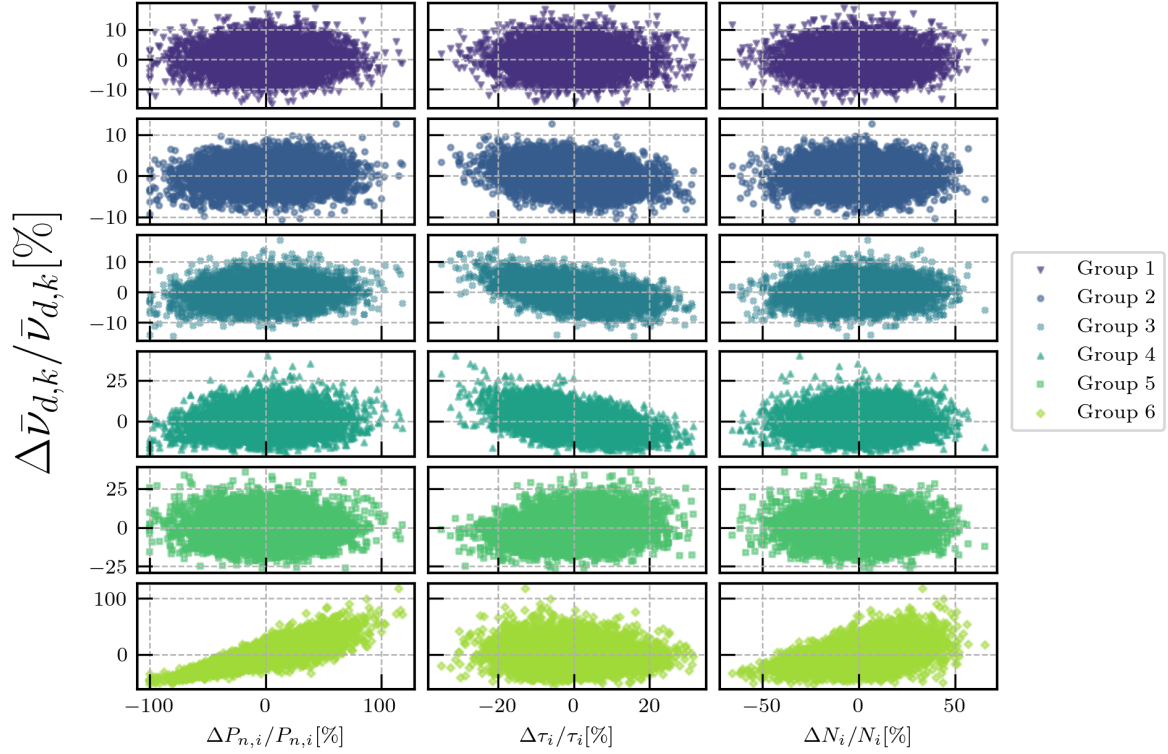


Figure 4.13: The difference in each DNP group yield from the ^{86}Ge relative changes in emission probability, half-life, and concentration after a thermal saturation irradiation of ^{235}U using the 0D scaled model. The group six yields are sensitive to the emission probability and concentration of ^{86}Ge . The group yields are also sensitive to the half-life of ^{86}Ge , as the changes in the ^{86}Ge half-life affects both the group half-lives and their yields.

4.3 Verification

The 0D scaled model verification uses DNP group parameters from thermal fission of ^{235}U . This verification uses the data from literature, introduced in Section 2.1.1, combined and recreated here with this work's results.

Tables 4.7, 4.8, 4.10, and 4.10 show the DNP group parameters from literature and how they compare to the 0D scaled model DNP group parameters. As discussed, the 0D scaled model excludes pulse irradiation data, so shorter-lived groups are less accurate. This appears in the yields: groups five and six differ significantly, while groups one and two match the literature more closely. The group half-lives generally agree with the literature, though groups three and five are outliers. This likely arises from missing pulse irradiation data, as both group half-lives are approximately twice as large as the values in the literature.

Table 4.11 compares the total delayed neutron yield for 100 fissions between the literature and the 0D scaled model. It can be seen that the data is not within uncertainty bounds of each other, and even differs between different approaches for the same dataset. Differences between microscopic and macroscopic delayed yields occur for many fissile nuclides, not only ^{235}U [113]. The 0D scaled model yield exceeds other literature values, but this is a data issue and not a fitting issue. The summed DNP data gives approximately the same yield as shown in Table 4.2. This indicates no issue with DNP group parameter formulation; rather, pulse irradiation data is needed to capture

Table 4.7: DNP group yields from thermal fission of ^{235}U (recreated from [129]).

Source	Group [-]					
	1	2	3	4	5	6
Keepin et al. [89]	0.00052	0.00346	0.00310	0.00624	0.00182	0.00066
Charlton et al. [35]*	0.00054	0.00344	0.00319	0.00607	0.00186	0.00066
ENDF/B-VI.8 [153]	0.00058	0.00302	0.00288	0.00646	0.00265	0.00111
Mills et al. [119]	0.00056	0.00325	0.00196	0.00658	0.00287	0.00122
Brady and England [22]	0.00068	0.00341	0.00292	0.00611	0.00310	0.00158
Saleh et al. [160]	0.00055	0.00363	0.00297	0.00593	0.00168	0.00063
Synetos and Williams [180]*	0.00051	0.00328	0.00214	0.00583	0.00270	0.00063
Tuttle [189]	0.00065	0.00362	0.00320	0.00692	0.00218	0.00044
Waldo et al.[194]	0.00055	0.00366	0.00327	0.00660	0.00192	0.00070
0D scaled model	0.00052	0.00297	0.00146	0.00650	0.00597	0.00500

* Modified to include a sixth group by Parish et al. [129].

Table 4.8: DNP group yield uncertainties from thermal fission of ^{235}U (recreated from [129]).

Source	Group [-]					
	1	2	3	4	5	6
Keepin et al. [89]	0.00006	0.00025	0.00041	0.00047	0.00019	0.00013
Charlton et al. [35]*	0.00003	0.00005	0.00024	0.00013	0.00014	0.00014
ENDF/B-VI.8 [153]	-	-	-	-	-	-
Mills et al. [119]	0.00004	0.00023	0.00020	0.00051	0.00035	0.00012
Brady and England [22]	0.00004	0.00019	0.00016	0.00034	0.00017	0.00009
Saleh et al. [160]	0.00005	0.00032	0.00027	0.00104	0.00015	0.00012
Synetos and Williams [180]*	0.00004	0.00020	0.00036	0.00037	0.00029	0.00012
Tuttle [189]	0.00007	0.00013	0.00041	0.00019	0.00021	0.00007
Waldo et al.[194]	0.00006	0.00041	0.00041	0.00038	0.00017	0.00009
0D scaled model	0.00002	0.00009	0.00006	0.00056	0.00051	0.00111

* Modified to include a sixth group by Parish et al. [129].

Table 4.9: DNP group half-lives from thermal fission of ^{235}U (recreated from [129]).

Source	Group [s]					
	1	2	3	4	5	6
Keepin et al. [89]	55.89897	22.72614	6.24457	2.30281	0.60802	0.23028
Charlton et al. [35]*	55.45177	22.43195	6.02737	2.18658	0.49159	0.23028
ENDF/B-VI.8 [153]	51.72740	21.19716	5.73797	2.28913	0.81595	0.24295
Mills et al. [119]	55.27489	22.80089	7.67860	2.77148	1.07381	0.28177
Brady and England [22]	52.11633	21.32761	5.68620	2.18727	0.70114	0.23462
Saleh et al. [160]	55.45177	22.65187	6.24457	2.31049	0.63013	0.23028
Synetos and Williams [180]*	56.35343	22.35959	7.00149	2.72893	0.77016	0.23028
Tuttle [189]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911
Waldo et al.[194]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911
0D scaled model	55.52977	23.54278	10.43942	3.14523	1.12586	0.25015

* Modified to include a sixth group by Parish et al. [129].

smaller yields of shorter-lived DNP groups. This can be seen in Table 4.7, where the group 5 and 6 yields are greatly

Table 4.10: DNP group half-life uncertainties from thermal fission of ^{235}U (recreated from [129]).

Source	Group [s]					
	1	2	3	4	5	6
Keepin et al. [89]	1.35239	0.67061	0.22503	0.09181	0.08000	0.02219
Charlton et al. [35]*	0.44361	0.50817	0.10482	0.06208	0.03138	0.02219
ENDF/B-VI.8 [153]	-	-	-	-	-	-
Mills et al. [119]	0.09918	0.11723	0.37861	0.11259	0.08521	0.01020
Brady and England [22]	-	-	-	-	-	-
Saleh et al. [160]	3.99253	0.14805	0.39380	0.03851	0.01719	0.02219
Synetos and Williams [180]*	0.45816	0.28851	0.77794	0.21488	0.09413	0.02219
Tuttle [189]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545
Waldo et al.[194]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545
0D scaled model	0.15676	0.12560	0.44990	0.12771	0.07629	0.02325

* Modified to include a sixth group by Parish et al. [129].

over-valued relative to other DNP group yields in the literature. The modified 0D scaled model, using Keepin et al. DNP group parameters for groups three through six, produces a yield more consistent with literature values.

Table 4.11: Total delayed neutron yields of 100 fissions from thermal fission of ^{235}U (based on [81]).

Approach	Data	Delayed Yield [-]
Microscopic	JEF-2.2 [123]	1.708 ± 0.12
Microscopic	JEF-3.1.1 [161]	1.477 ± 0.08
Microscopic	ENDF/B-VII.1 [33]	1.905 ± 0.10
Microscopic	ENDF/B-VI.8 [153]	1.670
Microscopic	Mills et al. [119]	1.645 ± 0.07
Microscopic	Brady and England [22]	1.780 ± 0.05
Microscopic	0D Scaled Model Data	2.254 ± 0.15
Microscopic	0D Scaled Model	2.256 ± 0.13
Hybrid	Modified 0D Scaled Model	1.531 ± 0.09
Macroscopic	JEF-2.2 [123]	1.655
Macroscopic	JEF-3.1.1 [161]	1.620
Macroscopic	ENDF/B-VII.0 [32]	1.585
Macroscopic	Keepin et al. [89]	1.580 ± 0.07
Macroscopic	Charlton et al. [35]	1.577 ± 0.03
Macroscopic	Saleh et al. [160]	1.645 ± 0.11
Macroscopic	Synetos and Williams [180]	1.510 ± 0.06
Macroscopic	Tuttle [189]	1.700 ± 0.05
Macroscopic	Waldo et al.[194]	1.670 ± 0.07

Table 4.12 compares average half-lives between the 0D scaled model and the literature. The 0D scaled model average half-life is too short, primarily due to overvalued group five and six yields that give those half-lives greater weight. The modified 0D scaled model, with groups three through six replaced by Keepin et al. DNP group parameters, aligns much more closely with literature. This again indicates the importance of pulse irradiation data to capture short-lived DNP behavior.

Table 4.12: Average half-life from thermal fission of ^{235}U .

Approach	Data	Average half-life [s]
Microscopic	ENDF/B-VI.8 [153]	7.66
Microscopic	Mills et al. [119]	8.63 ± 0.52
Microscopic	Brady and England [22]	7.90 ± 0.34
Microscopic	0D Scaled Model Data	6.39 ± 0.38
Microscopic	0D Scaled Model	6.39 ± 0.34
Hybrid	Modified 0D Scaled Model	8.74 ± 0.46
Macroscopic	Keepin et al. [89]	9.03 ± 0.63
Macroscopic	Charlton et al. [35]	8.92 ± 0.28
Macroscopic	Saleh et al. [160]	9.49 ± 0.90
Macroscopic	Synetos and Williams [180]	8.96 ± 0.56
Macroscopic	Tuttle [189]	8.84 ± 0.45
Macroscopic	Waldo et al.[194]	8.72 ± 0.73

Chapter 5

Proposed Work

Completion of the proposed work will result in a complete analysis, from start to finish, of improved DNP group parameters for thermal fission of ^{235}U and the effects of these improved parameters on transient simulations. MoSDeN, which will be completed with this work, will serve as a tool to investigate the DNP group parameters with MSR physical phenomena included. The proposed work is not exhaustive, and will not investigate every fissile nuclide, nor every irradiation energy. Work which is deemed interesting yet not critical is included in Section 5.4.

The critical work includes developing the 0D flow model, validating and verifying the results of the 0D flow model, generating improved group spectra along with the DNP group parameters, and finally using the DNP group parameters in transient simulations. The non-critical work includes development of the spatially resolved model, uncertainty tracking of MSR physical phenomena in the DNP group parameters, and investigation into the effects of varying the incident neutron energy spectrum on the DNP group parameters.

Figure 5.1 shows the critical work path, divided into development, simulation, writing, and review categories. The categories which require more focus are development and writing, whereas simulation and review are generally a more passive process. Most tasks take one month, but three tasks are assumed to take two months: development of the MSRE Moltres model, running the various MSRE transients, and writing the results and conclusions chapters. Development of the Moltres model is expected to take longer because there may be unexpected issues with the meshing of the upper and lower plena, data collection, or model development itself. The MSRE transient simulations are given two months of simulation time because there are 12 transient simulations for safety, security, and safeguards simulations. These transient simulations also require five transport solves: three for standard MSRE operation prior to a transient with higher and lower temperatures, one with one SQ removal of ^{235}U over one year, and one with one SQ removal of plutonium over one year. The diversion scenarios both require a depletion simulation, which also slightly increases the cost of the simulations. The results chapter is given two months because there will be many new results to discuss, whereas the other chapters will not have significant changes.

5.1 0D Flow Model

The new model that will be developed beyond the 0D scaled model is the 0D flow model. The 0D scaled model generates saturation irradiation results and scales the flux and chemical removal based on the fraction of salt in-core and ex-core, but it does not track decays of nuclides into DNPs during the decay of the sample after irradiation. However, the 0D scaled model offers a fast simulation, which does enable rapid development and simulation of many different parameters with ease.

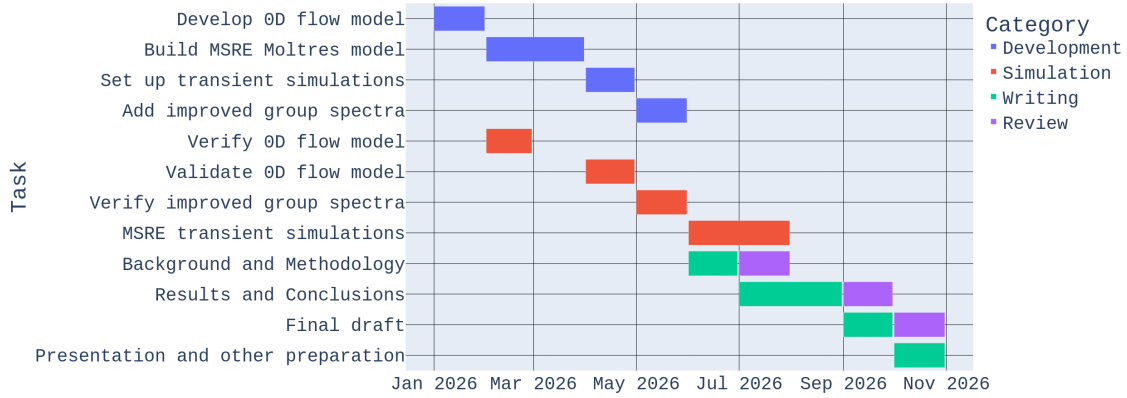


Figure 5.1: Gantt chart of proposed work and the time frame for each part of that work.

The 0D flow model improves the fidelity of the 0D scaled model by offering both saturation and pulse irradiation results, handles in-core and ex-core regions in time rather than simply scaling the flux, and enables tracking of decay parents for DNPs. There is also the option of using the 0D scaled model with the scaled flux approximation but with the decay chain modeled. The cost of this improved fidelity is an increased computational cost, but it is still not exceedingly expensive and can be run on a desktop. The neutron transport solve is relatively inexpensive, as the model only requires a fixed-source irradiation of the sample.

5.1.1 Least Squares Residual Treatment

A key aspect of the methodology is combining pulse and saturation irradiation DNP group parameters. One approach, used by Brady and Keepin et al., is combining two different fits after they have been generated. Brady uses the group 5 and 6 parameters from the pulse irradiation with the remaining parameters from the saturation irradiation to form the six DNP group parameters [24]. Keepin et al. fit four groups with each irradiation, and then renormalize at the second group [89].

Another approach is the Levenberg-Marquardt method, which combines pulse and saturation irradiation data [103, 112, 118]. This method of fitting DNP group parameters to both datasets simultaneously may yield more accurate results. This can also be accomplished using the trust region reflective (TRF) approach, which allows for bounds to be included for the parameters [25].

The proposed methods to compare are the current method, the Levenberg-Marquardt method, and the TRF method. The new methods minimize the residual of the form:

$$r = \min_x \left\| \frac{A_p x - b_p}{b_p + \epsilon} + \frac{A_s x - b_s}{b_s + \epsilon} \right\|_2^2, \quad (5.1)$$

where

- r = the residual of the problem [-]
- A_p = the pulse irradiation coefficient matrix [s^{-1}]
- b_p = the pulse irradiation delayed neutron count rate [$n \cdot s^{-1}$]
- A_s = the saturation irradiation coefficient matrix [s^{-1}]
- b_s = the saturation irradiation delayed neutron count rate [$n \cdot s^{-1}$]
- x = the solution vector containing group yields [-]
- ϵ = a small value to avoid division by zero
 $= 1 \times 10^{-12}$.

This formulation simultaneously treats the pulse and saturation irradiation data, generating improved DNP group parameters which fit both sets of delayed neutron count rates.

The results of this analysis will be used to evaluate which method can provide a solution with the least computational cost while also generating accurate group parameters. The group parameters will be evaluated by calculating the l^2 -norm of the residual for the least squares problem using the improved DNP group parameters over all times for both pulse and saturation irradiations.

5.1.2 Data Source Analysis

Huynh et al. compare the effects of cumulative fission yields and emission probabilities from Joint Evaluated Fission and Fusion (JEFF), ENDF, and Japanese Evaluated Nuclear Data Library (JENDL) on the total delayed neutron yield [81]. A similar comparison can also be found in the JEFF report 17 [123]. Mathieu et al. investigate the National Nuclear Data Center (NNDC) website, which sources data from Evaluated Nuclear Structure Data File (ENSDF), and various data compilations [113, 137, 8, 155].

These comparisons can be expanded to include the IAEA database, NNDC website, JEFF, ENDF, JENDL, ENSDF, and other data compilations for emission probabilities (delayed neutron branching ratios), half-lives, fission yields, and cross sections. Comparing these data sources clarifies the uncertainty caused by differences in the data. Table 5.1 shows the data sources used in the comparison. The "Concentration" column refers to data including fission yields and cross-sections that affect the DNP concentration term. Although half-lives also influence concentration, data sources without a concentration contribution provide only half-life and emission probability data. Audi et al., Pfeiffer and Kratz, and Rudstam et al. present data from the literature rather than official databases. Audi et al. use ENSDF, Atomic Mass Evaluation (AME), experimental data from the literature, and trends in neighboring nuclides. Pfeiffer and Kratz present experimental data, an update to the Kratz Herrmann formula (KHF), and data from a unified macroscopic-microscopic model within the quasi-particle random-phase approximation (QRPA). Rudstam et al. present experimental data.

Table 5.1: Various data sources as well as the data which they contain.

Data name	Year	Source	Half-life	Emission probability	Concentration
IAEA database	2018	[50]	Yes	Yes	No
JEFF 3.3	2017	[140]	Yes	Yes	Yes
ENDF/B-VIII.0	2018	[28]	Yes	Yes	Yes
JENDL-5	2021	[84]	Yes	Yes	Yes
ENSDF	2025	[55]	Yes	Yes	No
Audi et al.	2003	[8]	Yes	Yes	No
Pfeiffer and Kratz	2002	[137]	Yes	Yes	No
Rudstam et al.	1993	[155]	Yes	Yes	No

The expected outcome is that recently updated data and official databases will be more accurate than collections from the literature. Recent data should be more accurate due to improved equipment and measurement techniques. Official databases are expected to be more reliable, as they have many users reporting errors and have undergone extensive development over time. Investigating these data sources and comparing the DNP group parameters generated from each will be useful when analyzing uncertainty. If the differences are large, understanding the uncertainty propagation will be important. If the differences are negligible, this will simplify further uncertainty analysis.

5.1.3 Verification and Validation

Verifying and validating the DNP group parameters is essential to ensure their practical utility. Verification with the literature is important to ensure that the fundamental approach is working as expected. Since the literature does not contain DNP group parameters that include MSR physical phenomena, this verification with the literature will only be for the stationary solve.

After this, although the group parameters are verified with the literature, that does not necessarily mean they capture all physical phenomena of interest. Validation identifies potential weaknesses in the approach by highlighting discrepancies between predicted and experimental results.

Verification of the DNP group parameters will be of the same form as in Section 4.3. It has already been shown that these results should have a percent difference of 9% for thermal fission in ^{235}U for JEFF-3.1.1 data, so a similar percent difference can be expected for other datasets [113]. Results will vary based on the source of the data, but the DNP group parameters should closely match the results from the literature.

Validation of the DNP group parameters is conducted through the MSRE pump start-up and coast-down experimental data. This data assesses the accuracy of the group parameters in modeling reactivity changes from DNPs [142]. The validation requires modeling the MSRE core and upper and lower plena in Moltres [61]. The reactivity effects of the DNPs are calculated for both transients using static and improved DNP group parameters. The improved group parameters are expected to predict reactivity more accurately as they account for more physical phenomena in the MSRE. If neither set of reactivity responses aligns with the experimental results, then there is an issue in the kinetics simulation, as Moltres has previously demonstrated the ability to model these transients [144, 131]. These transients have also been evaluated using other tools, providing additional comparison [57, 47]. If the static group parameters predict reactivity more accurately than the improved group parameters, this suggests an issue with the improved group parameter formulation. The issue must be resolved before proceeding, as the improved DNP group parameters should provide higher accuracy in MSR modeling.

5.2 Improved Group Spectra

The delayed neutron energy spectra, χ_d is important for accurate prediction of reactor kinetics [44]. This can be seen in the equation for the effective delayed neutron fraction in Equation (2.34), recreated here as:

$$\beta_{eff} = \frac{\int \Phi^\dagger \chi_d \beta \bar{v} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}{\int \Phi^\dagger \chi \bar{v} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}.$$

Because the DNP group yields and half-lives are affected by the MSR physical phenomena, it follows that the DNP group spectra are also affected.

The methodology to evaluate the delayed neutron spectra is discussed in Section 2.1.2. The DNP group spectra will be generated using the half-life grouping and fractional grouping methods. The DNP group spectra will then be compared with the spectra generated from a stationary problem with no chemical removal to determine the effect of MSR phenomena on the spectra.

5.3 Transient Simulations

Transient simulations are critical for validating the DNP group parameters using MSRE transients and for investigating transients affecting security, safety, and safeguards design decisions. Section 3.5 discusses the proposed simulations in detail.

For safety design, the ULOF, UTOP, ULOHS, and FSOC are investigated for various reactor parameters to assess the impact of the improved DNP group parameters. For security design, sabotage-initiated transients are assumed to build off those studied for safety design. For safeguards design, comparisons are made between diversion scenarios. The diversion scenarios alter the fuel composition, affecting the effective delayed neutron fraction and reactor behavior during transients [195]. The improved DNP group parameters will be used to investigate the SNM removal which could remain undetected while using previous DNP group parameters.

The kinetics simulations will use Moltres and the MSRE to demonstrate how the DNP group parameters affect reactivity and temperature responses in various scenarios. These results can be compared with other analyses from the literature [75, 175]. The results will identify which transients are more sensitive to the improved data and highlight potential concerns. A PRK model can also be used for simple reactivity insertion tests with uncertainty tracking to more straightforwardly demonstrate differences between the DNP group parameters.

5.4 Future Work

This section contains various work that would be useful and interesting extensions of the proposed work. The three main categories of potential future work include development of a spatially resolved model for improved DNP group parameters, tracking of additional contributors to uncertainty, and sensitivity of DNP group parameters to varying incident neutron energy spectra. Additional miscellaneous work of interest could also include:

- Using a sinusoidal flux profile instead of a Heaviside function for the derivation given in Appendix A.
- Direct comparison of exponential stripping, least squares, and set half-life DNP group parameter fitting methods.
- More intermediate models between the 0D scaled and 0D flow models

- 0D scaled with explicit calculation of concentration rather than scaled flux (based on Appendix A.
- 0D scaled with cross-sections included.
- 0D scaled with simplified decay chains included.
- Investigate if there may be a way to structure DNP group parameters such that fuel salt movement is accounted for in the delayed neutron count rate (capturing circulation in the parameters themselves).
- Inclusion of the effects of photofission.
- Inclusion of the effects of photoneutrons.

5.4.1 Spatially Resolved Model

Although the 0D flow model provides useful insights, it does not provide a high level of fidelity for any specific geometry. A spatially resolved model can be developed to evaluate the accuracy of the 0D flow model. This model would verify whether the 0D flow model produces sufficiently accurate DNP group parameters. Section 3.2.3 discusses the spatially resolved model in detail. This spatially resolved model can compare multiple levels of fidelity for generating DNP group parameters: 0D scaled model using cumulative fission yields, 0D flow model with depletion, spatially resolved flow model using cumulative fission yields, and spatially resolved flow model with depletion. Each step increases computational cost, so if a simpler method generates DNP group parameters with negligible differences, it will be used preferentially.

The comparison between the 0D flow model and the spatially resolved model can be conducted as follows: the spatially resolved model is first executed to generate DNP group parameters and residence times; the residence times are supplied to the 0D flow model; and the resulting DNP group parameters are compared between the two models. If the results are within the uncertainty bounds of each other, then this indicates that the 0D flow model only needs residence time data for a given MSR to generate the DNP group parameters. If they do not match, then a spatially resolved approach may be necessary to properly generate the MSR DNP group parameters.

Sensitivity to Number of DNPs

The number of DNPs to track is of particular interest for the spatially resolved flow model. The 0D flow model's computational cost increases negligibly with more DNPs, while the spatially resolved flow model requires significantly more memory and operations to track each DNP over space and time. To improve computational efficiency, minimizing the number of DNPs is desirable. If the 0D flow model generates DNP group parameters with negligible differences compared to the spatially resolved flow model, minimizing the number of DNPs becomes less critical. However, it is of interest to optimize the spatially resolved flow model for potential future analyses which could include spatially resolved reprocessing and other effects.

The impact of each DNP is given in Equation (2.29), replicated here:

$$\bar{v}_{d,i} = P_{n,i} \lambda_i N_{i,0}.$$

Dividing $\bar{v}_{d,i}$ by the total delayed neutron yield, $\bar{v}_d$, provides the relative contribution of each DNP to the total delayed neutron yield. The DNPs with the most impact can be modeled preferentially to determine the minimum number of DNPs to model with negligible changes to the DNP group parameters. A full sensitivity study can be conducted to determine a reasonable number of DNPs to include in the spatially resolved flow model.

Reactor Generalization

If the DNP group parameters can be applied generally to different MSRs with similar residence times, then the amount of computational work required is drastically reduced. However, it is necessary that sufficient evidence is provided demonstrating that the improved parameters are identical for different MSRs.

Specifically, the MSRE and the MSFR can be simulated using the spatially resolved model [149, 27]. The models can be adjusted such that the in-core and ex-core residence times are identical for each. The results would be a set of group parameters for two distinct reactors but with the same residence times. Instead of calculating the neutron flux, both reactors will be given an incident neutron energy spectrum for irradiating the sample. The impact of the incident neutron energy spectra is discussed in Section 5.4.3, so differences caused by the spectra will not be studied in this section. Additionally, the fuel salt and fuel temperature in the MSFR will be replaced with those from the MSRE.

The two different reactor simulations would have all of the same input parameters aside from the geometry and thermal-hydraulic parameters. If the DNP group parameters have negligible differences, then the geometry and thermal-hydraulic parameters are not necessary when generating the group parameters. Instead, all that is needed are the other input parameters, meaning the work does not need to be replicated for every different reactor geometry and thermal-hydraulic configuration. If the DNP group parameters have non-negligible differences, then each MSR needs to have DNP group parameters generated for its own specific geometry and flow conditions. Further investigation may also be warranted to determine if there is potential for a reduced order model in which one or two parameters which could be used to capture these differences, thus reducing the total amount of work necessary to model all MSR designs.

5.4.2 MSR Uncertainties

Several sources of uncertainty are considered, including half-lives, emission probabilities, cross-sections, and fission yields. These uncertainties propagate into the DNP group and kinetics parameters [143]. However, uncertainties also arise from MSR physical phenomena, including flow rate and reprocessing rates, where the reprocessing rate is correlated with the flow rate. These uncertainties are important because they propagate to the DNP group parameters, which affect the reactor's response to different transients. It is important to understand this for safety, security, and safeguards design, as the design should account for these uncertainties. In particular, uncertainties of interest are in the residence times and chemical removal rates.

The residence time correlates linearly with the flow rate, which is dependent on the temperature. Although the flow rate would also have uncertainty based on variance in the pump, this uncertainty may be beyond the scope of future work in MoSDeN. The temperature affects fuel salt density, buoyant force, and viscosity, all influencing the flow rate and residence time. The temperature is influenced by the fission rate, which has uncertainties from the flux and fission cross section. For the 0D flow model, the residence time uncertainties can be conservatively approximated by treating the difference between the inlet and outlet temperatures as the temperature uncertainty. Assuming a constant mass flow rate, the density change due to the large temperature uncertainty provides a velocity uncertainty, which determines the variance in the residence time. If this conservative approach to temperature uncertainty results in negligible residence time uncertainty, it can be neglected for the spatially resolved flow model. Otherwise, the spatially resolved flow model can account for this uncertainty using the STOCHASTIC TOOLS module to directly determine residence time uncertainties. The STOCHASTIC TOOLS module is a toolbox designed for performing stochastic analysis for MOOSE-based applications. The STOCHASTIC TOOLS module enables setting a range of temperatures and simulating each temperature, which can be used to generate a histogram of residence

times as a function of core temperature.

The chemical removal uncertainty is challenging to treat, as there are different chemical processes for removing elements, such as sparging the fuel salt in the core, filtering elements with a nickel filter, and removing elements via liquid-liquid reductive extraction [158]. The removal efficiency of xenon in a gas separation system is given by [133, 158] as:

$$\epsilon = \frac{1 - e^{-k_L a A_C L \left(\frac{1}{Q_{salt}} + \frac{RT}{H} \frac{1}{Q_{He}} \right)}}{1 + \frac{RT}{H} \frac{Q_{salt}}{Q_{He}}}, \quad (5.2)$$

where

R = the universal gas constant [$L \cdot Pa \cdot mol^{-1} \cdot K^{-1}$]

T = the salt temperature [K]

H = Henry's law constant for solute gas [$Pa \cdot mol^{-1} \cdot L$]

Q_{salt} = the volumetric salt flow rate [$m^3 \cdot s^{-1}$]

Q_{He} = the volumetric helium flow rate [$m^3 \cdot s^{-1}$]

k_L = the liquid phase mass transfer coefficient [$m \cdot s^{-1}$]

a = the gas-liquid interfacial area per unit volume [m^{-1}]

A_C = the contactor cross-section [m^2]

L = the contractor length [m].

The liquid-liquid reductive extraction, such as with bismuth, has distribution coefficients and equilibrium constants dependent on the element, temperature, and concentration [120]. In general, an increase in the temperature leads to an increased distribution coefficient and a decreased equilibrium constant. The distribution coefficient represents the ratio of the bismuth phase concentration to the salt phase concentration, and can be thought of as an efficiency term that is multiplied with the mass flow rate to determine the removal rate. However, the distribution coefficient changes with temperature and flow rate differently for each element and fuel salt used.

Propagating the uncertainty in the chemical removal of elements from the fuel salt is challenging, as there are many variables to consider. These include the fuel salt composition, flow rate, redox potential, hydrostatic pressure, interfacial tension, and temperature [169, 186]. Thus, generic equations are not formulated for each possible combination of variables. Instead, only the chemical removal of noble gases and noble metals from the MSRE are investigated [107]. The noble gases are actively pumped out via the off-gas system while noble metals, including molybdenum, technetium, ruthenium, rhodium, palladium, silver, and antimony plate out via reduction by UF_6 . Some elements, specifically selenium, niobium, and tellurium, vary in their behavior. These elements only sometimes plates out based on the redox potential of the salt; a well reduced salt causes them to plate out, otherwise they act as salt seekers [87, 107].

The removal rate, λ_r , of noble gases from the pump bowl via the off-gas system is [85, 107]:

$$\lambda_r = \frac{\dot{V}}{V} \epsilon, \quad (5.3)$$

where

$\dot{V}$ = the salt flow rate through the gas-liquid separator loop [$m^3 \cdot s^{-1}$]

V = the total volume of salt in the primary loop [m^3]

ϵ = the removal efficiency [-].

For the MSRE, the values are a salt flow rate of 0.003153, a total salt volume of 1.996, and a removal efficiency of 0.026 [149]. For the 0D flow model, this removal occurs while the sample is in the ex-core region and stops while the sample is in the in-core region. This is because the removal occurs in the pump bowl and not inside the core. For the spatially resolved flow model, this removal also occurs in the ex-core region.

The removal rate, $\lambda_{i,l,r}$, of noble metal i to waste stream l via plate-out is [107]:

$$\lambda_{i,l,r} = \frac{h_{i,l} A_l}{V}, \quad (5.4)$$

where

$h_{i,l}$ = the mass transfer rate of element i to waste stream l [$m \cdot s^{-1}$]

A_l = the surface area of waste stream l [m^2].

The removal of noble metals does not occur in a single region, unlike the noble gas removal. Instead, there is a distribution of noble metals: 50% are in the hastalloy-N surfaces in the loop, 49% are in the ex-core region, and 1% are in the graphite surfaces in the in-core region [87]. Because 99% of the plate-out occurs in the ex-core region, the removal can be treated as occurring only in the ex-core region. The surface areas and mass transfer rates for the noble metals are given in Table 5.2, recreated from [107].

Table 5.2: Noble metal mass transfer rates, surface areas, and resulting removal rates (recreated from [107] with data from [87]).

Component	$h_{i,l}$ [$m \cdot s^{-1}$]	A_l [m^2]	$\lambda_{i,l,r}$ [s^{-1}]
Heat Exchanger	4.657×10^{-5}	2.93×10^1	6.827×10^{-4}
Fuel loop piping, core and pump volute	1.041×10^{-4}	6.60×10^0	3.441×10^{-4}
Core wall cooling annulus	4.318×10^{-5}	1.43×10^1	3.095×10^{-4}
Core graphite (exposed to salt)	5.334×10^{-6}	1.36×10^2	3.637×10^{-4}
Miscellaneous	-	-	1.695×10^{-4}
Bubbles	4.233×10^{-4}	3.205×10^1	6.7978×10^{-3}
Total	-	-	8.6674×10^{-3}

5.4.3 Incident Neutron Energy Spectra

The incident neutron energy spectrum directly affects the DNP group parameters [22]. However, it would be computationally expensive to generate new DNP group parameters for every unique incident neutron energy spectrum. Some coarser resolution is required, so it is of interest to determine a reasonable categorization of the DNP group parameters for different energy spectra.

This sensitivity study could investigate different levels of fidelity for organizing the energy spectra: fine, middling, coarse, and single-value spectra. The approach can use the 0D flow model, with changes only to the neutron source

energy spectra. The fine, middling, and coarse spectra include the full incident neutron energy spectra but with different bin sizes. The single-value spectra include several categories: neutron energy classification, mean energy, and median energy. The neutron energy classification is provided in Table 5.3, recreated from [110]. The study will also examine hand-crafted spectra where the mean and median remain unchanged, but the spectrum itself varies.

Table 5.3: Classification of neutron energy categories by energy range from [110].

Category	Energy Range [MeV]
Cold	$< 3 \times 10^{-9}$
Thermal	$3 \times 10^{-9} - 4 \times 10^{-7}$
Epithermal	$4 \times 10^{-7} - 1 \times 10^{-4}$
Intermediate	$1 \times 10^{-4} - 2 \times 10^{-1}$
Fast	$2 \times 10^{-1} - 1 \times 10^1$
Relativistic	$> 1 \times 10^1$

This sensitivity study can determine the necessary level of detail in the incident neutron energy spectrum for generating DNP group parameters. Organizing by category or mean energy is expected to suffice for most cases. For cases where this approach is inadequate, a more robust method can be selectively applied.

Appendix A

Derivation of the Delayed Neutron Count Rate in the 0D Flow Model

This derivation begins with Equations (AA.1) and (AA.2) provided by Wilson and England [197]:

$$R(t) = \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}, \quad (\text{AA.1})$$

and

$$\dot{n}_d(T, t) = \bar{v}_d \int_0^T F(\tau) R(T - \tau + t) d\tau, \quad (\text{AA.2})$$

where

T = the length of the fission history [s]

t = the decay time from the end of the fission history [s]

$R(t)$ = the probability of delayed neutron emission at time t [$n \cdot s^{-1}$]

$F(\tau)$ = the fission rate history [$\frac{\text{fissions}}{s}$].

Because the results are normalized to a single delayed neutron, the $\bar{v}_d$ term is divided out.

Although the saturation, or infinite, irradiation derivation makes use of assuming the time is infinite, due to the spatial dependence of this problem, the in-core and ex-core timing must be handled properly. For example, if the fission rate history uses a Fourier decomposition of a square wave, then it would not be possible to extract spatial information from this formulation. Specifically, it is important where in the primary loop the sample initially exists. If the sample is initially in the core inlet, then the final portion of the fission rate history would consist purely of decay. If the sample is initially in the center of the core, then the final portion of the fission rate history would consist of irradiation.

For this derivation, the sample is assumed to exist as a point at the core inlet. This means the full cycle consists of an in-core residence time of fission followed by an ex-core residence time of decay, after which the cycle repeats

until the net irradiation time is met. The fission rate history is therefore given as:

$$\begin{aligned} \frac{F(\tau)}{F_0} = & H(\tau) - H(\tau - T) + \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - j \cdot \tau_{ex}) \\ & - H(\tau - j \cdot \tau_{in} - (j-1) \cdot \tau_{ex}), \end{aligned} \quad (\text{AA.3})$$

where

H = the Heaviside function [-]

$$J = \left\lceil \frac{T}{\tau_{in} + \tau_{ex}} \right\rceil$$

= the minimum number of cycles to meet the irradiation time, T [-]

F_0 = the magnitude of the constant fission rate [$\frac{\text{fissions}}{s}$].

Plugging this into Equation (AA.2) for group k yields:

$$\begin{aligned} \dot{n}_{d,k}(T, t) = & \int_0^T H(\tau) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau - \int_0^T H(\tau - T) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau \\ & + \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - j \cdot \tau_{ex}) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau \\ & - \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - (j-1) \cdot \tau_{ex}) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau, \end{aligned} \quad (\text{AA.4})$$

which can be simplified to:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k} \lambda_k} = & \int_0^T H(\tau) e^{-\lambda_k(T-\tau+t)} d\tau - \int_0^T H(\tau - T) e^{-\lambda_k(T-\tau+t)} d\tau \\ & + \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - j \cdot \tau_{ex}) e^{-\lambda_k(T-\tau+t)} d\tau \\ & - \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - (j-1) \cdot \tau_{ex}) e^{-\lambda_k(T-\tau+t)} d\tau \end{aligned} \quad (\text{AA.5})$$

Solving the integrals yields:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k} \lambda_k} = & \frac{1}{\lambda_k} \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T)} \right) + 0 \\ & + \sum_{j=1}^J \frac{H(T - (j \cdot \tau_{in} + j \cdot \tau_{ex}))}{\lambda_k} \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-j \cdot \tau_{ex})} \right) \\ & - \sum_{j=1}^J \frac{H(T - (j \cdot \tau_{in} + (j-1) \cdot \tau_{ex}))}{\lambda_k} \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-(j-1) \cdot \tau_{ex})} \right). \end{aligned} \quad (\text{AA.6})$$

The Heaviside functions in the summations resolve to unity while the cycle is within range of the net irradiation

time, and become 0 when the cycle time goes beyond the net irradiation time. The problem then simplifies to:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k}} = & \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T)} \right) \\ & + \sum_{j=1}^J \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-j \cdot \tau_{ex})} \right) \\ & - \sum_{j=1}^J \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-(j-1) \cdot \tau_{ex})} \right). \end{aligned} \quad (\text{AA.7})$$

If the net irradiation time is assumed to be infinite, then the delayed neutron count rate becomes:

$$\dot{n}_{d,k}(T, t) = F_0 \bar{v}_{d,k} e^{-\lambda_k t}, \quad (\text{AA.8})$$

which matches the result from Wilson and England [197]. However, as previously mentioned, this infinite approach will not work for the spatially dependent analysis. For example, if an in-core residence time is selected which is relatively small compared to the ex-core residence time, Equation (AA.8) would not vary at all, even though the fission rate history would be significantly different. Instead, the problem takes on the form:

$$\frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k} e^{-\lambda_k t}} = 1 - e^{-\lambda_k T} + \left(1 - e^{\lambda_k \tau_{ex}} \right) \sum_{j=1}^J e^{-\lambda_k(T-j \cdot \tau_{in}-(j-1) \cdot \tau_{ex})}. \quad (\text{AA.9})$$

By isolating the delayed neutron count rate and summing over all groups, the finalized equation can be formulated as:

$$\dot{n}_d(T, t) = F_0 \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} - \bar{v}_{d,k} e^{-\lambda_k t} e^{-\lambda_k T} \quad (\text{AA.10})$$

$$+ \bar{v}_{d,k} e^{-\lambda_k t} \left(1 - e^{\lambda_k \tau_{ex}} \right) \sum_{j=1}^J e^{-\lambda_k(j \cdot \tau_{in}+(j-1) \cdot \tau_{ex}-T)}, \quad (\text{AA.11})$$

which can be rewritten as:

$$\dot{n}_d(T, t) = F_0 \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} \left(1 - e^{-\lambda_k T} + \left(1 - e^{\lambda_k \tau_{ex}} \right) \sum_{j=1}^J e^{-\lambda_k(T-j \cdot \tau_{in}-(j-1) \cdot \tau_{ex})} \right). \quad (\text{AA.12})$$

If the in-core residence time is zero, then the delayed neutron count rate goes to zero, which is the expected behavior.

Appendix B

Derivation of the Equilibrium Concentration from the Cumulative Fission Yield

This derivation begins with the Bateman equation, given in Equation (2.57) and recreated here as:

$$\frac{dN_j}{dt} = \sum_{i \neq j} [(\gamma_{i \rightarrow j} \sigma_{f,i} \phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \phi) N_i] - (\lambda_j + \sigma_j \phi + \lambda_{r,j}) N_j + \dot{N}_{f,j},$$

where

- N_j = the atomic density of isotope j $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- $\gamma_{i \rightarrow j}$ = the fractional fission product yield of j from the fission of isotope i
- $\sigma_{f,i}$ = the microscopic fission cross-section of isotope i $[\text{cm}^2]$
- ϕ = the spectrum-averaged scalar flux in the fuel region $\left[\frac{\text{neutrons}}{\text{cm}^2 \text{s}} \right]$
- $\lambda_{i \rightarrow j}$ = the decay constant of decay $i \rightarrow j$ $[\text{s}^{-1}]$
- $\sigma_{i \rightarrow j}$ = the microscopic transmutation cross-section of reaction $i \rightarrow j$ $[\text{cm}^2]$
- N_i = the atomic density of isotope i $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- λ_j = the decay constant of isotope j $[\text{s}^{-1}]$
- $\lambda_{r,j}$ = the removal constant for removal of isotope j $[\text{s}^{-1}]$
- σ_j = the microscopic total transmutation cross-section of isotope j $[\text{cm}^2]$
- $\dot{N}_{f,j}$ = the feed rate for isotope j $\left[\frac{\text{atoms}}{\text{cm}^3 \cdot \text{s}} \right]$.

This can be rewritten to the simplified form given in Equation (2.61), recreated here as:

$$\frac{dN_j}{dt} = S_j(t) - \lambda_{net,j} N_j(t),$$

where

$$\begin{aligned}
 S_j &= \text{the source term for nuclide } j, \\
 &\quad \text{containing fission, decay, and transmutation sources } \left[\frac{\text{atoms}}{\text{cm}^3 \text{s}} \right] \\
 \lambda_{net,j} &= \text{the loss term for nuclide } j, \\
 &\quad \text{containing decay and transmutation losses } [\text{s}^{-1}].
 \end{aligned}$$

This can be rewritten using the scaled flux method, which is useful for flowing fuel reactors as it scales the flux term by the fraction of fuel salt in the core. This changes Equation (2.60) to Equation (BB.1), assuming chemical removal only in the ex-core region, given as:

$$\frac{dN_j}{dt} = f_{in}S_j(t) - \lambda_j N_j(t) - f_{ex}\lambda_{r,j}N_j(t), \quad (\text{BB.1})$$

where

$$\begin{aligned}
 f_{in} &= \text{the fraction of fuel salt in the core} \\
 f_{ex} &= \text{the fraction of fuel salt not in the core} \\
 \lambda_{r,j} &= \text{the chemical removal rate of } j.
 \end{aligned}$$

The source term, S_j , can then be written using the cumulative fission yield as shown in Equation (BB.2). The cumulative fission yield is the total number of some nuclide formed per fission event, including decay chains that lead to that nuclide. Because the cumulative fission yield captures decays into the nuclide j , the $\lambda_{i \rightarrow j}$ term is not included. Additionally, parasitic absorption is neglected in this derivation, which leads to:

$$S_j(t) = CFY_j \phi \Sigma_f, \quad (\text{BB.2})$$

where

$$CFY_i = \text{the cumulative fission yield of nuclide } i.$$

Finally, the fission reaction rate term, $\phi \Sigma_f$, is assumed to be equal to 1. Equation (BB.3) can be used to calculate the concentration as a function of time using cumulative fission yields, given as:

$$\frac{dN_j}{dt} = f_{in}CFY_j - \lambda_j N_j(t) - f_{ex}\lambda_{r,j}N_j(t) \quad (\text{BB.3})$$

Assuming a stationary fuel with no recirculation or chemical removal, the equation simplifies greatly. The solution to that equation, assuming no initial concentration of nuclide j , is given as:

$$N_j(t) = \frac{CFY_j}{\lambda_j} (1 - e^{-\lambda_j t}) \quad (\text{BB.4})$$

As previously mentioned, the cumulative fission yield gives the total amount of the nuclide generated including from decay chains, so the time dependent result is non-physical. For a saturation irradiation however, the sample is irradiated for a sufficiently long time such that all DNPs reach equilibrium. This means that the equation should be

taken as time approaches infinity, which leads to the form [81]:

$$N_j(t) = \frac{CFY_j}{\lambda_j}. \quad (\text{BB.5})$$

For the case of a nuclide in a circulating fuel reactor with a non-zero chemical removal rate, the concentration as a function of time comes out as:

$$N_j(t) = \frac{f_{in}CFY_j}{f_{ex}\lambda_{r,j} + \lambda_j} \left(1 - e^{-f_{ex}\lambda_{r,j}t - \lambda_j t} \right) \quad (\text{BB.6})$$

Equation (BB.6) can also be taken as time goes to infinity, resulting in:

$$N_j(t_\infty) = \frac{f_{in}CFY_j}{f_{ex}\lambda_{r,j} + \lambda_j} \quad (\text{BB.7})$$

Appendix C

Discussion of Turbulent Modeling in Moltres

The proposed spatially resolved model will use Moltres to simulate circulation of a fissile sample in molten salt. The spatially resolved model requires turbulent modeling to capture vortices and other thermal-hydraulic effects that influence the residence time of fissile material in different regions. In contrast, the 0D scaled and 0D flow models approximate these effects by varying residence times and applying sensitivity analyses to assess the impact of prolonged residence in the ex-core or in-core regions on the DNP group parameters.

The proposed transient simulations using Moltres could also include the turbulence effects. Turbulence would not affect the DNP group parameters, but it would affect the concentration distribution for each group. Since only the MSRE is studied in these transients, turbulent modeling may not be necessary to capture the effects of interest. The in-core region of the MSRE consists of flow through channels, which would not have large residence time differences from turbulence. Turbulence would cause mixing in the upper and lower plena, so the Moltres simulation without turbulence will assume perfect mixing in those regions.

In the proposed transient simulations, neglecting turbulence may affect residence time in the upper and lower plena, slightly increasing the ex-core residence time relative to the in-core residence time. Because Moltres can model turbulence using the Spalart-Allmaras turbulence model, it is straightforward to include in the transient simulations, though it does introduce more variables [176, 131]. The purpose of the transient simulations is to demonstrate the difference between simulations using improved DNP group parameters compared to existing group parameters. Modeling turbulence would not significantly change this difference, but would improve the simulation accuracy.

For these reasons, modeling turbulence in the Moltres transient simulations will be treated as an optional extension. If sufficient time is available, turbulent modeling will be included. If a small amount of additional time is available, a sensitivity analysis of the difference between transients with and without turbulence will be conducted. If insufficient time is available, this analysis will be moved to the future work section.

Appendix D

An Analysis of Pulse and Saturation Data

The effects of using pulse or saturation irradiation data can be demonstrated through a simple example problem. Instead of using several hundred DNPs, only four will be considered: A , B , C , and D . Each of these DNPs will have an emission probability of unity and will only be formed via fission, having a cumulative fission yield and independent fission yield of unity as well. The only difference between these DNPs will be in their half-lives, given as: $\lambda_A = 0.001$, $\lambda_B = 0.01$, $\lambda_C = 0.1$, and $\lambda_D = 1 \text{ s}^{-1}$.

The concentration of each nuclide can be calculated using Equation (3.6), recreated here:

$$N_i = \frac{CFY_i}{\lambda_i}.$$

This gives concentrations of $N_A = 1000$, $N_B = 100$, $N_C = 10$, and $N_D = 1$ atoms.

Because all of the DNPs come directly from fission, the independent fission yield can be used to calculate the concentration after a single normalized fission event. This gives concentrations of $N_A = 0.1$, $N_B = 0.1$, $N_C = 0.1$, and $N_D = 0.1$ atoms.

Next, to complete the comparison, these four DNPs can be fit into two DNP groups to determine what parameters best fit the delayed neutron count rates produced. The total delayed neutron yield should be four because four delayed neutrons are released per fission event, and the average half-life is calculated to be approximately 192.5 seconds. Because the yield is so large for this example problem, the bounds of the least squares solve are increased to accommodate for larger yields only for this problem. The results comparing the data of these four DNPs to the different group fits are given in Table D.1.

Table D.1: A comparison of the raw DNP data to the different irradiation methods and the corresponding yields and average half-lives.

Method	Delayed Yield [-]	Average Half-life [s]
Data	4.00	192.5
4 Group Saturation	4.00	192.5
3 Group Saturation	3.97	188.2
2 Group Saturation	2.55	261.0
1 Group Saturation	1.52	361.8
4 Group Pulse	4.00	192.5
3 Group Pulse	4.06	78.0
2 Group Pulse	3.44	57.34
1 Group Pulse	1.30	148.4

Because each DNP is equally important, the 3 group structure captures 75% of the total delayed neutrons, while the 2 group structure is only able to capture 50% of the total delayed neutrons. The delayed neutron results show that both the pulse and saturation irradiations generally capture the delayed neutrons, provided there are sufficiently many groups available. As the number of groups decreases, the delayed yield decreases.

The average half-life shows that the saturation irradiation tends to over-estimate the average half-life as the number of groups becomes too few. The pulse irradiation oscillates a bit, but tends to under-estimate the average half-life.

Although this example does not exactly represent reality, it provides a useful understanding of the effects of both the pulse and saturation irradiations on the DNP group parameters. The results here provide further motivation to combine the pulse and saturation irradiation data in the DNP group parameter formulation, as discussed in Section 5.1.1.

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